Supplementary material

Dynamic Lifecycle Particle Swarm Optimization: A New Swarm-based Metaheuristic Approach

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Section numbers (e.g. Section 3.1) refer to the sections of the paper. The tables and figures of this material are numbered A1, A2, ...

A Detailed results

This supplementary material gives the per-function results summarised in Section 3 of the paper, together with the supplementary tables and figures referred to in that section. All values are computed over independent runs under the protocol of Section 3. Every run (best value, best solution, convergence curve, run time, seed and platform) is recorded and available from the authors, so that all statistics can be recomputed.

A.1 Velocity functions v_1 and v_2 (Section 3.1)

This part supports Section 3.1. Table **??** and Figure **??** describe the tails of v_1 and v_2 in the pack (part (B)). Figure **??** shows the time per value of the two functions and Table **??** the share of the run time of DLPSO spent in the velocity function (part (A)). Figure **??** shows the mean Friedman ranks for every steepness, and Tables **??–??** ($D = 10$), Tables **??–??** ($D = 30$) and Figure **??** give the per-function results for the default steepness $\alpha = \frac{1}{10}$ (part (C)).

Table A1: Tails of v_1 and v_2 in the pack (5000 animals, 1000 generations): value at birth and at the end of life, and share of the (animal, generation) pairs below 10^{-2}, 10^{-4} and 10^{-8} of the peak

	$v(0) = v(1)$ (birth, end of life)			$v < 10^{-2}$		$v < 10^{-4}$		$v < 10^{-8}$	
Steepness	v_1	v_2	v_2/v_1	v_1	v_2	v_1	v_2	v_1	v_2
$\alpha = 1/5$	0.280	0.280	1.000	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
$\alpha = 1/10$ (default)	$6.2 \cdot 10^{-3}$	0.027	4.301	6.2%	0.0%	0.0%	0.0%	0.0%	0.0%
$\alpha = 1/20$	$1.5 \cdot 10^{-9}$	$1.8 \cdot 10^{-4}$	$1.2 \cdot 10^{5}$	53.8%	41.8%	34.5%	0.0%	6.2%	0.0%
$\alpha = 1/40$	$4.6 \cdot 10^{-36}$	$8.2 \cdot 10^{-9}$	$1.8 \cdot 10^{27}$	76.9%	71.0%	67.3%	48.5%	53.8%	1.7%

The profile is exactly 0 at birth and at the end of life from $1/\alpha = 122$ (v_1) and $1/\alpha = 1491$ (v_2) in double precision.

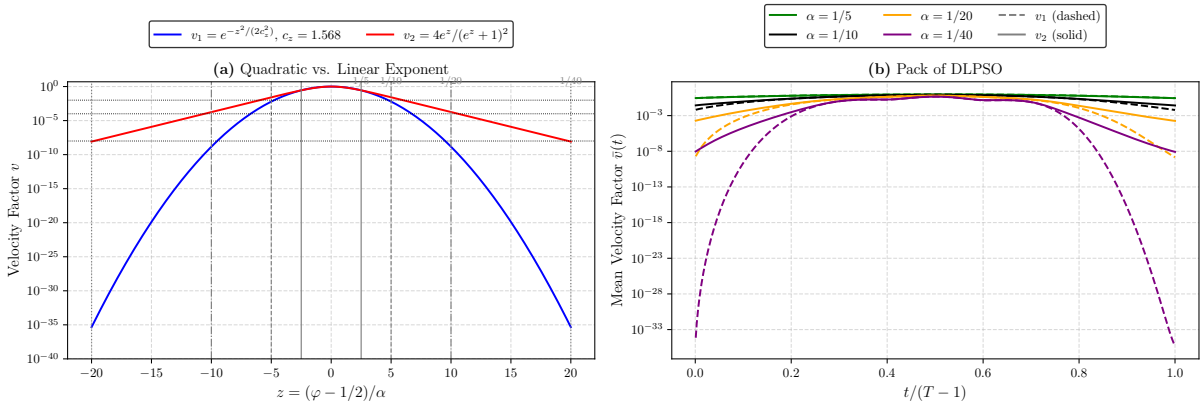


Fig. A1: (a) v_1 and v_2 as functions of $z = (\varphi - \frac{1}{2})/\alpha$: the quadratic exponent of v_1 gives a parabola, the linear exponent of v_2 two straight lines. The vertical lines mark birth and end of life, $|z| = 1/(2\alpha)$, for the values of α of Table **??**. (b) Mean velocity factor $\bar{v}(t)$ of the pack of DLPSO (default development types, dashed: v_1, solid: v_2)

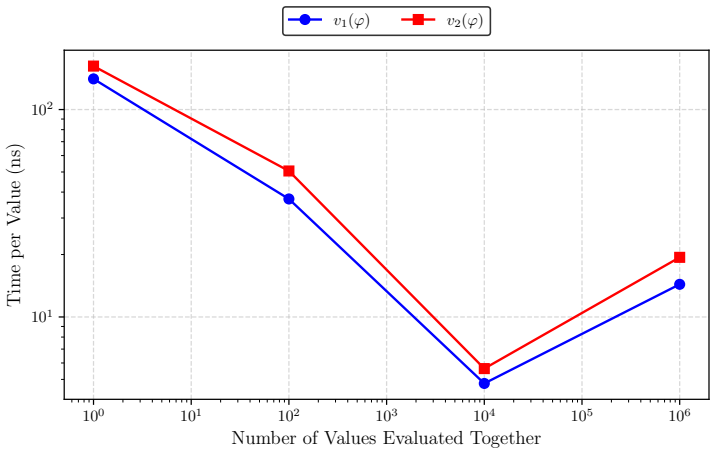


Fig. A2: Time per value of the velocity functions v_1 and v_2 as a function of the number of values evaluated together

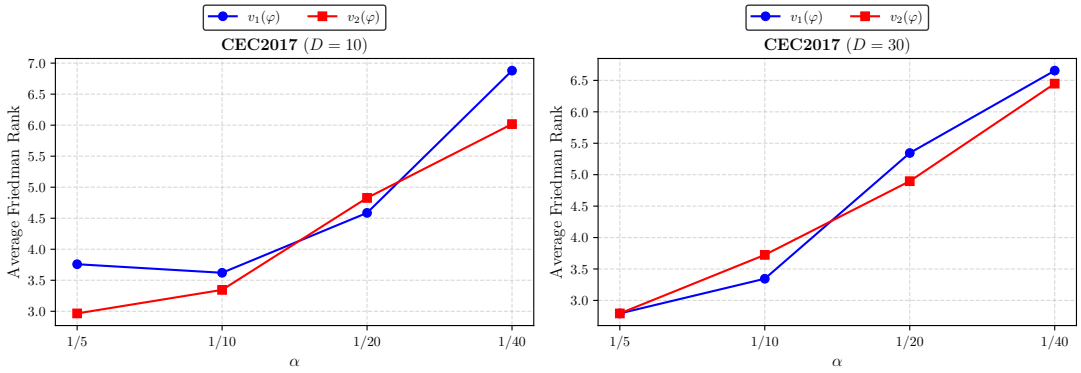
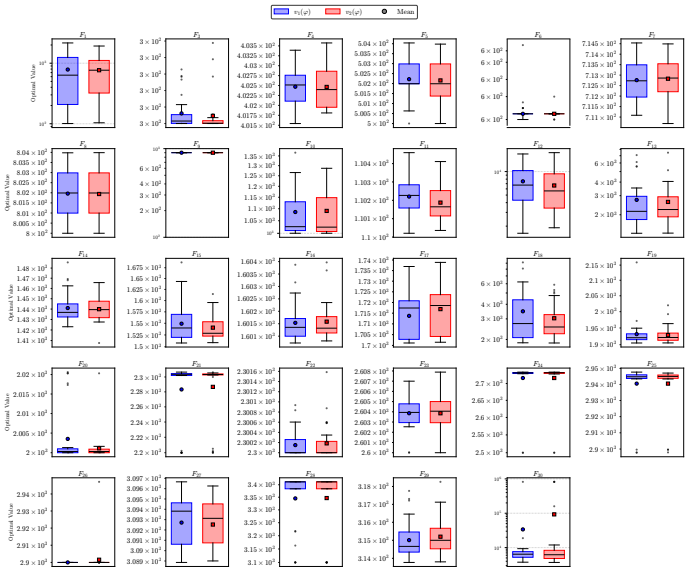


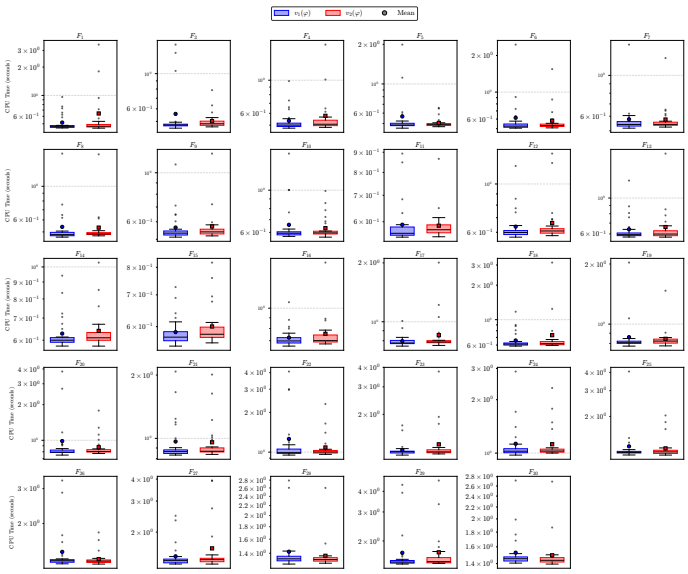
Fig. A3: Mean Friedman ranks of DLPSO with v_1 and with v_2 as functions of the steepness of the profile (CEC2017, $D = 10$ and $D = 30$, lower is better)

Table A2: Results of DLPSO with the velocity profiles v_1 (Gaussian) and v_2 ($\alpha = 1/10$) on CEC2017, $D = 10$: statistics over 30 runs of the best objective value $f(x_{best})$ (MaxFE $= 10000 \cdot D$, 100 animals) (part 1/3)

Profile	Metric	F_1	F_3	F_4	F_5	F_6	F_7	F_8	F_9	F_{10}	F_{11}
v_1 (Gaussian)	Mean	786.027446	300	402.47005	502.217894	600	712.759045	801.960382	900	1086.109557	1102.203403
	Std	618.4	9.83E-08	0.4618	1.042	1.53E-12	1.038	1.071	2.94E-14	96.87	0.969
	Min	100.779028	300	401.543817	500.001832	600	711.085453	800	900	1000.187369	1100.206842
	Q_1	208.151162	300	402.106837	501.989918	600	711.952764	800.994965	900	1010.719113	1101.580439
	Median	634.612607	300	402.514498	501.989918	600	712.728762	801.989918	900	1026.065172	1102.266513
	Q_3	1246.007124	300	402.759498	502.984877	600	713.492276	802.984877	900	1130.121006	1102.845923
	Max	2162.103552	300	403.390706	504.039011	600	714.538224	803.979836	900	1367.204075	1104.597216
	IQR	1038	5.31E-08	0.6527	0.995	1.14E-13	1.54	1.99	0	119.4	1.265
	Time (s)	0.52	0.56	0.53	0.57	0.61	0.58	0.54	0.57	0.66	0.59
v_2 (proposed)	Mean	765.898294	300	402.464817	502.156303	600	712.828718	801.939722	900	1090.726874	1101.875207
	Std	514.8	1.18E-07	0.5559	1.022	3.36E-13	0.9125	1.14	2.08E-14	97.17	1.022
	Min	103.699884	300	401.810474	500	600	710.69188	800	900	1000.249822	1100.36013
	Q_1	321.376495	300	401.952747	501.373095	600	712.207826	800.994963	900	1007.048536	1101.149688
	Median	766.231668	300	402.397784	501.989919	600	712.859637	801.989918	900	1024.23088	1101.645789
	Q_3	1111.049923	300	402.860417	502.984877	600	713.542352	802.984877	900	1148.328212	1102.529461
	Max	1922.654669	300	403.57464	503.97991	600	714.491782	803.986905	900	1286.808917	1104.111414
	IQR	789.7	1.69E-08	0.9077	1.612	0	1.335	1.99	0	141.3	1.38
	Time (s)	0.64	0.50	0.58	0.51	0.58	0.58	0.53	0.57	0.63	0.59



(a) Best value $f(x_{best})$ (30 runs)



(b) CPU time (30 runs)

Fig. A4: Distribution over the runs of the best value $f(x_{best})$ and of the CPU time (seconds) of DLPSO with the velocity functions v_1 and v_2 on CEC2017 ($D = 10$, $\alpha = \frac{1}{10}$, box: first quartile Q_1 to third quartile Q_3, line: median, marker: mean, whiskers: most extreme runs within 1.5 IQR of the box, points: other runs)

Table A3: Results of DLPSO with the velocity profiles v_1 (continued, part 2/3)

Profile	Metric	F_{12}	F_{13}	F_{14}	F_{15}	F_{16}	F_{17}	F_{18}	F_{19}	F_{20}	F_{21}
v_1 (Gaussian)	Mean	7499.868505	2744.063867	1440.968053	1549.199737	1601.535459	1713.744189	3483.1608	1931.405162	2003.519738	2283.217814
	Std	4110	1441	14.91	39.74	0.7183	9.872	1862	44.53	7.073	41.42
	Min	1621.096665	1350.736062	1423.07604	1507.995575	1600.725858	1701.12245	1872.201253	1905.051751	2000	2200.000015
	Q_1	4296.371555	1807.278882	1432.401831	1519.871471	1600.997787	1702.804031	2073.39659	1914.980568	2000	2302.09353
	Median	6711.954229	2150.565847	1436.807336	1539.924109	1601.354538	1717.407718	2735.397704	1920.708772	2000.312433	2303.627062
	Q_3	10306.424512	2957.907933	1444.984148	1569.444743	1601.709311	1720.795348	4370.935921	1932.365737	2000.994959	2304.548085
	Max	16808.689582	7090.591205	1485.650747	1687.531368	1603.873206	1736.937301	9142.390603	2159.138393	2020.624141	2306.582426
	IQR	6010	1151	12.58	49.57	0.7115	17.99	2298	17.39	0.995	2.455
	Time (s)	0.64	0.64	0.63	0.58	0.72	0.80	0.66	0.86	0.99	0.97
v_2 (proposed)	Mean	6633.026652	2625.854803	1439.886078	1540.647042	1601.583871	1716.893519	3036.428561	1928.604623	2001.118462	2286.680856
	Std	3959	1212	11.23	28.62	0.7339	11.07	1141	25.39	3.606	38.05
	Min	1906.860296	1355.013638	1407.703907	1508.716755	1600.809172	1701.469634	1864.91209	1905.254145	2000	2200.000062
	Q_1	3405.064234	1913.571451	1431.734541	1522.528615	1601.128009	1704.146031	2238.286307	1913.104636	2000	2302.760879
	Median	5652.829482	2235.16605	1439.461282	1528.56742	1601.321383	1718.57017	2556.767863	1921.391319	2000.312173	2303.354184
	Q_3	9360.333438	2928.860285	1447.575136	1553.372901	1601.785573	1723.679396	3263.53256	1934.551729	2000.902337	2304.117917
	Max	17466.614629	7467.147442	1465.713407	1614.620444	1603.968344	1738.882998	5881.929418	2019.871032	2020.312182	2306.327524
	IQR	5955	1015	15.84	30.84	0.6576	19.53	1025	21.45	0.9023	1.357
	Time (s)	0.66	0.66	0.64	0.60	0.75	0.85	0.73	0.84	0.87	0.96

Table A4: Results of DLPSO with the velocity profiles v_1 (continued, part 3/3)

Profile	Metric	F_{22}	F_{23}	F_{24}	F_{25}	F_{26}	F_{27}	F_{28}	F_{29}	F_{30}
v_1 (Gaussian)	Mean	2300.149651	2603.860311	2716.626419	2940.512967	2900	3092.714343	3345.202557	3150.075154	33821.451833
	Std	0.2699	1.536	57.91	14.08	5.81E-11	2.371	115	10.32	1.46E+05
	Min	2300	2600.000004	2500.000599	2897.742869	2900	3088.860547	3100	3137.719579	3759.094604
	Q_1	2300	2602.944291	2731.029777	2943.701358	2900	3090.617382	3383.734041	3143.439417	5217.321743
	Median	2300.000001	2603.940407	2732.143305	2944.782561	2900	3093.818765	3411.821808	3146.603731	6417.383884
	Q_3	2300.258822	2604.781421	2732.694692	2946.00285	2900	3094.626776	3411.821808	3154.594913	7624.790182
	Max	2300.936225	2606.99968	2734.46565	2947.690134	2900	3096.659365	3411.821808	3177.523444	8.2059E+05
	IQR	0.2588	1.837	1.665	2.301	1.59E-11	4.009	28.09	11.16	2407
	Time (s)	1.26	1.18	1.13	1.25	1.44	1.46	1.42	1.69	1.53
v_2 (proposed)	Mean	2300.183768	2603.856374	2716.657027	2940.515321	2901.571645	3092.52478	3346.6498	3151.98322	92688.102296
	Std	0.3592	1.759	57.92	13.86	8.464	2.385	123.6	10.35	2.44E+05
	Min	2300	2600.00002	2500.000875	2897.940444	2900	3089.005527	3100	3138.042498	3697.04549
	Q_1	2300	2602.93494	2730.808672	2943.811316	2900	3090.751899	3383.745226	3145.262705	4817.673977
	Median	2300.000003	2604.059857	2732.611592	2944.993963	2900	3093.126069	3411.821808	3149.984013	6195.517904
	Q_3	2300.222986	2605.003977	2733.131769	2945.877123	2900	3094.523083	3411.821808	3156.66341	8329.036134
	Max	2301.585856	2607.888162	2734.525721	2947.088442	2947.149355	3096.257544	3411.821808	3182.596847	8.2060E+05
	IQR	0.223	2.069	2.323	2.066	1.82E-12	3.771	28.08	11.4	3511
	Time (s)	1.09	1.29	1.12	1.21	1.32	1.62	1.37	1.71	1.50

Mean: mean over the runs (the result used in the comparisons and figures); Std: standard deviation; Min, Max: best and worst run; Q_1, Median, Q_3: first quartile, median and third quartile over the runs; IQR $= Q_3 - Q_1$: interquartile range; Time: mean run time (seconds).

Table A5: Results of DLPSO with the velocity profiles v_1 (Gaussian) and v_2 ($\alpha = 1/10$) on CEC2017, $D = 30$: statistics over 30 runs of the best objective value $f(x_{best})$ (MaxFE $= 10000 \cdot D$, 100 animals) (part 1/3)

Profile	Metric	F_1	F_3	F_4	F_5	F_6	F_7	F_8	F_9	F_{10}	F_{11}
v_1 (Gaussian)	Mean	1815.861761	680.344527	497.405455	519.791174	600.000609	748.969898	819.246429	900.28021	3804.022851	1139.362602
	Std	1476	288.6	16.84	4.082	6.66E-04	4.425	4.478	0.4254	404.5	29.84
	Min	129.138499	355.363131	458.56164	512.934469	600	740.934539	808.029794	900	3109.653858	1109.616077
	Q_1	452.960149	456.994472	484.69427	516.373344	600.000138	745.652743	816.914301	900	3486.400637	1114.750739
	Median	1634.775182	588.440032	499.792855	519.917375	600.000329	749.279926	819.231856	900.089528	3901.60657	1120.29542
	Q_3	2356.544333	899.430608	513.310729	521.802616	600.000839	751.053368	822.884084	900.268585	4180.268772	1172.165396
	Max	6379.726673	1374.848704	518.207774	527.893271	600.002419	759.783583	826.908978	901.727747	4532.154315	1191.691657
	IQR	1904	442.4	28.62	5.429	7.01E-04	5.401	5.97	0.2686	693.9	57.41
	Time (s)	1.83	1.86	1.98	1.98	2.08	2.20	1.99	2.15	2.71	2.23
v_2 (proposed)	Mean	2556.141267	644.851638	491.67588	520.513364	600.000815	750.230354	821.605908	900.294909	3651.371542	1135.891452
	Std	2204	267.2	15.79	4.35	1.12E-03	4.593	5.793	0.4866	330.3	28.64
	Min	121.596577	303.909149	458.562699	512.934463	600.000014	739.067459	810.944605	900	3022.589754	1109.411218
	Q_1	429.792113	423.639688	483.673099	517.053278	600.000106	747.127567	818.819944	900	3478.298645	1112.368663
	Median	2661.623167	565.455178	487.263952	520.338491	600.000412	750.13609	819.940668	900.089528	3615.699592	1117.277
	Q_3	4099.685718	802.369746	508.225423	522.884053	600.000779	754.071617	824.522806	900.268585	3818.259235	1170.466225
	Max	7826.222256	1236.312729	516.443797	529.848757	600.004305	759.494282	835.280972	902.085881	4778.069659	1182.816907
	IQR	3670	378.7	24.55	5.831	6.73E-04	6.944	5.703	0.2686	340	58.1
	Time (s)	1.84	1.86	2.02	1.99	2.10	2.22	2.00	2.17	2.74	2.24

Table A6: Results of DLPSO with the velocity profiles v_1 (continued, part 2/3)

Profile	Metric	F_{12}	F_{13}	F_{14}	F_{15}	F_{16}	F_{17}	F_{18}	F_{19}	F_{20}	F_{21}
v_1 (Gaussian)	Mean	40486.804103	10415.661615	2425.514285	2893.286007	1696.587397	1749.289059	87860.021533	6641.132091	2088.023861	2321.157228
	Std	2.911e+04	5790	1163	2521	115.6	15.96	8.807e+04	4020	53.32	5.103
	Min	15754.432918	1637.589006	1561.179111	1508.548666	1603.607055	1727.318886	19944.161989	2087.458205	2026.892306	2310.338344
	Q_1	22450.827899	6259.905528	1691.55503	1586.453733	1615.067371	1737.180743	41555.862756	3282.050389	2049.636296	2317.977808
	Median	33909.812473	9282.050338	1918.660869	2009.230869	1626.448811	1745.621672	66386.692958	5557.790406	2067.58788	2321.072021
	Q_3	43736.999381	13769.668263	2538.816765	2627.351441	1761.945725	1762.488937	97499.55204	8865.991768	2112.89118	2325.299558
	Max	1.5579E+05	25110.452312	7128.664054	14165.565074	2029.026778	1780.699332	5.0211E+05	16595.000697	2226.162044	2332.642981
	IQR	2.129e+04	7510	847.3	1041	146.9	25.31	5.594e+04	5584	63.25	7.322
	Time (s)	2.45	2.38	2.44	2.25	2.78	3.19	2.37	3.71	3.26	3.80
v_2 (proposed)	Mean	39014.312957	10933.014986	2950.930121	2863.92521	1746.488037	1748.446942	77507.466624	6419.946265	2086.965553	2321.029265
	Std	2.619e+04	6490	1988	2150	130.1	13.32	4.162e+04	3862	61.25	4.201
	Min	9591.834129	1885.872479	1545.11714	1505.885226	1607.286067	1729.747099	24779.710469	2041.591076	2009.981574	2313.295797
	Q_1	23171.656937	5994.628219	1771.053047	1622.12927	1623.796612	1736.416522	43937.836644	3172.872805	2031.910472	2317.995089
	Median	33800.762673	10317.513432	1944.947563	1998.364079	1724.572365	1746.649679	73992.812703	5762.453871	2057.529765	2320.049124
	Q_3	42452.631845	13455.916304	3401.374188	2971.678225	1860.674324	1759.49502	1.0221E+05	8309.957019	2151.962314	2323.857576
	Max	1.5131E+05	29085.427953	9307.155221	11863.880847	1967.165119	1781.443852	1.9801E+05	16736.655959	2181.176855	2329.752908
	IQR	1.928e+04	7461	1630	1350	236.9	23.08	5.827e+04	5137	120.1	5.862
	Time (s)	2.47	2.40	2.44	2.26	2.78	3.22	2.37	3.70	3.28	3.81

Table A7: Results of DLPSO with the velocity profiles v_1 (continued, part 3/3)

Profile	Metric	F_{22}	F_{23}	F_{24}	F_{25}	F_{26}	F_{27}	F_{28}	F_{29}	F_{30}
v_1 (Gaussian)	Mean	2300.245763	2673.373843	2839.171124	2887.271491	3811.504508	3211.161685	3180.61322	3395.240709	6005.478007
	Std	0.7373	6.687	5.013	0.8151	87.32	6.087	40.67	36.32	1108
	Min	2300	2653.759696	2830.463238	2883.441985	3675.100995	3200.283796	3100.000428	3342.556073	5142.250303
	Q_1	2300	2668.621241	2835.815391	2887.049636	3744.712155	3207.61533	3187.033621	3375.054195	5332.397396
	Median	2300	2673.430231	2838.816386	2887.320219	3804.550153	3210.221167	3192.897312	3388.502217	5893.003286
	Q_3	2300	2676.769694	2842.0662	2887.646806	3872.998237	3213.397332	3206.743249	3406.506947	6299.869231
	Max	2302.459765	2686.089972	2848.663208	2888.706679	4003.401473	3230.004783	3249.294602	3508.532352	11245.274432
	IQR	4.55E-13	8.148	6.251	0.5972	128.3	5.782	19.71	31.45	967.5
	Time (s)	4.62	5.49	4.82	5.26	6.44	6.75	6.44	6.28	6.42
v_2 (proposed)	Mean	2300.491099	2672.091493	2840.093194	2887.353362	3834.394275	3210.103311	3182.30373	3402.981881	5947.556331
	Std	0.9822	7.417	4.28	0.8895	84.8	6.592	38.76	38.25	1120
	Min	2300	2658.64298	2829.074113	2883.57313	3715.028794	3197.869979	3100.000011	3334.067427	5126.964162
	Q_1	2300	2665.202177	2837.521594	2887.053098	3765.14777	3206.628217	3189.334977	3381.885448	5296.347189
	Median	2300	2672.34895	2840.422001	2887.286159	3832.621257	3208.71479	3192.045763	3398.096416	5675.186994
	Q_3	2300	2678.026281	2842.014387	2887.742533	3895.860101	3211.464039	3207.260216	3416.94738	6148.868312
	Max	2302.459765	2683.676751	2852.789584	2888.980943	4060.562972	3229.801135	3240.126519	3512.130775	11234.011694
	IQR	4.55E-13	12.82	4.493	0.6894	130.7	4.836	17.93	35.06	852.5
	Time (s)	4.61	5.51	4.78	5.31	6.45	6.78	6.40	6.30	6.37

Mean: mean over the runs (the result used in the comparisons and figures); Std: standard deviation; Min, Max: best and worst run; Q_1, Median, Q_3: first quartile, median and third quartile over the runs; IQR $= Q_3 - Q_1$: interquartile range; Time: mean run time (seconds).

Table A8: Share of the run time of DLPSO spent in the velocity profile (CEC2017, mean over the 29 functions and 30 runs)

D	Profile	Mean run time (s)	Mean time in the profile (s)	Share of the run time	Profile evaluations per run
10	v_1 (Gaussian)	0.881	0.0131	1.72%	100 000
	v_2 (proposed)	0.879	0.0200	2.62%	100 000
30	v_1 (Gaussian)	3.523	0.0391	1.32%	300 000
	v_2 (proposed)	3.532	0.0580	1.98%	300 000

A.2 Sensitivity analysis (Section 3.2)

This part supports Section 3.2. Tables **??–??** give, for every value of every parameter, the mean Friedman rank, the counts $+/=/-$ of the default and the Holm-adjusted p-value of the signed-rank test. Tables **??–??** and Figures **??–??** describe the convergence speed for every value of every parameter.

Table A9: Sensitivity of DLPSO to its parameters (one parameter changed at a time, 30 runs): mean Friedman rank within the parameter, $+/=/-$ of the default (rank-sum tests) and p_{Holm} of the signed-rank test of the default against the value (part 1/4)

		Classical, $D=10$			CEC2017, $D=10$			CEC2017, $D=30$		
Parameter	Value	Mean rank	Default $+/=/-$	p_{Holm}	Mean rank	Default $+/=/-$	p_{Holm}	Mean rank	Default $+/=/-$	p_{Holm}
Proportions $\eta=(\eta_E,\eta_B,\eta_L)$	default	9.94	–	–	9.14	–	–	8.03	–	–
	$\eta=(0.00,0.00,1.00)$	6.22	1/9/8	1.0000	9.48	8/16/5	1.0000	4.72	3/10/16	1.0000
	$\eta=(0.00,0.25,0.75)$	5.97	1/9/8	0.5477	9.34	5/20/4	1.0000	5.50	0/20/9	1.0000
	$\eta=(0.00,0.50,0.50)$	8.72	1/14/3	1.0000	9.62	3/22/4	1.0000	7.60	0/24/5	1.0000
	$\eta=(0.00,0.75,0.25)$	7.25	2/14/2	1.0000	8.59	4/23/2	1.0000	5.91	0/24/5	0.1032
	$\eta=(0.00,1.00,0.00)$	6.56	2/11/5	1.0000	8.07	5/23/1	1.0000	6.16	0/22/7	1.0000
	$\eta=(0.25,0.00,0.75)$	10.19	5/12/1	1.0000	8.59	0/27/2	1.0000	9.62	2/25/2	1.0000
	$\eta=(0.25,0.25,0.50)$	11.22	2/15/1	1.0000	9.34	0/28/1	1.0000	10.03	1/28/0	1.0000
	$\eta=(0.25,0.50,0.25)$	9.92	1/17/0	1.0000	8.09	0/29/0	1.0000	10.03	2/27/0	0.4893
	$\eta=(0.25,0.75,0.00)$	8.19	1/15/2	1.0000	8.84	0/29/0	1.0000	9.45	2/25/2	1.0000
	$\eta=(0.50,0.00,0.50)$	12.67	7/11/0	1.0000	10.17	6/22/1	1.0000	12.00	9/19/1	0.7473
	$\eta=(0.50,0.25,0.25)$	11.17	1/17/0	1.0000	9.31	6/22/1	1.0000	11.24	7/22/0	0.9879
	$\eta=(0.50,0.50,0.00)$	8.33	2/13/3	1.0000	8.47	2/25/2	1.0000	10.34	5/23/1	1.0000
	$\eta=(0.75,0.00,0.25)$	9.56	2/16/0	1.0000	8.38	4/25/0	1.0000	11.72	11/17/1	1.0000
	$\eta=(0.75,0.25,0.00)$	8.50	2/12/4	1.0000	9.45	7/20/2	1.0000	10.55	8/20/1	1.0000
	$\eta=(1.00,0.00,0.00)$	7.17	3/9/6	1.0000	8.12	7/19/3	1.0000	10.45	8/19/2	1.0000
	$\eta=(0.33,0.33,0.33)$	11.42	0/17/1	1.0000	10.00	3/26/0	1.0000	9.62	3/26/0	1.0000
		Friedman test: Classical, $D=10$: $p=0.0002$; CEC2017, $D=10$: $p=0.9362$; CEC2017, $D=30$: $p=<0.0001$								
Prime times b_E,b_B,b_L (spread s)	$s=0.05$, $(b_E,b_B,b_L)=(0.45,0.5,0.55)$	2.22	2/11/5	0.9114	3.05	3/24/2	0.6312	2.10	0/23/6	0.6487
	$s=0.1$, $(b_E,b_B,b_L)=(0.4,0.5,0.6)$	2.67	1/16/1	1.0000	3.19	0/29/0	0.6312	2.83	1/26/2	0.9126
	default	3.03	–	–	2.98	–	–	3.03	–	–
	$s=0.2$, $(b_E,b_B,b_L)=(0.3,0.5,0.7)$	3.47	1/17/0	0.4957	2.88	0/27/2	0.5454	3.21	5/24/0	0.9126
	$s=0.25$, $(b_E,b_B,b_L)=(0.25,0.5,0.75)$	3.61	5/12/1	1.0000	2.90	4/24/1	0.6312	3.83	8/21/0	0.9126
		Friedman test: Classical, $D=10$: $p=0.0501$; CEC2017, $D=10$: $p=0.9419$; CEC2017, $D=30$: $p=0.0012$								
Individual deviation $\delta_i\sim\mathscr{U}(-\Delta_b,\Delta_b)$	$\Delta_b=0$	2.69	0/18/0	1.0000	2.71	0/29/0	0.4165	2.26	0/26/3	1.0000
	$\Delta_b=0.025$	1.83	0/18/0	1.0000	2.67	0/29/0	0.9048	2.28	0/29/0	1.0000
	default	2.39	–	–	2.36	–	–	2.60	–	–
	$\Delta_b=0.1$	3.08	0/18/0	1.0000	2.26	1/27/1	0.9048	2.86	1/27/1	1.0000
		Friedman test: Classical, $D=10$: $p=0.0284$; CEC2017, $D=10$: $p=0.4402$; CEC2017, $D=30$: $p=0.2242$								

Table A10: Sensitivity of DLPSO to its parameters (continued, part 2/4)

Parameter	Value	Classical, $D=10$ Mean rank	Classical, $D=10$ Default $+/=/-$	Classical, $D=10$ p_{Holm}	CEC2017, $D=10$ Mean rank	CEC2017, $D=10$ Default $+/=/-$	CEC2017, $D=10$ p_{Holm}	CEC2017, $D=30$ Mean rank	CEC2017, $D=30$ Default $+/=/-$	CEC2017, $D=30$ p_{Holm}
Stage thresholds P_1, P_2	$(P_1,P_2)=(0.1,0.9)$	5.14	11/6/1	0.3327	3.66	10/17/2	1.0000	4.24	16/11/2	0.4618
	default	3.42	–	–	3.17	–	–	3.41	–	–
	$(P_1,P_2)=(0.3,0.7)$	2.36	1/8/9	1.0000	3.50	2/22/5	0.8090	2.97	3/20/6	1.0000
	$(P_1,P_2)=(0.4,0.6)$	2.83	1/6/11	1.0000	4.10	5/21/3	0.1209	3.10	6/15/8	1.0000
	$(P_1,P_2)=(0.1,0.7)$	3.64	4/11/3	1.0000	3.36	8/19/2	1.0000	4.66	17/10/2	0.3799
	$(P_1,P_2)=(0.3,0.9)$	3.61	7/10/1	1.0000	3.21	5/21/3	0.8231	2.62	2/20/7	1.0000
		Friedman test: Classical, $D=10$: $p=0.0003$; CEC2017, $D=10$: $p=0.4061$; CEC2017, $D=30$: $p=<0.0001$								
Steepness of ν_2 (α)	$\alpha=1/5$	3.31	4/13/1	0.5790	1.86	0/27/2	0.5090	1.55	1/26/2	0.1674
	default	2.75	–	–	2.00	–	–	2.14	–	–
	$\alpha=1/20$	2.22	1/8/9	0.6076	2.76	2/27/0	0.0055	2.79	3/26/0	0.1674
	$\alpha=1/40$	1.72	2/6/10	0.6076	3.38	11/17/1	0.0031	3.52	8/21/0	0.0038
		Friedman test: Classical, $D=10$: $p=0.0017$; CEC2017, $D=10$: $p=<0.0001$; CEC2017, $D=30$: $p=<0.0001$								
Maximal loudness $A_i^0 \sim \mathscr{U}(0, A_{\max})$	$A_{\max}=0.25$	1.72	1/15/2	1.0000	2.43	0/28/1	0.9491	2.10	0/29/0	1.0000
	$A_{\max}=0.5$	2.75	0/17/1	1.0000	2.43	0/29/0	0.9491	2.83	0/29/0	1.0000
	$A_{\max}=0.75$	2.69	0/18/0	1.0000	2.71	0/29/0	0.9491	2.28	0/29/0	1.0000
	default	2.83	–	–	2.43	–	–	2.79	–	–
		Friedman test: Classical, $D=10$: $p=0.0311$; CEC2017, $D=10$: $p=0.7943$; CEC2017, $D=30$: $p=0.0728$								
Loudness at the end of life (ρ)	$\rho=0$	2.25	1/16/1	1.0000	3.12	0/29/0	1.0000	3.10	0/29/0	1.0000
	$\rho=0.25$	3.22	0/18/0	1.0000	2.93	0/29/0	1.0000	3.52	0/29/0	0.4966
	default	2.86	–	–	2.84	–	–	2.83	–	–
	$\rho=0.75$	3.00	0/18/0	1.0000	3.22	0/29/0	1.0000	2.55	0/29/0	1.0000
	$\rho=1$	3.67	0/18/0	0.4335	2.88	0/29/0	1.0000	3.00	0/29/0	1.0000
		Friedman test: Classical, $D=10$: $p=0.0988$; CEC2017, $D=10$: $p=0.8574$; CEC2017, $D=30$: $p=0.1751$								
Mass $w_i \sim \mathscr{U}(1-\delta, 1+\delta)$	$\delta=0$	2.50	1/16/1	0.6705	3.34	0/29/0	1.0000	3.48	1/27/1	1.0000
	$\delta=0.1$	3.19	0/18/0	0.4924	3.45	0/29/0	1.0000	3.45	0/29/0	1.0000
	default	2.33	–	–	3.09	–	–	3.62	–	–
	$\delta=0.3$	3.39	2/16/0	0.0359	2.81	0/28/1	1.0000	3.31	1/28/0	1.0000
	$\delta=0.4$	4.75	11/7/0	0.0042	3.98	2/25/2	1.0000	3.34	5/23/1	0.6866
	$\delta=0.5$	4.83	10/8/0	0.1295	4.33	4/23/2	0.3785	3.79	7/21/1	1.0000
		Friedman test: Classical, $D=10$: $p=<0.0001$; CEC2017, $D=10$: $p=0.0178$; CEC2017, $D=30$: $p=0.9292$								

Table A11: Sensitivity of DLPSO to its parameters (continued, part 3/4)

Parameter	Value	Classical, $D=10$ Mean rank	Classical, $D=10$ Default $+/=/-$	Classical, $D=10$ p_{Holm}	CEC2017, $D=10$ Mean rank	CEC2017, $D=10$ Default $+/=/-$	CEC2017, $D=10$ p_{Holm}	CEC2017, $D=30$ Mean rank	CEC2017, $D=30$ Default $+/=/-$	CEC2017, $D=30$ p_{Holm}
Noise $r \sim \mathscr{U}(\underline{r}, \overline{r})$	$r \sim \mathscr{U}(0.25, 0.75)$	3.17	12/3/3	0.0770	3.07	17/7/5	0.0259	3.41	22/5/2	<0.0001
	$r \sim \mathscr{U}(0,1)$	2.22	11/4/3	0.0770	2.66	17/7/5	0.0167	2.97	22/5/2	<0.0001
	default	1.67	–	–	1.81	–	–	1.21	–	–
	$r \sim \mathscr{U}(-0.75, 1.75)$	2.94	14/4/0	0.0039	2.47	15/9/5	0.0259	2.41	21/8/0	<0.0001
		Friedman test: Classical, $D=10$: $p=0.0016$; CEC2017, $D=10$: $p=0.0024$; CEC2017, $D=30$: $p=<0.0001$								
Maximum velocity factor $v_{\max}$	$v_{\max}=0.25$	2.28	1/7/10	1.0000	3.14	1/27/1	1.0000	3.24	0/28/1	1.0000
	$v_{\max}=0.5$	3.00	1/12/5	1.0000	3.19	1/28/0	0.8090	3.17	0/29/0	1.0000
	default	2.89	–	–	2.90	–	–	3.10	–	–
	$v_{\max}=2$	3.17	6/11/1	1.0000	3.02	2/27/0	1.0000	2.83	2/26/1	1.0000
	$v_{\max}=4$	3.67	9/9/0	0.6162	2.76	2/24/3	1.0000	2.66	1/27/1	1.0000
		Friedman test: Classical, $D=10$: $p=0.1196$; CEC2017, $D=10$: $p=0.8321$; CEC2017, $D=30$: $p=0.5800$								
Initial step factor κ_0	$\kappa_0=0.1$	2.00	2/15/1	0.7623	2.24	2/26/1	0.2046	1.93	1/28/0	0.9098
	default	1.92	–	–	1.86	–	–	2.00	–	–
	$\kappa_0=1$	2.08	0/18/0	0.7623	1.90	1/28/0	0.4279	2.07	0/29/0	0.9098
		Friedman test: Classical, $D=10$: $p=0.8809$; CEC2017, $D=10$: $p=0.2540$; CEC2017, $D=30$: $p=0.8712$								
Memory of the step factor H	$H=1$	1.89	3/9/6	0.7987	2.45	1/25/3	0.2022	2.38	3/26/0	0.4107
	default	2.17	–	–	1.84	–	–	1.76	–	–
	$H=10$	1.94	0/18/0	0.3081	1.71	0/28/1	0.7186	1.86	0/27/2	0.4107
		Friedman test: Classical, $D=10$: $p=0.6778$; CEC2017, $D=10$: $p=0.0090$; CEC2017, $D=30$: $p=0.0405$								
Standard deviation σ_κ of $\log_{10}\kappa_i^t$ around the memory	$\sigma_\kappa=0.1$	2.47	7/11/0	0.4924	2.10	2/27/0	0.8554	1.93	1/25/3	1.0000
	default	1.89	–	–	1.97	–	–	1.97	–	–
	$\sigma_\kappa=0.5$	1.64	1/17/0	0.7337	1.93	1/28/0	0.4910	2.10	1/28/0	1.0000
		Friedman test: Classical, $D=10$: $p=0.0355$; CEC2017, $D=10$: $p=0.7788$; CEC2017, $D=30$: $p=0.7855$								
Travel probability of the old animals μ_i^t	$\mu_i^t=0.1$ (constant)	2.61	8/10/0	0.0770	2.38	5/24/0	0.6550	2.60	1/28/0	0.0061
	$\mu_i^t=0.5$ (constant)	1.56	0/17/1	0.6397	1.66	0/29/0	0.6550	1.71	0/29/0	0.3505
	default	1.83	–	–	1.97	–	–	1.69	–	–
		Friedman test: Classical, $D=10$: $p=0.0046$; CEC2017, $D=10$: $p=0.0190$; CEC2017, $D=30$: $p=0.0003$								

Table A12: Sensitivity of DLPSO to its parameters (continued, part 4/4)

Parameter	Value	Classical, $D=10$ Mean rank	Classical, $D=10$ Default $+/=/-$	Classical, $D=10$ p_{Holm}	CEC2017, $D=10$ Mean rank	CEC2017, $D=10$ Default $+/=/-$	CEC2017, $D=10$ p_{Holm}	CEC2017, $D=30$ Mean rank	CEC2017, $D=30$ Default $+/=/-$	CEC2017, $D=30$ p_{Holm}
Resting radius of the old animals ε	$\varepsilon=0.25$	2.53	7/11/0	0.0072	2.07	3/26/0	0.2104	2.17	0/29/0	1.0000
	default	1.36	–	–	1.59	–	–	1.91	–	–
	$\varepsilon=1$	2.11	0/18/0	0.3927	2.34	0/29/0	0.0029	1.91	0/29/0	1.0000
				Friedman test: Classical, $D=10$: $p=0.0017$; CEC2017, $D=10$: $p=0.0119$; CEC2017, $D=30$: $p=0.5088$						
Population size N	$N=30$	2.47	6/3/9	0.8846	3.50	13/14/2	<0.0001	3.34	13/14/2	0.0038
	$N=50$	2.08	1/6/11	0.8846	2.21	5/18/6	0.4727	2.14	2/21/6	0.3579
	default	2.33	–	–	1.98	–	–	1.86	–	–
	$N=200$	3.11	14/3/1	0.0624	2.31	15/10/4	0.4727	2.66	14/13/2	0.1310
				Friedman test: Classical, $D=10$: $p=0.1000$; CEC2017, $D=10$: $p=<0.0001$; CEC2017, $D=30$: $p=<0.0001$						

Table A13: Convergence speed of DLPSO for every value of each parameter (t_{50}, t_{90} and final progress; mean over the functions of a suite, 30 runs) (part 1/3)

Parameter	Value	Classical, $D=10$ t_{50}	Classical, $D=10$ t_{90}	Classical, $D=10$ final	CEC2017, $D=10$ t_{50}	CEC2017, $D=10$ t_{90}	CEC2017, $D=10$ final	CEC2017, $D=30$ t_{50}	CEC2017, $D=30$ t_{90}	CEC2017, $D=30$ final
Proportions $\eta=(\eta_E,\eta_B,\eta_L)$	default	0.18	0.80	0.93	0.14	0.88	0.93	0.10	0.84	0.92
	$\eta=(0.00,0.00,1.00)$	0.20	0.72	0.95	0.16	0.92	0.91	0.14	0.60	0.96
	$\eta=(0.00,0.25,0.75)$	0.20	0.70	0.95	0.16	0.90	0.92	0.14	0.60	0.96
	$\eta=(0.00,0.50,0.50)$	0.20	0.80	0.94	0.16	0.96	0.91	0.14	0.78	0.93
	$\eta=(0.00,0.75,0.25)$	0.20	0.70	0.98	0.16	0.88	0.92	0.14	0.74	0.93
	$\eta=(0.00,1.00,0.00)$	0.20	0.72	0.96	0.16	0.88	0.92	0.14	0.76	0.93
	$\eta=(0.25,0.00,0.75)$	0.18	0.88	0.92	0.14	0.80	0.94	0.10	0.88	0.92
	$\eta=(0.25,0.25,0.50)$	0.18	0.84	0.92	0.14	0.90	0.92	0.10	0.88	0.91
	$\eta=(0.25,0.50,0.25)$	0.18	0.84	0.92	0.14	0.88	0.92	0.10	0.88	0.91
	$\eta=(0.25,0.75,0.00)$	0.18	0.80	0.93	0.12	0.88	0.92	0.10	0.84	0.91
	$\eta=(0.50,0.00,0.50)$	0.18	0.94	0.91	0.12	0.90	0.92	0.10	–	0.89
	$\eta=(0.50,0.25,0.25)$	0.16	0.88	0.91	0.12	0.90	0.92	0.10	–	0.90
	$\eta=(0.50,0.50,0.00)$	0.18	0.80	0.93	0.12	0.84	0.93	0.10	0.92	0.91
	$\eta=(0.75,0.00,0.25)$	0.16	0.82	0.92	0.12	0.88	0.92	0.10	–	0.89
	$\eta=(0.75,0.25,0.00)$	0.16	0.84	0.92	0.12	0.88	0.92	0.10	–	0.89
	$\eta=(1.00,0.00,0.00)$	0.16	0.78	0.92	0.12	0.82	0.93	0.10	–	0.89
	$\eta=(0.33,0.33,0.33)$	0.18	0.90	0.91	0.12	0.90	0.93	0.10	0.90	0.91
Prime times b_E,b_B,b_L (spread s)	$s=0.05$, $(b_E,b_B,b_L)=(0.45,0.5,0.55)$	0.18	0.66	0.99	0.14	0.82	0.95	0.12	0.46	0.99
	$s=0.1$, $(b_E,b_B,b_L)=(0.4,0.5,0.6)$	0.18	0.68	0.97	0.14	0.76	0.96	0.12	0.50	0.97
	default	0.18	0.72	0.96	0.14	0.76	0.96	0.10	0.56	0.96
	$s=0.2$, $(b_E,b_B,b_L)=(0.3,0.5,0.7)$	0.18	0.76	0.95	0.12	0.74	0.96	0.10	0.62	0.95
	$s=0.25$, $(b_E,b_B,b_L)=(0.25,0.5,0.75)$	0.20	0.76	0.95	0.12	0.72	0.97	0.08	0.70	0.94
Individual deviation $\delta_i\sim\mathscr{U}(-\Delta_b,\Delta_b)$	$\Delta_b=0$	0.18	0.68	0.98	0.12	0.72	0.97	0.10	0.46	0.99
	$\Delta_b=0.025$	0.18	0.64	0.99	0.12	0.76	0.96	0.10	0.46	0.98
	default	0.18	0.66	0.99	0.12	0.70	0.98	0.10	0.46	0.98
	$\Delta_b=0.1$	0.18	0.70	0.98	0.12	0.72	0.97	0.10	0.50	0.98

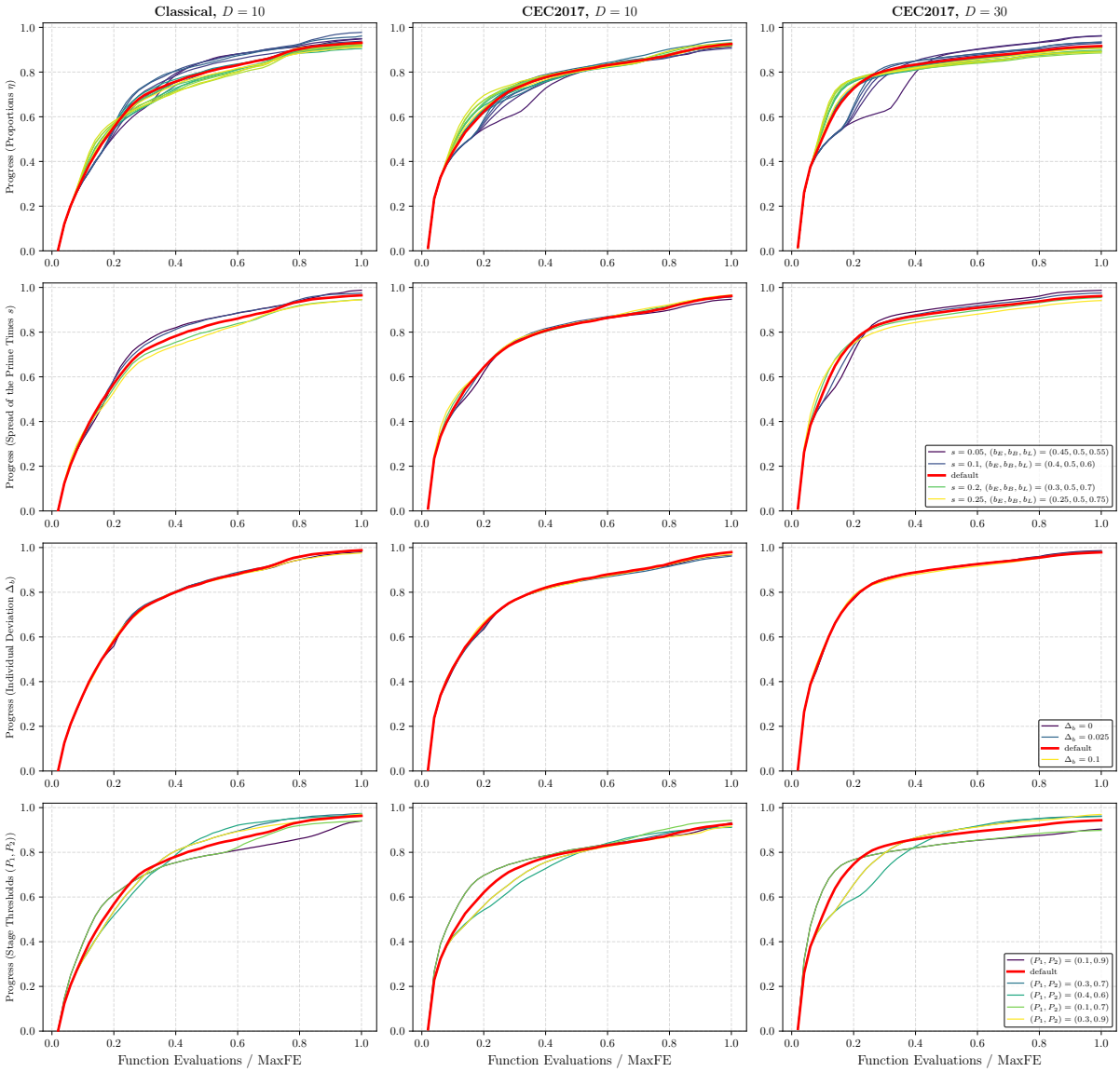


Fig. A5: Normalised convergence progress of DLPSO for every value of the parameters η, s, Δ_b, (P_1, P_2) (rows) on the classical functions and on CEC2017 with $D = 10$ and $D = 30$ (columns), averaged over the functions of the suite and over 30 runs (red: default, 1: best final value among the values of the parameter) (part 1/4)

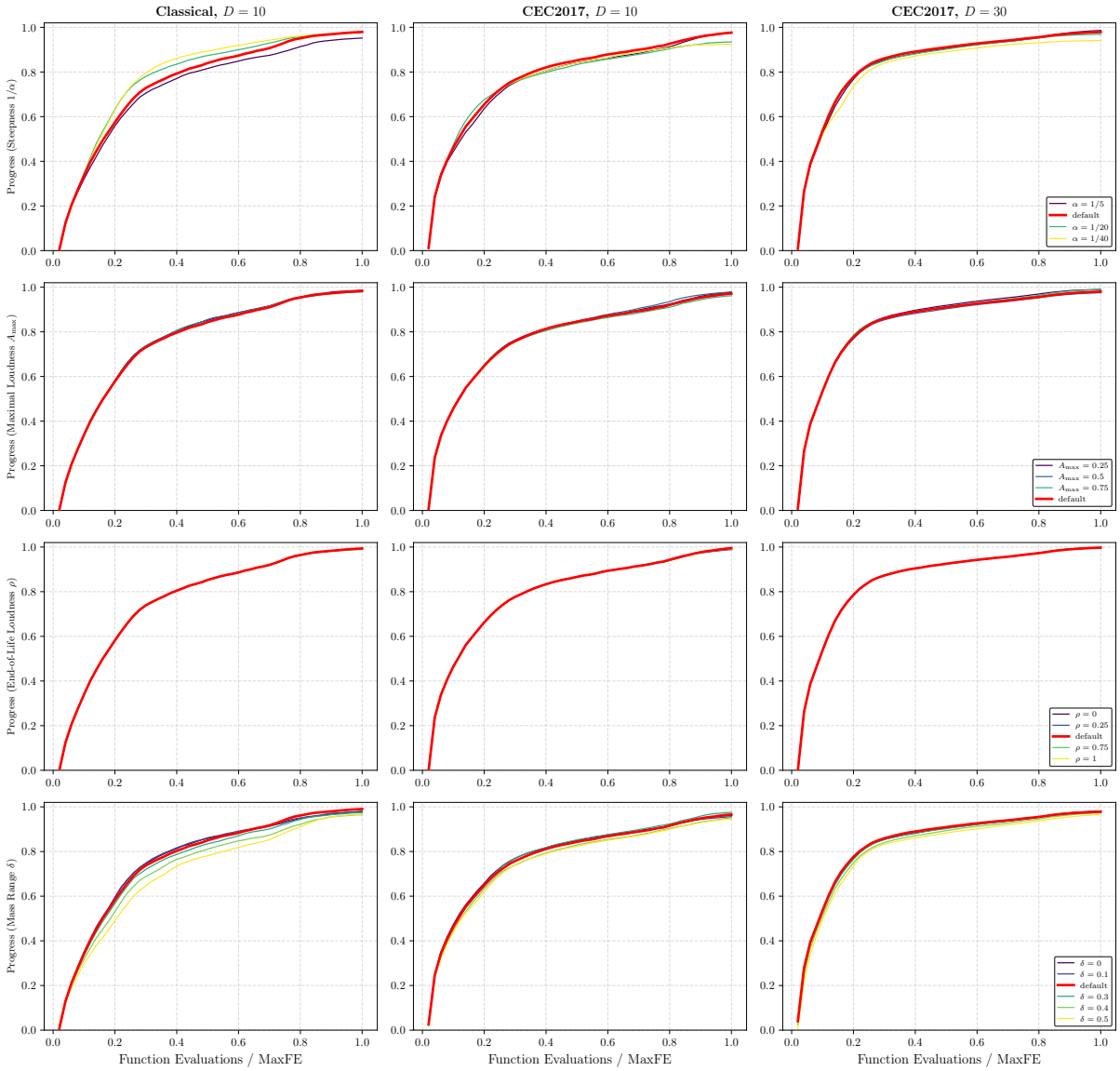


Fig. A6: Normalised convergence progress of DLPSO for every value of the parameters α, $A_{\max}$, ρ, δ (rows) on the classical functions and on CEC2017 with $D = 10$ and $D = 30$ (columns), averaged over the functions of the suite and over 30 runs (red: default, 1: best final value among the values of the parameter) (part 2/4)

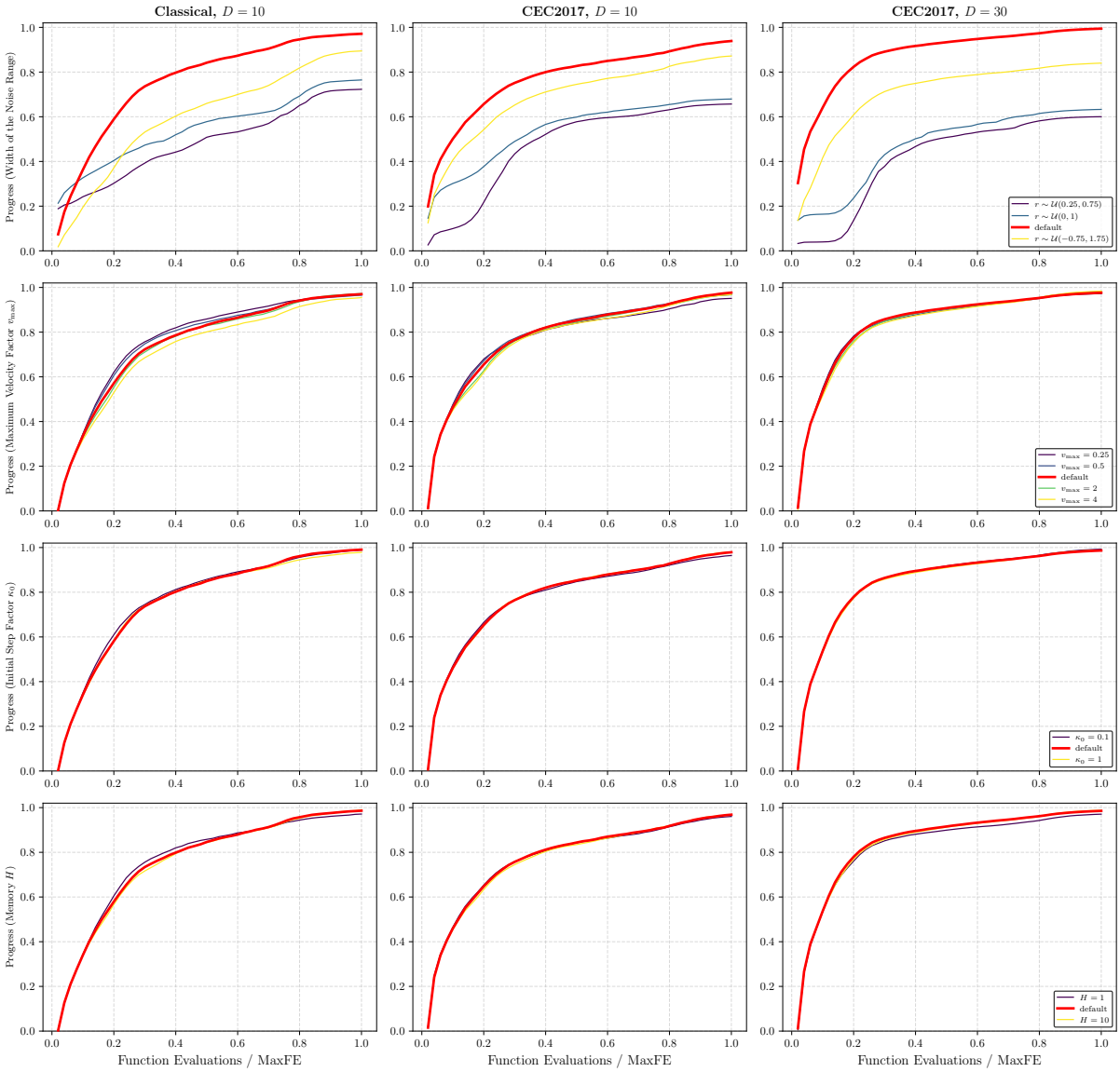


Fig. A7: Normalised convergence progress of DLPSO for every value of the parameters noise width c_r, $v_{\max}$, κ_0, H (rows) on the classical functions and on CEC2017 with $D = 10$ and $D = 30$ (columns), averaged over the functions of the suite and over 30 runs (red: default, 1: best final value among the values of the parameter) (part 3/4)

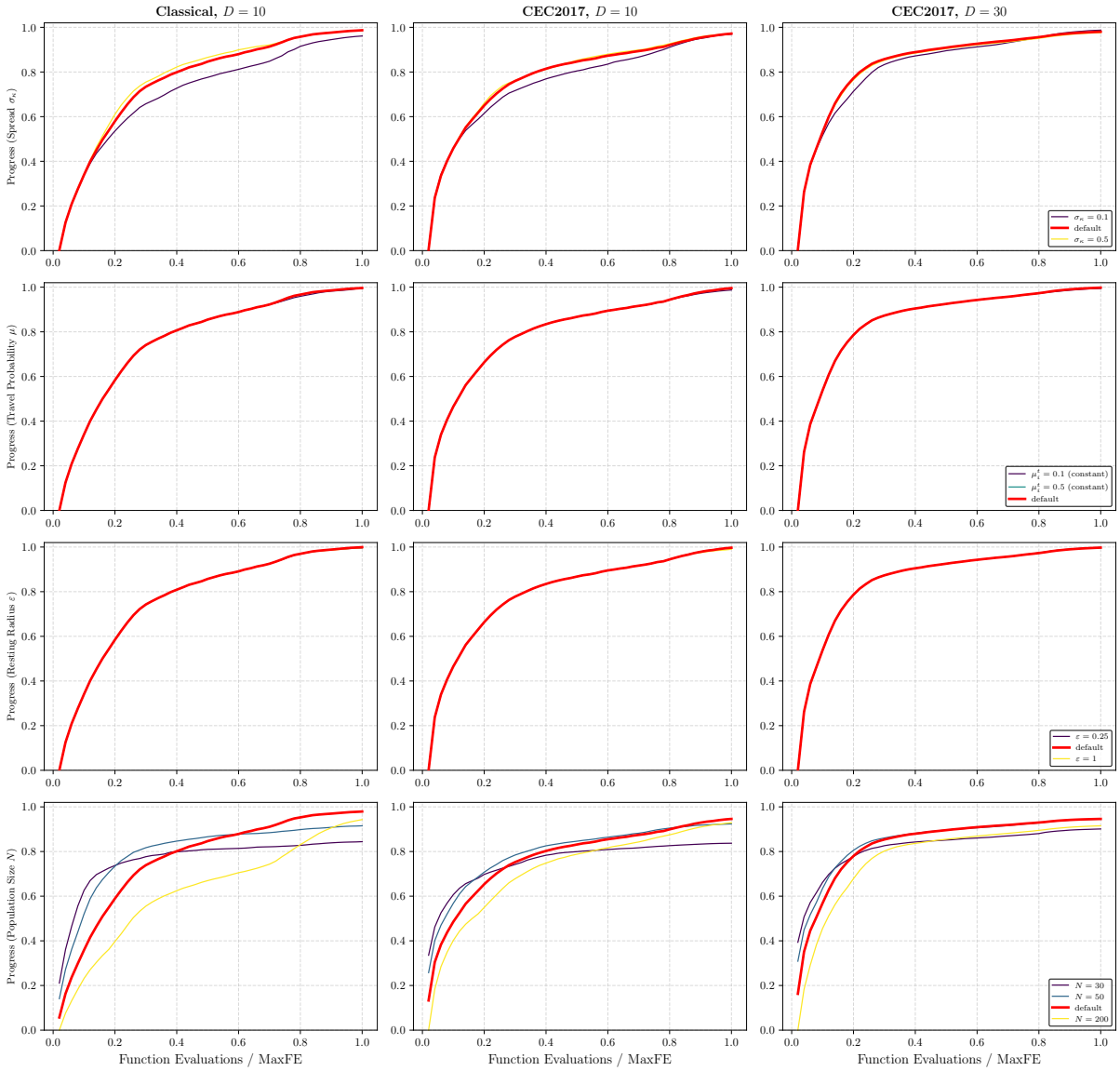


Fig. A8: Normalised convergence progress of DLPSO for every value of the parameters σ_κ, μ_i^t, ε, N (rows) on the classical functions and on CEC2017 with $D = 10$ and $D = 30$ (columns), averaged over the functions of the suite and over 30 runs (red: default, 1: best final value among the values of the parameter) (part 4/4)

Table A14: Convergence speed of DLPSO for every value of each parameter (continued, part 2/3)

		Classical, $D = 10$			CEC2017, $D = 10$			CEC2017, $D = 30$		
Parameter	Value	t_{50}	t_{90}	final	t_{50}	t_{90}	final	t_{50}	t_{90}	final
Stage thresholds P_1, P_2	$(P_1, P_2) = (0.1, 0.9)$	0.14	0.90	0.94	0.10	0.92	0.93	0.08	0.96	0.90
	default	0.18	0.72	0.96	0.14	0.88	0.93	0.10	0.66	0.94
	$(P_1, P_2) = (0.3, 0.7)$	0.20	0.62	0.97	0.16	0.88	0.91	0.12	0.54	0.96
	$(P_1, P_2) = (0.4, 0.6)$	0.20	0.56	0.97	0.16	0.86	0.91	0.12	0.54	0.96
	$(P_1, P_2) = (0.1, 0.7)$	0.14	0.76	0.94	0.10	0.78	0.94	0.08	–	0.90
	$(P_1, P_2) = (0.3, 0.9)$	0.20	0.64	0.97	0.16	0.94	0.92	0.12	0.54	0.97
Steepness of v_2 (α)	$\alpha = 1/5$	0.18	0.78	0.95	0.14	0.76	0.98	0.10	0.48	0.99
	default	0.18	0.68	0.98	0.12	0.72	0.98	0.10	0.46	0.98
	$\alpha = 1/20$	0.16	0.62	0.98	0.12	0.80	0.93	0.10	0.50	0.97
	$\alpha = 1/40$	0.16	0.54	0.98	0.12	0.76	0.92	0.10	0.56	0.94
Maximal loudness $A_i^0 \sim \mathscr{U}(0, A_{\max})$	$A_{\max} = 0.25$	0.18	0.66	0.99	0.12	0.72	0.98	0.10	0.42	0.99
	$A_{\max} = 0.5$	0.18	0.66	0.98	0.12	0.70	0.98	0.10	0.50	0.98
	$A_{\max} = 0.75$	0.18	0.66	0.98	0.12	0.78	0.96	0.10	0.44	0.99
	default	0.18	0.68	0.98	0.12	0.74	0.97	0.10	0.46	0.98
Loudness at the end of life (ρ)	$\rho = 0$	0.18	0.64	0.99	0.12	0.64	0.99	0.10	0.40	1.00
	$\rho = 0.25$	0.18	0.64	0.99	0.12	0.64	0.99	0.10	0.40	1.00
	default	0.18	0.64	0.99	0.12	0.64	0.99	0.10	0.40	1.00
	$\rho = 0.75$	0.18	0.64	0.99	0.12	0.64	0.99	0.10	0.40	1.00
	$\rho = 1$	0.18	0.64	0.99	0.12	0.64	0.99	0.10	0.40	1.00
Mass $w_i \sim \mathscr{U}(1-\delta, 1+\delta)$	$\delta = 0$	0.16	0.66	0.98	0.12	0.74	0.96	0.10	0.48	0.98
	$\delta = 0.1$	0.16	0.64	0.98	0.12	0.76	0.96	0.10	0.48	0.98
	default	0.16	0.66	0.99	0.12	0.76	0.97	0.10	0.46	0.98
	$\delta = 0.3$	0.18	0.70	0.97	0.12	0.72	0.98	0.10	0.50	0.98
	$\delta = 0.4$	0.20	0.76	0.96	0.14	0.80	0.95	0.10	0.54	0.97
	$\delta = 0.5$	0.22	0.78	0.97	0.14	0.82	0.95	0.12	0.60	0.97
Noise $r \sim \mathscr{U}(\underline{r}, \bar{r})$	$r \sim \mathscr{U}(0.25, 0.75)$	0.50	–	0.72	0.38	–	0.66	0.48	–	0.60
	$r \sim \mathscr{U}(0, 1)$	0.38	–	0.76	0.32	–	0.68	0.40	–	0.63
	default	0.16	0.70	0.97	0.10	0.84	0.94	0.06	0.34	0.99
	$r \sim \mathscr{U}(-0.75, 1.75)$	0.28	–	0.89	0.18	–	0.87	0.14	–	0.84

Table A15: Convergence speed of DLPSO for every value of each parameter (continued, part 3/3)

		Classical, $D = 10$			CEC2017, $D = 10$			CEC2017, $D = 30$		
Parameter	Value	t_{50}	t_{90}	final	t_{50}	t_{90}	final	t_{50}	t_{90}	final
Maximum velocity factor $v_{\max}$	$v_{\max} = 0.25$	0.16	0.64	0.97	0.12	0.80	0.95	0.10	0.48	0.97
	$v_{\max} = 0.5$	0.16	0.70	0.97	0.12	0.70	0.97	0.10	0.52	0.97
	default	0.18	0.72	0.97	0.12	0.72	0.98	0.10	0.46	0.98
	$v_{\max} = 2$	0.18	0.74	0.97	0.14	0.72	0.97	0.10	0.50	0.98
	$v_{\max} = 4$	0.20	0.78	0.95	0.14	0.76	0.97	0.10	0.54	0.98
Initial step factor κ_0	$\kappa_0 = 0.1$	0.16	0.64	0.99	0.12	0.74	0.96	0.10	0.44	0.99
	default	0.18	0.66	0.99	0.12	0.70	0.98	0.10	0.44	0.99
	$\kappa_0 = 1$	0.18	0.68	0.98	0.12	0.70	0.98	0.10	0.46	0.99
Memory of the step factor H	$H = 1$	0.16	0.68	0.97	0.12	0.76	0.96	0.10	0.52	0.97
	default	0.18	0.66	0.99	0.12	0.76	0.97	0.10	0.44	0.99
	$H = 10$	0.18	0.68	0.99	0.14	0.74	0.97	0.10	0.46	0.99
Standard deviation σ_κ of $\log_{10} \kappa_i^t$ around the memory	$\sigma_\kappa = 0.1$	0.18	0.78	0.96	0.12	0.78	0.97	0.10	0.54	0.99
	default	0.18	0.66	0.99	0.12	0.74	0.97	0.10	0.46	0.98
	$\sigma_\kappa = 0.5$	0.16	0.62	0.99	0.12	0.72	0.97	0.10	0.48	0.98
Travel probability of the old animals μ_i^t	$\mu_i^t = 0.1$ (constant)	0.16	0.64	0.99	0.12	0.64	0.99	0.10	0.38	0.99
	$\mu_i^t = 0.5$ (constant)	0.16	0.64	1.00	0.12	0.64	1.00	0.10	0.38	1.00
	default	0.16	0.64	1.00	0.12	0.64	1.00	0.10	0.38	1.00
Resting radius of the old animals ε	$\varepsilon = 0.25$	0.16	0.64	1.00	0.12	0.64	1.00	0.10	0.38	1.00
	default	0.16	0.64	1.00	0.12	0.64	1.00	0.10	0.38	1.00
	$\varepsilon = 1$	0.16	0.64	1.00	0.12	0.64	0.99	0.10	0.38	1.00
Population size N	$N = 30$	0.08	–	0.84	0.06	–	0.84	0.04	0.98	0.90
	$N = 50$	0.10	0.82	0.92	0.08	0.78	0.92	0.06	0.54	0.94
	default	0.16	0.68	0.98	0.12	0.82	0.95	0.08	0.54	0.95
	$N = 200$	0.26	0.90	0.94	0.18	0.88	0.93	0.12	0.84	0.92

Progress of a configuration c on a function: $(L_0 - L_c(t))/(L_0 - L_{\min})$, where $L_c(t)$ is the mean over the runs of $\log_{10}(f(x_{best}(t)) - f^* + 10^{-8})$ after the fraction t of MaxFE, L_0 the largest initial value and $L_{\min}$ the smallest final value over the values of the parameter; smaller t_{50}, t_{90} mean faster convergence.

A.3 Convergence behaviour of DLPSO (Section 3.3)

This part supports Section 3.3. Figures **??** and **??** show the convergence of DLPSO on the classical functions and on CEC2017 with $D = 10$, and Tables **??**–**??** give the Wilcoxon tests of the improvement of DLPSO on every function.

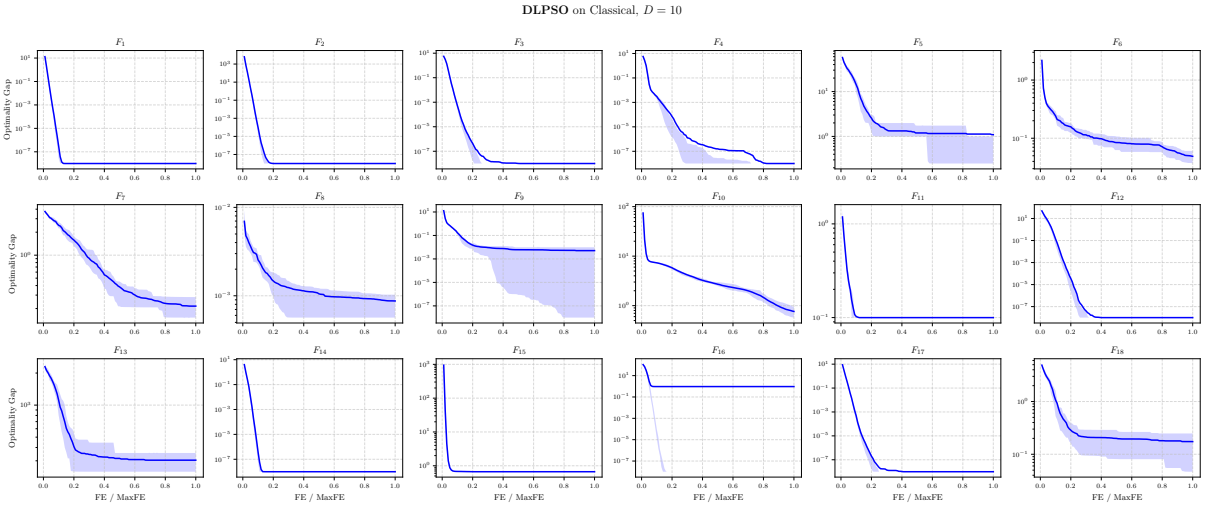


Fig. A9: Convergence of DLPSO on the classical functions F1–F18 ($D = 10$, $N = 50$, 30 runs): mean optimality gap $f(x_{best}) - f^*$ as a function of the fraction of MaxFE used, with the band between the first and the third quartile of the runs

Table A16: Wilcoxon tests of the improvement of DLPSO on every function in the second half and in the last quarter of the budget (part 1/3)

Classical, $D = 10$										
	Second half ($L(T/2)$ vs $L(T)$)					Last quarter ($L(3T/4)$ vs $L(T)$)				
Function	Gain	Runs	R^+	p_{Holm}	Sig.	Gain	Runs	R^+	p_{Holm}	Sig.
F_1	0.00	20/30	210	0.0003	Yes	0.00	1/30	1	0.9519	No
F_2	0.00	27/30	378	< 0.0001	Yes	0.00	6/30	21	0.1091	No
F_3	0.00	29/30	435	< 0.0001	Yes	0.00	21/30	231	0.0003	Yes
F_4	0.15	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_5	0.00	30/30	465	< 0.0001	Yes	0.00	28/30	406	< 0.0001	Yes
F_6	0.25	26/30	351	< 0.0001	Yes	0.22	26/30	351	< 0.0001	Yes
F_7	0.20	30/30	465	< 0.0001	Yes	0.01	30/30	465	< 0.0001	Yes
F_8	0.03	30/30	465	< 0.0001	Yes	0.00	29/30	435	< 0.0001	Yes
F_9	0.00	30/30	465	< 0.0001	Yes	0.00	28/30	406	< 0.0001	Yes
F_{10}	0.53	30/30	465	< 0.0001	Yes	0.36	30/30	465	< 0.0001	Yes
F_{11}	0.00	28/30	406	< 0.0001	Yes	0.00	27/30	378	< 0.0001	Yes
F_{12}	0.00	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{13}	0.00	30/30	465	< 0.0001	Yes	0.00	26/30	351	< 0.0001	Yes
F_{14}	0.00	18/30	171	0.0004	Yes	0.00	1/30	1	0.9519	No
F_{15}	0.00	28/30	406	< 0.0001	Yes	0.00	25/30	325	< 0.0001	Yes
F_{16}	0.00	6/30	21	0.0143	Yes	0.00	0/30	0	1.0000	No
F_{17}	0.00	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{18}	0.00	30/30	465	< 0.0001	Yes	0.00	29/30	435	< 0.0001	Yes

Table A17: Wilcoxon tests of the improvement of DLPSO on every function in the second half and in the last quarter of the budget (continued, part 2/3)

CEC2017, $D = 10$										
	Second half ($L(T/2)$ vs $L(T)$)					Last quarter ($L(3T/4)$ vs $L(T)$)				
Function	Gain	Runs	R^+	p_{Holm}	Sig.	Gain	Runs	R^+	p_{Holm}	Sig.
F_1	0.02	30/30	465	< 0.0001	Yes	0.01	30/30	465	< 0.0001	Yes
F_3	7.34	30/30	465	< 0.0001	Yes	5.09	30/30	465	< 0.0001	Yes
F_4	0.23	30/30	465	< 0.0001	Yes	0.12	30/30	465	< 0.0001	Yes
F_5	0.21	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_6	3.31	30/30	465	< 0.0001	Yes	2.10	30/30	465	< 0.0001	Yes
F_7	0.02	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_8	0.17	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_9	0.13	30/30	465	< 0.0001	Yes	0.00	28/30	406	< 0.0001	Yes
F_{10}	0.15	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{11}	0.24	30/30	465	< 0.0001	Yes	0.16	30/30	465	< 0.0001	Yes
F_{12}	0.05	30/30	465	< 0.0001	Yes	0.02	30/30	465	< 0.0001	Yes
F_{13}	0.42	30/30	465	< 0.0001	Yes	0.14	30/30	465	< 0.0001	Yes
F_{14}	0.36	30/30	465	< 0.0001	Yes	0.20	30/30	465	< 0.0001	Yes
F_{15}	0.70	30/30	465	< 0.0001	Yes	0.32	30/30	465	< 0.0001	Yes
F_{16}	0.28	30/30	465	< 0.0001	Yes	0.17	30/30	465	< 0.0001	Yes
F_{17}	0.06	30/30	465	< 0.0001	Yes	0.01	30/30	465	< 0.0001	Yes
F_{18}	0.33	30/30	465	< 0.0001	Yes	0.25	30/30	465	< 0.0001	Yes
F_{19}	0.67	30/30	465	< 0.0001	Yes	0.38	30/30	465	< 0.0001	Yes
F_{20}	0.84	30/30	465	< 0.0001	Yes	0.01	30/30	465	< 0.0001	Yes
F_{21}	0.00	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{22}	0.00	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{23}	0.00	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{24}	0.00	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{25}	0.00	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{26}	0.00	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{27}	0.00	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{28}	0.00	30/30	465	< 0.0001	Yes	0.00	19/30	190	0.0001	Yes
F_{29}	0.01	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{30}	0.33	30/30	465	< 0.0001	Yes	0.12	30/30	465	< 0.0001	Yes

Table A18: Wilcoxon tests of the improvement of DLPSO on every function in the second half and in the last quarter of the budget (continued, part 3/3)

CEC2017, $D = 30$										
	Second half ($L(T/2)$ vs $L(T)$)					Last quarter ($L(3T/4)$ vs $L(T)$)				
Function	Gain	Runs	R^+	p_{Holm}	Sig.	Gain	Runs	R^+	p_{Holm}	Sig.
F_1	0.02	30/30	465	< 0.0001	Yes	0.01	30/30	465	< 0.0001	Yes
F_3	1.67	30/30	465	< 0.0001	Yes	0.85	30/30	465	< 0.0001	Yes
F_4	0.02	30/30	465	< 0.0001	Yes	0.01	30/30	465	< 0.0001	Yes
F_5	0.05	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_6	0.57	30/30	465	< 0.0001	Yes	0.37	30/30	465	< 0.0001	Yes
F_7	0.03	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_8	0.10	30/30	465	< 0.0001	Yes	0.01	30/30	465	< 0.0001	Yes
F_9	0.00	30/30	465	< 0.0001	Yes	0.00	22/30	253	< 0.0001	Yes
F_{10}	0.06	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{11}	0.07	30/30	465	< 0.0001	Yes	0.01	30/30	465	< 0.0001	Yes
F_{12}	0.23	30/30	465	< 0.0001	Yes	0.07	30/30	465	< 0.0001	Yes
F_{13}	0.00	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{14}	0.47	30/30	465	< 0.0001	Yes	0.12	30/30	465	< 0.0001	Yes
F_{15}	0.00	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{16}	0.29	30/30	465	< 0.0001	Yes	0.14	30/30	465	< 0.0001	Yes
F_{17}	0.03	30/30	465	< 0.0001	Yes	0.01	30/30	465	< 0.0001	Yes
F_{18}	0.32	30/30	465	< 0.0001	Yes	0.20	30/30	465	< 0.0001	Yes
F_{19}	0.00	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{20}	0.03	30/30	465	< 0.0001	Yes	0.01	30/30	465	< 0.0001	Yes
F_{21}	0.01	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{22}	0.00	30/30	465	< 0.0001	Yes	0.00	29/30	435	< 0.0001	Yes
F_{23}	0.00	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{24}	0.01	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{25}	0.00	30/30	465	< 0.0001	Yes	0.00	23/30	276	< 0.0001	Yes
F_{26}	0.00	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{27}	0.00	29/30	435	< 0.0001	Yes	0.00	29/30	435	< 0.0001	Yes
F_{28}	0.00	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{29}	0.01	30/30	465	< 0.0001	Yes	0.00	30/30	465	< 0.0001	Yes
F_{30}	0.03	30/30	465	< 0.0001	Yes	0.01	30/30	465	< 0.0001	Yes

Gain: median over the runs of $L(t) - L(T)$ (decades of improvement after t); Runs: runs that improved after t; R^+: sum of ranks of the runs that improved (the best-so-far value cannot increase, so that $R^- = 0$); p_{Holm}: two-sided signed-rank test, Holm over the functions of the suite; Sig.: significant improvement at the level 0.05.

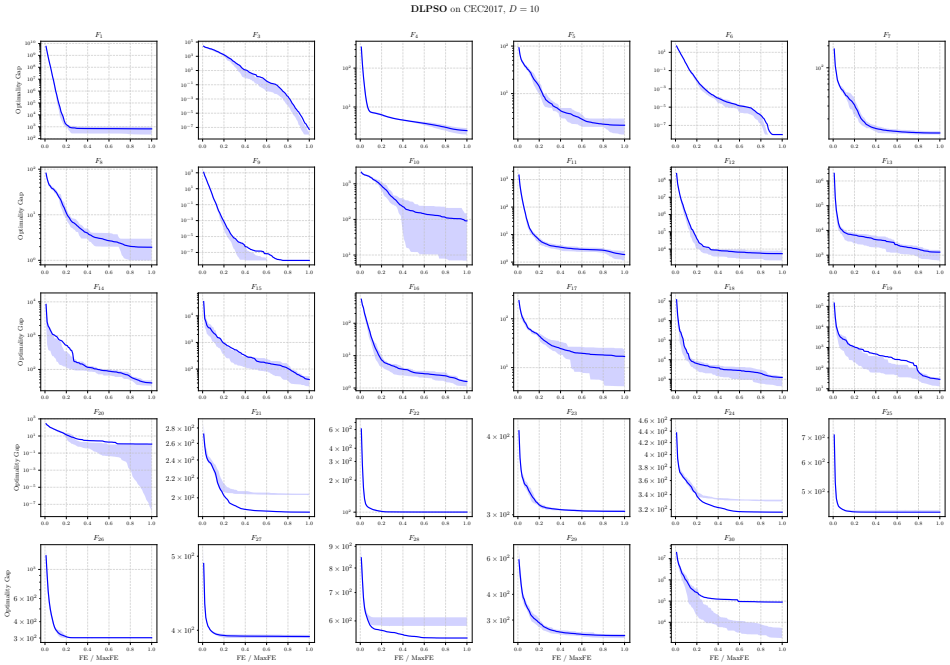
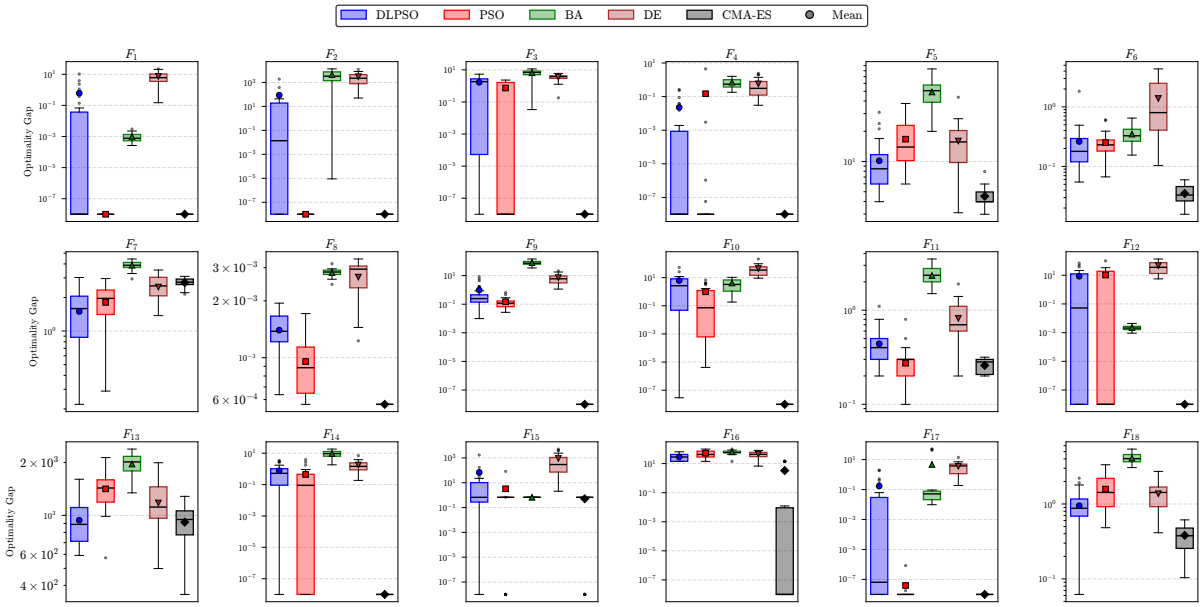


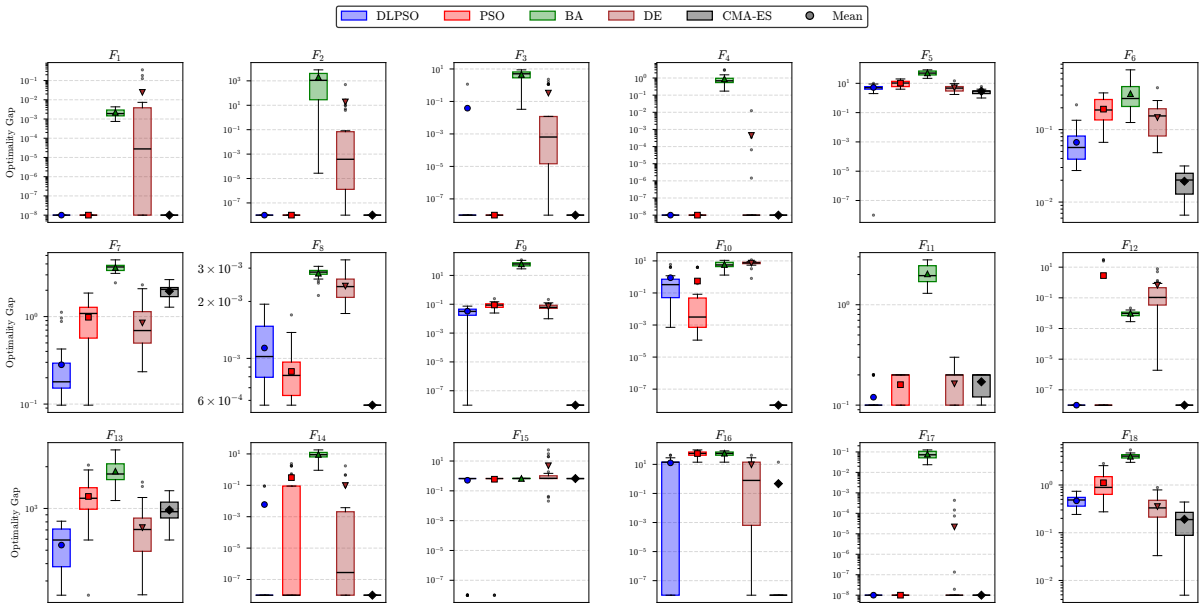
Fig. A10: Convergence of DLPSO on CEC2017 ($D = 10$, $N = 100$, 30 runs): mean distance to the optimum $f(x_{best}) - F^*$ with the band between the first and the third quartile

A.4 Classical benchmark functions (Section 3.4)

This part supports Section 3.4. Tables **??**–**??** give the per-function results of the five algorithms for every population size. Figure **??** shows the distributions of the optimality gaps with 10 and 20 particles, Figures **??** and **??** the distributions of the CPU times, and Figures **??**–**??** the convergence curves for 10, 20, 30 and 50 particles.

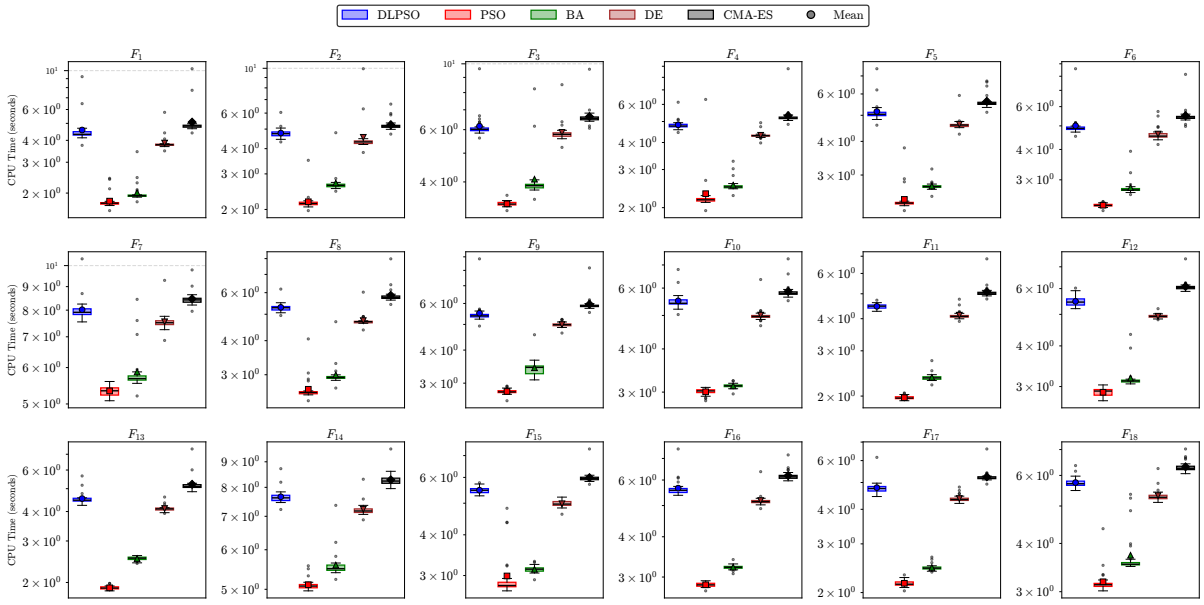


(a) 10 particles

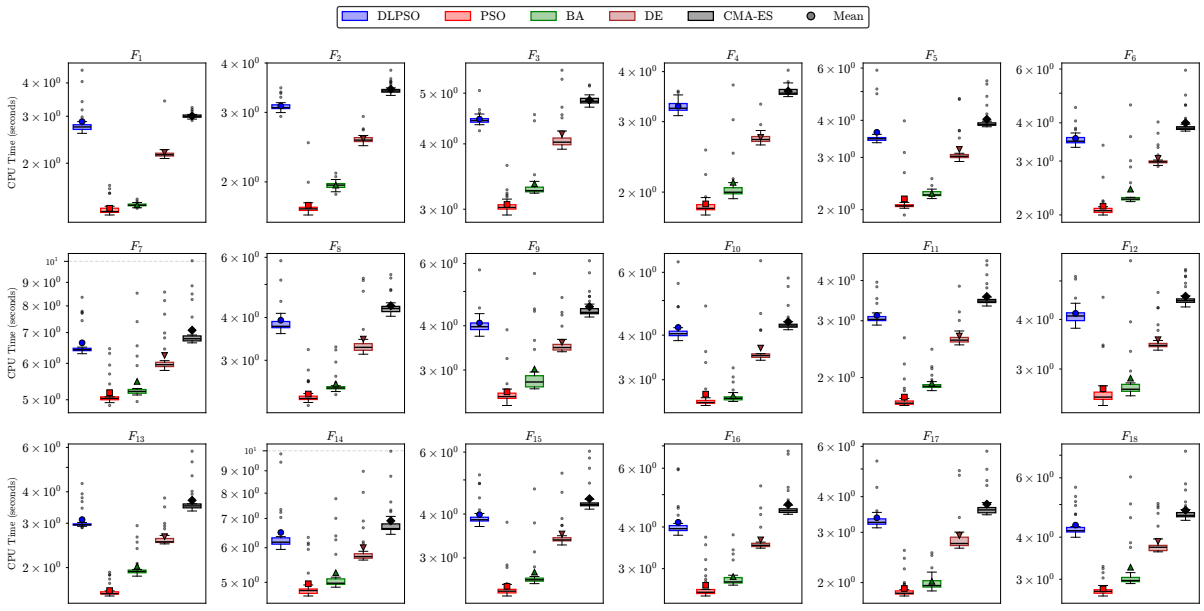


(b) 20 particles

Fig. A11: Distribution over the runs of the optimality gap $|f(x_{best}) - f^*|$ on the classical functions with 10 and 20 particles (same budget, 30 runs, box: first quartile Q_1 to third quartile Q_3, line: median, marker: mean, whiskers: most extreme runs within 1.5 IQR of the box, points: other runs)

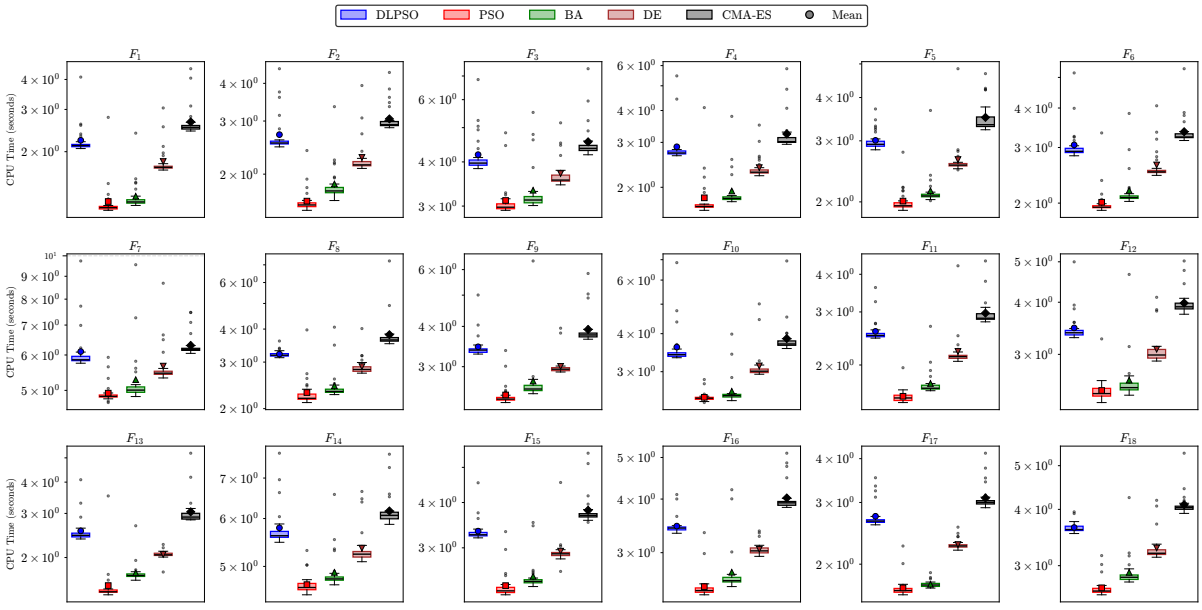


(a) 10 particles

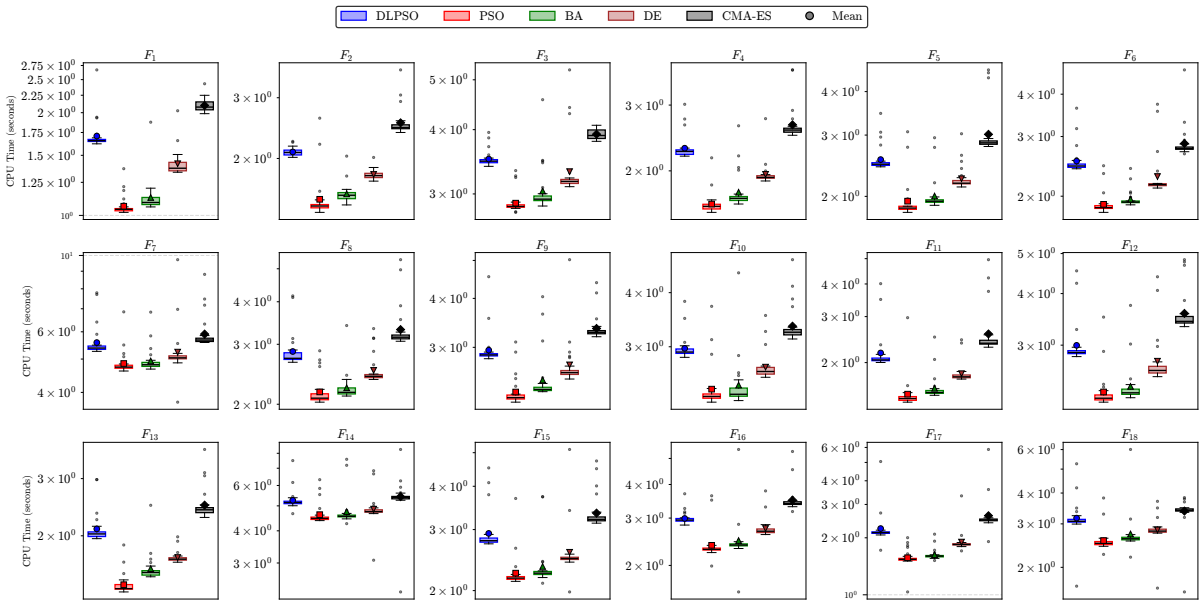


(b) 20 particles

Fig. A12: Distribution over the runs of the CPU time (seconds) on the classical functions with 10 and 20 particles (same budget, 30 runs, box: first quartile Q_1 to third quartile Q_3, line: median, marker: mean, whiskers: most extreme runs within 1.5 IQR of the box, points: other runs)



(a) 30 particles



(b) 50 particles

Fig. A13: Distribution over the runs of the CPU time (seconds) on the classical functions with 30 and 50 particles (same budget, 30 runs, box: first quartile Q_1 to third quartile Q_3, line: median, marker: mean, whiskers: most extreme runs within 1.5 IQR of the box, points: other runs)

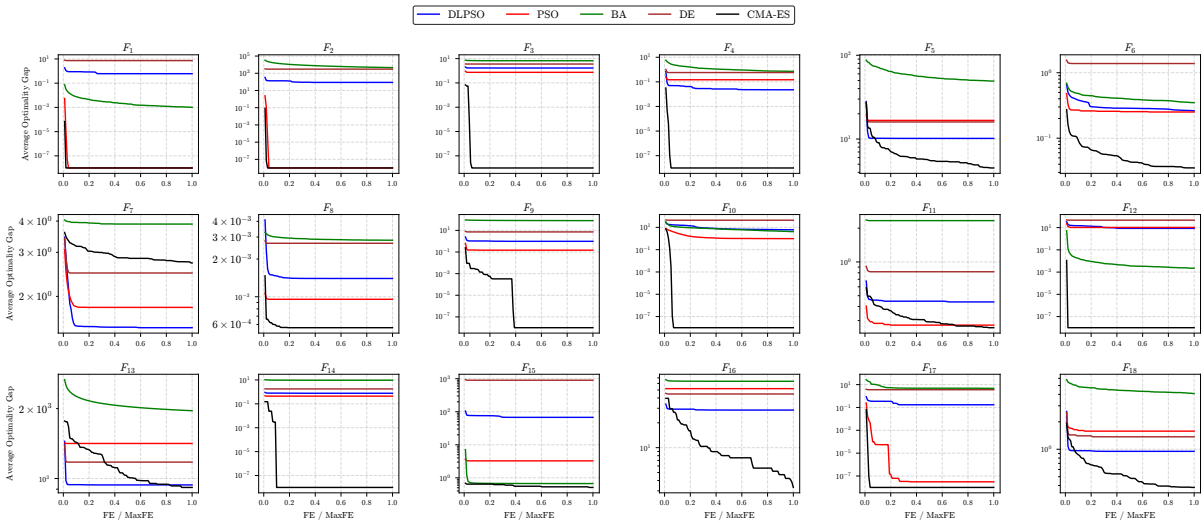


Fig. A14: Convergence curves of DLPSO, PSO, BA, DE and CMA-ES on all the classical functions F1–F18 with 10 particles (mean over 30 runs of the best-so-far optimality gap $|f(x_{best}) - f^*|$ as a function of the fraction of MaxFE used)

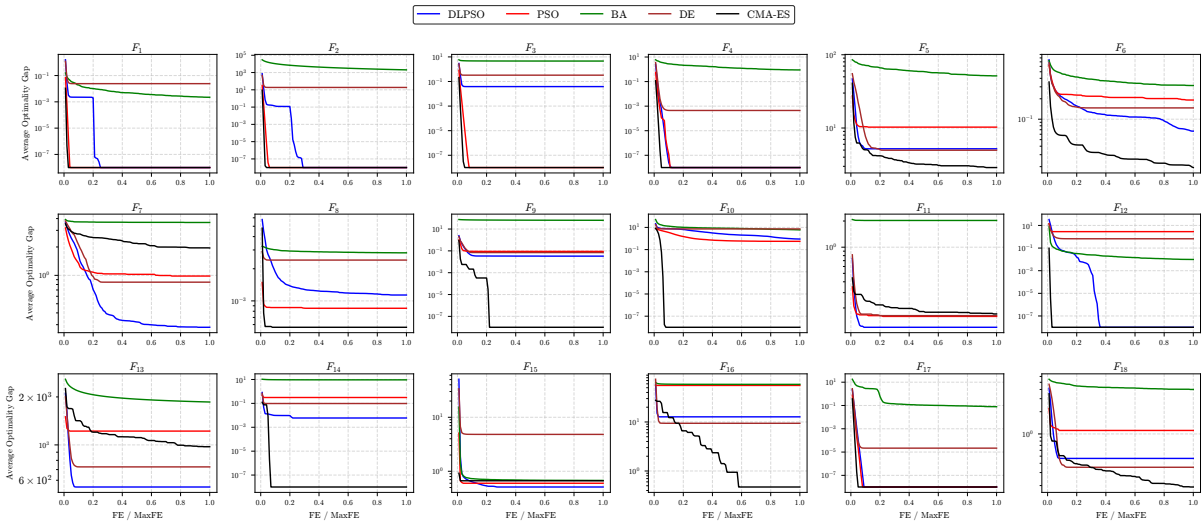


Fig. A15: Convergence curves of DLPSO, PSO, BA, DE and CMA-ES on all the classical functions F1–F18 with 20 particles (mean over 30 runs of the best-so-far optimality gap $|f(x_{best}) - f^*|$ as a function of the fraction of MaxFE used)

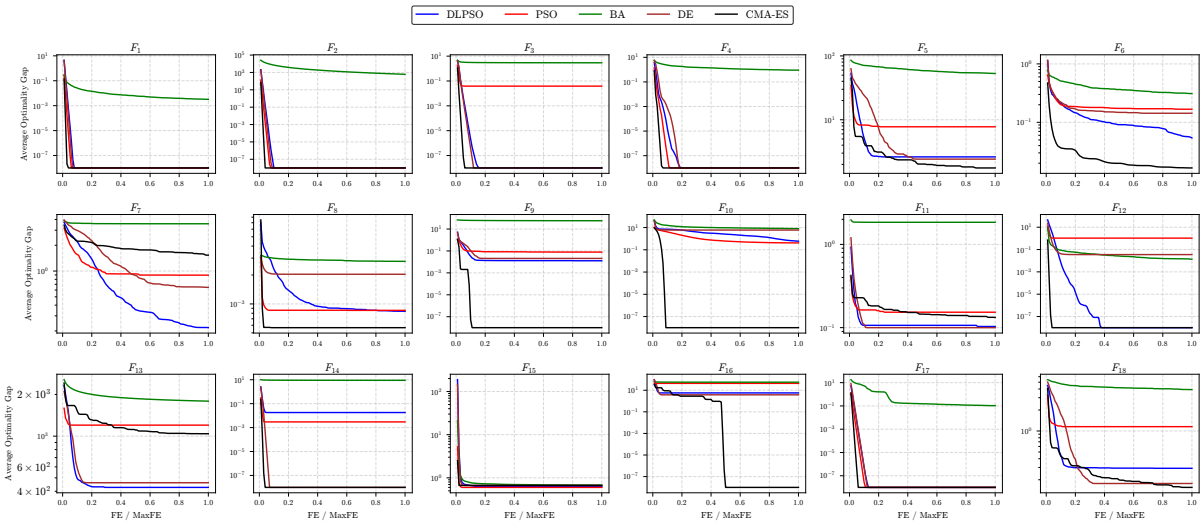


Fig. A16: Convergence curves of DLPSO, PSO, BA, DE and CMA-ES on all the classical functions F1–F18 with 30 particles (mean over 30 runs of the best-so-far optimality gap $|f(x_{best}) - f^*|$ as a function of the fraction of MaxFE used)

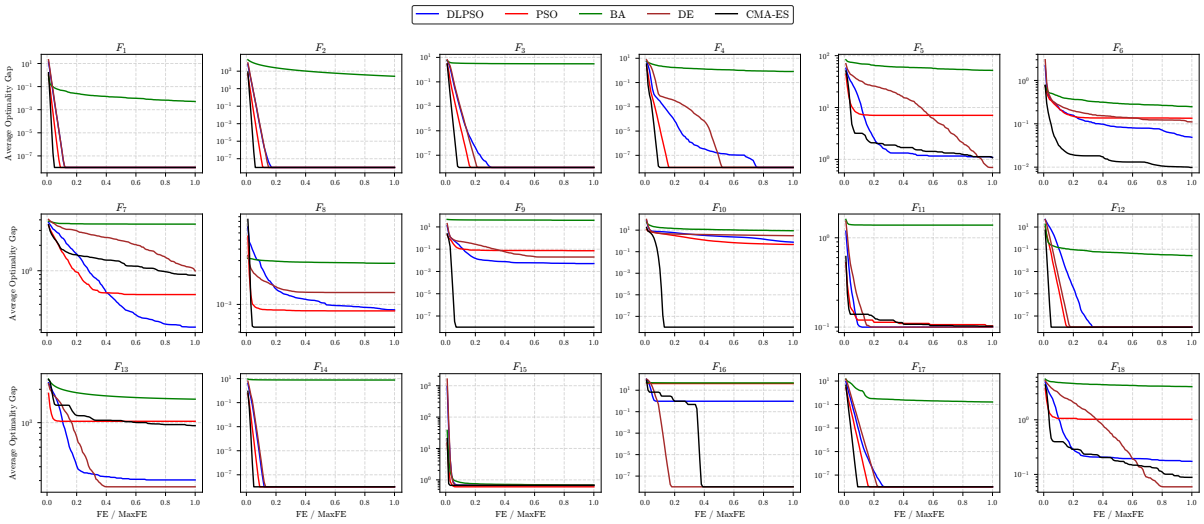


Fig. A17: Convergence curves of DLPSO, PSO, BA, DE and CMA-ES on all the classical functions F1–F18 with 50 particles (mean over 30 runs of the best-so-far optimality gap $|f(x_{best}) - f^*|$ as a function of the fraction of MaxFE used)

Table A19: Results on the classical functions F1–F18 ($D = 10$, 30 runs, MaxFE $= 10^4 D$): optimality gap $|f(x_{best}) - f^*|$ (part 1/8)

No.par = 10										
Algorithm	Metric	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9
DLPSO	Mean	0.6077	86.96	1.681	0.02266	10.17	0.2623	1.499	1.40E-03	0.9729
	Std	1.977	343.9	1.516	0.06299	6.218	0.3151	0.7528	3.39E-04	1.87
	Min	3.13E-190	7.87E-183	4.00E-15	4.57E-187	3.98	0.05466	0.2199	6.37E-04	9.86E-03
	Q_1	1.17E-144	5.31E-137	5.31E-05	2.09E-15	5.97	0.1193	0.8785	1.21E-03	0.1403
	Median	1.65E-99	0.01375	1.851	5.94E-11	8.457	0.1789	1.59	1.38E-03	0.2473
	Q_3	0.03702	19.09	2.755	8.58E-04	11.71	0.2943	2.055	1.66E-03	0.4542
	Max	10.35	1901	5.384	0.2635	30.86	1.83	3.024	1.95E-03	8.384
	IQR	0.03702	19.09	2.755	8.58E-04	5.742	0.175	1.176	4.47E-04	0.314
	Time (s)	4.58	4.78	6.17	4.83	5.16	5.01	8.02	5.30	5.51
PSO	Mean	0	0	0.7415	0.1481	16.68	0.2513	1.808	9.55E-04	0.1478
	Std	0	0	0.8928	0.797	7.846	0.1211	0.679	3.24E-04	0.1309
	Min	0	0	4.00E-15	3.82E-17	5.97	0.06671	0.2891	5.66E-04	0.0271
	Q_1	0	0	4.00E-15	1.99E-15	10.2	0.1809	1.408	6.48E-04	0.06701
	Median	0	0	7.55E-15	1.83E-14	13.93	0.2303	1.964	8.85E-04	0.1181
	Q_3	0	0	1.646	4.20E-13	22.88	0.2781	2.332	1.14E-03	0.1686
	Max	9.88E-324	3.95E-323	2.317	4.44	37.81	0.6072	2.958	1.71E-03	0.6531
	IQR	0	0	1.646	4.18E-13	12.69	0.09722	0.9235	4.91E-04	0.1016
	Time (s)	1.80	2.18	3.38	2.32	2.44	2.36	5.34	2.65	2.79
BA	Mean	1.01E-03	4549	6.644	0.7159	49.18	0.3487	3.891	2.83E-03	79.77
	Std	6.42E-04	4032	2.7	0.4655	13.93	0.1197	0.3241	1.33E-04	30.46
	Min	2.63E-04	8.96E-06	0.03356	0.1804	19.95	0.155	2.936	2.45E-03	34.22
	Q_1	5.33E-04	1409	4.992	0.3721	38.86	0.2646	3.719	2.76E-03	62.78
	Median	7.65E-04	3361	6.942	0.5472	50.71	0.3271	3.892	2.85E-03	73.87
	Q_3	1.35E-03	7684	8.488	1.012	57.72	0.4209	4.129	2.91E-03	98.1
	Max	3.01E-03	1.357e+04	11.51	1.605	83.51	0.6488	4.431	3.15E-03	145.1
	IQR	8.21E-04	6275	3.495	0.6399	18.86	0.1562	0.4101	1.51E-04	35.32
	Time (s)	2.01	2.70	4.09	2.54	2.73	2.79	5.87	3.00	3.43
DE	Mean	7.316	3035	3.58	0.5703	15.98	1.382	2.48	2.67E-03	7.202
	Std	5.577	2886	1.282	0.6319	7.988	1.188	0.6518	5.56E-04	5.16
	Min	0.1488	50.46	0.1817	0.03058	3.092	0.1033	1.375	1.23E-03	1.142
	Q_1	3.491	792.2	2.935	0.1225	9.762	0.4062	2.07	2.35E-03	3.105
	Median	6.082	2261	3.818	0.3087	15.65	0.8015	2.54	2.95E-03	5.979
	Q_3	10.44	4491	4.208	0.7864	20.34	2.492	3.035	3.06E-03	9.425
	Max	22.18	1.217e+04	6.033	2.413	43.63	4.348	3.526	3.34E-03	21.63
	IQR	6.952	3699	1.272	0.6639	10.57	2.086	0.9652	7.16E-04	6.32
	Time (s)	3.86	4.55	5.88	4.32	4.63	4.63	7.55	4.74	4.99
CMA-ES	Mean	1.17E-16	1.12E-16	1.45E-11	3.48E-12	4.51	0.03509	2.72	5.66E-04	6.29E-17
	Std	4.50E-17	3.73E-17	2.24E-12	6.97E-13	1.112	0.01303	0.2574	7.16E-17	7.41E-17
	Min	1.88E-17	4.29E-17	8.15E-12	2.11E-12	2.985	0.01571	2.144	5.66E-04	0
	Q_1	9.47E-17	7.99E-17	1.31E-11	2.89E-12	3.98	0.02627	2.613	5.66E-04	0
	Median	1.11E-16	1.08E-16	1.50E-11	3.36E-12	3.98	0.03302	2.743	5.66E-04	5.55E-17
	Q_3	1.34E-16	1.37E-16	1.61E-11	4.07E-12	4.975	0.04586	2.925	5.66E-04	1.11E-16
	Max	3.01E-16	1.95E-16	1.82E-11	4.86E-12	7.96	0.05956	3.095	5.66E-04	3.33E-16
	IQR	3.90E-17	5.71E-17	3.00E-12	1.18E-12	0.995	0.01958	0.3121	1.00E-16	1.11E-16
	Time (s)	5.08	5.25	6.64	5.32	5.63	5.50	8.47	5.86	5.96

Table A20: Results on the classical functions F1–F18 (continued, part 2/8)

No.par = 10										
Algorithm	Metric	F_{10}	F_{11}	F_{12}	F_{13}	F_{14}	F_{15}	F_{16}	F_{17}	F_{18}
DLPSO	Mean	6.13	0.4399	8.532	938.4	0.7696	67.6	28.54	0.1718	0.9482
	Std	10.21	0.2169	16.49	268.3	0.9783	309.5	14.87	0.4795	0.5061
	Min	2.94E-08	0.1999	1.97E-78	592.2	1.50E-32	1.97E-31	14.14	6.21E-93	0.06197
	Q_1	0.04789	0.2999	1.43E-56	712.3	0.09125	0.2767	14.15	1.14E-67	0.6881
	Median	2.651	0.3999	0.05237	889.2	0.5217	0.6667	28.27	5.99E-08	0.8771
	Q_3	8.152	0.4999	12.53	1109	1.035	10.26	42.41	0.02956	1.166
	Max	52.56	1.1	70.57	1605	3.498	1726	64.52	1.953	2.209
	IQR	8.104	0.2	12.53	396.5	0.9434	9.982	28.26	0.02956	0.4778
	Time (s)	5.51	4.47	5.48	4.54	7.66	5.48	5.66	4.79	5.75
PSO	Mean	0.9813	0.2732	10.36	1415	0.4422	3.258	51.83	3.11E-08	1.58
	Std	1.654	0.1315	21.24	315.2	0.8555	14.7	23.18	1.50E-07	0.8407
	Min	4.18E-06	0.09987	4.79E-189	574	1.50E-32	1.97E-31	14.14	4.77E-32	0.4824
	Q_1	6.16E-04	0.1999	6.69E-184	1189	1.50E-32	0.6667	28.27	1.15E-18	0.9214
	Median	0.07196	0.2999	1.28E-178	1431	0.08953	0.6667	42.41	1.11E-15	1.426
	Q_3	1.202	0.2999	18.94	1594	0.4543	0.6667	70.68	2.32E-12	2.199
	Max	6.653	0.7999	102.4	2132	3.91	82.41	98.96	8.40E-07	3.345
	IQR	1.201	0.1	18.94	404.6	0.4543	6.44E-10	42.41	2.32E-12	1.278
	Time (s)	3.01	1.97	2.88	1.90	5.11	2.99	2.84	2.16	3.18
BA	Mean	4.144	2.337	2.24E-03	1959	9.499	0.6693	63.65	4.716	4.089
	Std	3.34	0.475	9.10E-04	288.9	4.397	1.98E-03	16.22	14.06	0.5921
	Min	0.1837	1.5	9.17E-04	1343	1.819	0.6669	14.17	9.74E-03	3.063
	Q_1	1.084	2	1.64E-03	1790	6.443	0.6679	56.57	0.02101	3.659
	Median	3.274	2.35	2.09E-03	2013	9.469	0.6686	56.6	0.05124	4.03
	Q_3	6.78	2.775	2.76E-03	2166	12.93	0.6703	70.72	0.07792	4.576
	Max	10.4	3.5	4.38E-03	2382	18.33	0.6739	99.01	49.94	5.409
	IQR	5.695	0.775	1.12E-03	376.7	6.483	2.41E-03	14.15	0.05692	0.9173
	Time (s)	3.13	2.37	3.18	2.53	5.59	3.13	3.22	2.46	3.73
DE	Mean	44.07	0.8173	46.73	1177	1.774	899.7	44.69	3.567	1.37
	Std	39.54	0.3629	36.97	311.3	1.394	1307	18.69	2.922	0.5362
	Min	8.998	0.1999	5.597	498.8	0.1857	2.054	6.69	0.187	0.4133
	Q_1	14.46	0.5999	13.3	963.2	0.8686	71.89	30	1.143	0.9211
	Median	33.99	0.6999	35.75	1115	1.448	292.7	42.95	3.749	1.425
	Q_3	58.62	1.1	75.82	1454	2.443	1079	58.75	4.966	1.685
	Max	208.8	1.9	136.1	1992	7.018	5134	80.33	14.02	2.724
	IQR	44.16	0.5	62.52	490.5	1.574	1007	28.76	3.823	0.7642
	Time (s)	5.01	4.13	4.93	4.12	7.23	4.98	5.18	4.37	5.32
CMA-ES	Mean	1.43E-16	0.2591	1.13E-16	915.3	1.25E-16	0.5111	3.299	3.11E-11	0.3809
	Std	4.74E-17	0.0453	4.18E-17	207.8	4.78E-17	0.282	5.979	5.19E-12	0.1272
	Min	4.36E-17	0.1999	4.09E-17	355.3	4.15E-17	1.47E-16	0	1.97E-11	0.1037
	Q_1	1.18E-16	0.2073	8.97E-17	774.8	1.02E-16	0.6667	0	2.74E-11	0.2551
	Median	1.44E-16	0.2819	1.04E-16	947.5	1.23E-16	0.6667	0	3.12E-11	0.3763
	Q_3	1.65E-16	0.2999	1.40E-16	1061	1.42E-16	0.6667	9.10E-03	3.44E-11	0.477
	Max	2.86E-16	0.3167	2.10E-16	1283	3.02E-16	0.6667	14.14	4.32E-11	0.6134
	IQR	4.63E-17	0.09254	4.99E-17	286.2	4.03E-17	1.11E-16	9.10E-03	6.99E-12	0.2219
	Time (s)	5.87	5.09	6.09	5.22	8.28	5.99	6.19	5.24	6.32

Table A21: Results on the classical functions F1–F18 (continued, part 3/8)

No.par = 20										
Algorithm	Metric	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9
DLPSO	Mean	2.97E-84	2.04E-92	0.0385	2.18E-12	5.207	0.06672	0.2815	1.14E-03	0.03289
	Std	1.60E-83	1.10E-91	0.2074	1.17E-11	2.175	0.03879	0.2531	3.81E-04	0.01892
	Min	5.30E-148	2.42E-150	4.00E-15	2.02E-149	0	0.02724	0.09716	5.66E-04	0
	Q_1	3.25E-121	4.72E-119	4.00E-15	3.63E-101	3.98	0.03906	0.1522	7.95E-04	0.01723
	Median	5.38E-115	1.89E-110	4.00E-15	3.69E-79	4.975	0.05665	0.1797	1.02E-03	0.032
	Q_3	1.59E-108	1.96E-107	4.00E-15	4.74E-71	5.97	0.08151	0.2935	1.48E-03	0.04546
	Max	8.90E-83	6.12E-91	1.155	6.53E-11	9.95	0.2199	1.12	1.93E-03	0.07382
	IQR	1.59E-108	1.96E-107	0	4.74E-71	1.99	0.04245	0.1413	6.84E-04	0.02823
	Time (s)	2.87	3.12	4.46	3.28	3.65	3.56	6.65	3.93	4.07
PSO	Mean	1.56E-222	2.90E-221	4.23E-15	1.47E-15	10.28	0.1917	0.9832	8.53E-04	0.08931
	Std	0	0	8.86E-16	1.69E-15	4.732	0.07284	0.4713	2.66E-04	0.04209
	Min	7.53E-232	5.00E-231	4.00E-15	2.90E-180	3.98	0.06674	0.09716	5.66E-04	0.02464
	Q_1	3.20E-228	7.73E-227	4.00E-15	6.11E-16	5.97	0.136	0.5675	6.37E-04	0.06093
	Median	7.05E-227	7.64E-225	4.00E-15	9.71E-16	10.45	0.1867	1.086	8.13E-04	0.08731
	Q_3	2.64E-225	7.10E-223	4.00E-15	2.05E-15	13.68	0.2604	1.277	9.56E-04	0.1076
	Max	3.55E-221	4.83E-220	7.55E-15	8.94E-15	19.9	0.3211	1.858	1.70E-03	0.2435
	IQR	2.64E-225	7.10E-223	0	1.44E-15	7.711	0.1244	0.7094	3.19E-04	0.04667
	Time (s)	1.36	1.75	3.06	1.87	2.17	2.14	5.18	2.39	2.60
BA	Mean	2.22E-03	2041	4.59	0.8881	51.47	0.3132	3.636	2.82E-03	63.34
	Std	9.81E-04	2326	2.663	0.628	18.35	0.1405	0.3991	1.78E-04	21.98
	Min	7.47E-04	2.72E-05	0.03291	0.172	21.92	0.1253	2.432	2.15E-03	28.55
	Q_1	1.46E-03	28.37	2.846	0.542	37.26	0.2086	3.366	2.78E-03	45.74
	Median	1.91E-03	1106	5.047	0.6848	50.01	0.2687	3.693	2.86E-03	61.95
	Q_3	2.93E-03	4060	6.557	0.9608	69.02	0.3927	3.852	2.92E-03	74.74
	Max	4.29E-03	7976	8.827	2.992	83.19	0.6682	4.45	3.07E-03	119.7
	IQR	1.47E-03	4032	3.711	0.4188	31.77	0.1841	0.4862	1.32E-04	28.99
	Time (s)	1.40	1.96	3.36	2.12	2.27	2.43	5.48	2.56	3.02
DE	Mean	0.02422	18.84	0.3266	4.45E-04	4.977	0.1467	0.8467	2.41E-03	0.06989
	Std	0.07348	88.58	0.6203	2.22E-03	2.574	0.07404	0.5061	3.84E-04	0.03823
	Min	6.74E-25	2.97E-21	4.73E-13	0	1.735	0.04805	0.2342	1.73E-03	9.86E-03
	Q_1	2.64E-10	1.29E-06	1.46E-05	4.26E-214	2.986	0.08162	0.4977	2.10E-03	0.05163
	Median	2.78E-05	3.76E-04	6.45E-04	1.29E-164	4.718	0.1545	0.6924	2.40E-03	0.06109
	Q_3	3.83E-03	0.06852	0.01182	2.22E-16	6.002	0.1941	1.14	2.62E-03	0.08451
	Max	0.3601	494.9	2.315	0.01237	14.65	0.3786	2.305	3.32E-03	0.2144
	IQR	3.83E-03	0.06851	0.0118	2.22E-16	3.015	0.1125	0.642	5.24E-04	0.03288
	Time (s)	2.20	2.57	4.18	2.74	3.19	3.06	6.24	3.46	3.59
CMA-ES	Mean	2.79E-16	2.58E-16	1.33E-11	2.71E-12	2.885	0.01938	1.951	5.66E-04	2.96E-16
	Std	7.11E-17	9.39E-17	2.03E-12	4.80E-13	0.9386	7.22E-03	0.3199	8.14E-17	1.38E-16
	Min	1.08E-16	7.37E-17	9.60E-12	1.74E-12	0.995	6.60E-03	1.281	5.66E-04	0
	Q_1	2.52E-16	1.83E-16	1.20E-11	2.36E-12	1.99	0.01289	1.692	5.66E-04	2.22E-16
	Median	2.90E-16	2.52E-16	1.36E-11	2.75E-12	2.985	0.02015	2.046	5.66E-04	2.78E-16
	Q_3	3.17E-16	3.45E-16	1.49E-11	3.00E-12	2.985	0.02496	2.149	5.66E-04	4.44E-16
	Max	4.13E-16	4.48E-16	1.71E-11	3.88E-12	5.97	0.03149	2.647	5.66E-04	5.55E-16
	IQR	6.59E-17	1.62E-16	2.94E-12	6.38E-13	0.995	0.01208	0.4569	7.49E-17	2.22E-16
	Time (s)	3.01	3.43	4.85	3.58	4.03	4.00	7.08	4.33	4.53

Table A22: Results on the classical functions F1–F18 (continued, part 4/8)

No.par = 20										
Algorithm	Metric	F_{10}	F_{11}	F_{12}	F_{13}	F_{14}	F_{15}	F_{16}	F_{17}	F_{18}
DLPSO	Mean	0.8929	0.1199	1.77E-41	544.3	5.97E-03	0.5111	12.72	5.86E-21	0.4659
	Std	1.496	0.04	9.47E-41	170.3	0.02233	0.282	12.83	3.15E-20	0.1358
	Min	7.18E-04	0.09987	1.63E-72	236.9	1.50E-32	1.97E-31	5.68E-14	1.41E-73	0.2417
	Q_1	0.05104	0.09987	5.77E-61	380	1.50E-32	0.6667	5.68E-14	3.58E-65	0.3606
	Median	0.3305	0.09987	6.88E-58	592.2	1.50E-32	0.6667	14.14	1.25E-60	0.4857
	Q_3	0.7191	0.09987	1.04E-55	710.6	1.50E-32	0.6667	14.14	3.74E-58	0.5501
	Max	5.994	0.1999	5.28E-40	809.3	0.08953	0.6667	42.41	1.76E-19	0.7392
	IQR	0.668	3.54E-12	1.04E-55	330.6	0	2.85E-07	14.14	3.74E-58	0.1895
	Time (s)	4.21	3.12	4.14	3.10	6.50	3.99	4.13	3.37	4.34
PSO	Mean	0.5555	0.1599	2.831	1221	0.3101	0.6	55.6	2.22E-106	1.115
	Std	1.348	0.04899	8.523	388.7	0.6471	0.2	19.97	1.10E-105	0.6717
	Min	1.14E-04	0.09987	1.08E-132	236.9	1.50E-32	3.20E-31	14.14	8.40E-120	0.2741
	Q_1	7.18E-04	0.09987	2.77E-129	987	1.50E-32	0.6667	42.41	1.23E-116	0.6361
	Median	3.15E-03	0.1999	9.64E-127	1184	1.50E-32	0.6667	56.55	3.28E-115	0.8911
	Q_3	0.04842	0.1999	3.35E-124	1411	0.08953	0.6667	70.68	2.98E-113	1.498
	Max	3.987	0.1999	29.93	2055	2.4	0.6667	98.96	6.15E-105	2.822
	IQR	0.04771	0.1	3.35E-124	424.4	0.08953	0	28.27	2.98E-113	0.8617
	Time (s)	2.73	1.74	2.68	1.62	4.96	2.49	2.68	1.91	2.81
BA	Mean	5.981	2.03	9.82E-03	1858	9.297	0.6766	58.49	0.07841	4
	Std	2.715	0.4329	3.64E-03	342.3	4.479	8.57E-03	17.02	0.03114	0.5294
	Min	1.305	1.3	2.80E-03	1140	0.9099	0.6686	14.18	0.02371	2.984
	Q_1	4.408	1.7	6.88E-03	1615	6.446	0.6719	42.46	0.05152	3.68
	Median	5.588	1.95	9.63E-03	1772	8.906	0.6738	56.63	0.07255	4.034
	Q_3	8.159	2.45	0.01169	2099	12.47	0.6779	70.74	0.1073	4.336
	Max	10.86	2.8	0.02148	2644	18.06	0.7137	84.9	0.1258	5.442
	IQR	3.751	0.75	4.81E-03	484.5	6.022	5.95E-03	28.28	0.05578	0.656
	Time (s)	2.70	1.92	2.85	2.02	5.26	2.73	2.84	2.02	3.26
DE	Mean	7.282	0.1632	0.6639	727.8	0.09708	4.819	9.395	2.17E-05	0.3537
	Std	1.891	0.06574	1.62	347.4	0.3235	11.72	11.72	8.14E-05	0.2216
	Min	0.8087	0.09987	1.89E-06	238.4	1.32E-24	0.02053	7.11E-09	1.55E-125	0.0331
	Q_1	6.74	0.09987	0.03382	490.8	3.42E-11	0.6667	6.36E-04	1.11E-69	0.2123
	Median	7.528	0.1999	0.1062	705.2	2.78E-07	0.6911	0.7861	1.26E-37	0.3313
	Q_3	8.262	0.1999	0.4579	852.1	2.07E-03	1.025	14.16	1.35E-14	0.4783
	Max	11.6	0.2999	7.805	1549	1.728	55.74	42.41	4.32E-04	0.8863
	IQR	1.522	0.1	0.4241	361.2	2.07E-03	0.3581	14.16	1.35E-14	0.266
	Time (s)	3.68	2.69	3.56	2.65	6.00	3.51	3.66	2.92	3.89
CMA-ES	Mean	3.80E-16	0.1707	2.67E-16	973.9	2.73E-16	0.6667	0.4712	2.59E-11	0.1912
	Std	1.22E-16	0.042	8.29E-17	205.1	9.92E-17	1.03E-16	2.538	4.15E-12	0.1115
	Min	1.60E-16	0.1	1.05E-16	592.2	7.79E-17	0.6667	0	1.69E-11	4.91E-03
	Q_1	2.89E-16	0.1206	2.10E-16	853.9	2.27E-16	0.6667	0	2.33E-11	0.08818
	Median	3.75E-16	0.1999	2.64E-16	947.6	2.59E-16	0.6667	0	2.68E-11	0.1887
	Q_3	4.70E-16	0.1999	3.35E-16	1110	3.23E-16	0.6667	4.26E-14	2.92E-11	0.2671
	Max	6.77E-16	0.2012	4.39E-16	1342	4.80E-16	0.6667	14.14	3.30E-11	0.4399
	IQR	1.81E-16	0.07924	1.24E-16	256.6	9.64E-17	1.11E-16	4.26E-14	5.84E-12	0.1789
	Time (s)	4.36	3.57	4.57	3.70	6.91	4.42	4.67	3.76	4.81

Table A23: Results on the classical functions F1–F18 (continued, part 5/8)

No.par = 30										
Algorithm	Metric	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9
DLPSO	Mean	1.58E-74	3.08E-69	4.00E-15	7.40E-18	2.62	0.05416	0.2182	8.38E-04	0.01272
	Std	6.16E-74	1.22E-68	0	3.99E-17	1.347	0.02487	0.2321	2.69E-04	0.01017
	Min	9.12E-112	3.66E-109	4.00E-15	1.13E-94	0	0.02546	0.09716	5.66E-04	0
	Q_1	3.35E-86	2.26E-83	4.00E-15	2.41E-68	1.99	0.03709	0.09716	5.66E-04	7.40E-03
	Median	2.76E-80	3.51E-80	4.00E-15	6.06E-58	2.985	0.04907	0.1522	7.95E-04	9.86E-03
	Q_3	7.48E-78	1.06E-75	4.00E-15	1.45E-52	2.985	0.06502	0.2277	1.02E-03	0.01478
	Max	3.35E-73	6.29E-68	4.00E-15	2.22E-16	4.975	0.1332	1.253	1.68E-03	0.04679
	IQR	7.48E-78	1.06E-75	0	1.45E-52	0.995	0.02793	0.1306	4.57E-04	7.38E-03
	Time (s)	2.23	2.70	4.19	2.88	3.01	3.06	6.11	3.22	3.46
PSO	Mean	1.71E-160	5.25E-157	0.0385	1.64E-15	7.761	0.1677	0.8948	8.57E-04	0.07856
	Std	3.74E-160	2.32E-156	0.2074	1.45E-15	3.655	0.0765	0.6385	1.97E-04	0.03951
	Min	1.27E-165	8.31E-165	4.00E-15	3.17E-120	1.99	0.05628	0.09716	5.66E-04	9.86E-03
	Q_1	5.82E-164	1.40E-160	4.00E-15	6.11E-16	4.975	0.1013	0.2341	6.81E-04	0.04619
	Median	4.77E-163	1.63E-159	4.00E-15	1.30E-15	6.965	0.159	1.051	8.13E-04	0.07503
	Q_3	1.10E-161	2.30E-158	4.00E-15	1.87E-15	9.95	0.2114	1.195	1.02E-03	0.1021
	Max	1.29E-159	1.28E-155	1.155	6.83E-15	15.92	0.3888	2.195	1.20E-03	0.1795
	IQR	1.09E-161	2.29E-158	0	1.26E-15	4.975	0.1101	0.9604	3.41E-04	0.05588
	Time (s)	1.24	1.63	3.11	1.82	2.01	2.01	4.93	2.30	2.45
BA	Mean	3.17E-03	609.9	2.922	0.908	53.16	0.3099	3.564	2.77E-03	57.05
	Std	1.15E-03	905.1	2.527	0.4507	17.64	0.1461	0.4296	1.71E-04	21.39
	Min	1.34E-03	7.37E-05	0.02722	0.04366	19.71	0.1622	2.733	2.12E-03	22.19
	Q_1	2.37E-03	3.34E-04	0.08416	0.5844	41.67	0.2169	3.183	2.70E-03	38.98
	Median	3.14E-03	113.2	2.342	0.7967	52.62	0.261	3.707	2.81E-03	56.71
	Q_3	3.91E-03	1044	4.541	1.261	64.76	0.3368	3.815	2.89E-03	73.26
	Max	6.51E-03	3602	7.484	2.219	87.28	0.7945	4.364	2.97E-03	97.78
	IQR	1.55E-03	1044	4.456	0.6768	23.09	0.1199	0.6317	1.87E-04	34.28
	Time (s)	1.30	1.86	3.33	1.93	2.15	2.19	5.28	2.44	2.71
DE	Mean	1.89E-12	3.40E-11	3.40E-15	2.13E-90	2.418	0.1431	0.6444	2.04E-03	0.02081
	Std	1.02E-11	1.83E-10	1.32E-15	8.80E-90	1.434	0.05238	0.4499	3.58E-04	0.01564
	Min	7.42E-159	6.63E-158	4.44E-16	2.36E-194	0	0.02956	0.1327	1.03E-03	0
	Q_1	3.02E-156	4.33E-154	4.00E-15	1.07E-132	1.99	0.1129	0.3536	1.79E-03	9.86E-03
	Median	2.46E-155	3.56E-153	4.00E-15	2.53E-122	2.031	0.1527	0.4928	2.10E-03	0.01846
	Q_3	2.32E-154	1.83E-151	4.00E-15	1.48E-113	2.985	0.1744	0.7587	2.32E-03	0.02955
	Max	5.68E-11	1.02E-09	4.00E-15	4.64E-89	4.975	0.2407	1.868	2.72E-03	0.05416
	IQR	2.29E-154	1.83E-151	0	1.48E-113	0.995	0.06155	0.4051	5.33E-04	0.01969
	Time (s)	1.82	2.27	3.71	2.39	2.65	2.64	5.67	2.88	3.01
CMA-ES	Mean	3.93E-16	3.96E-16	1.28E-11	2.54E-12	1.758	0.01657	1.533	5.66E-04	3.66E-16
	Std	1.06E-16	1.22E-16	2.33E-12	2.97E-13	0.9844	6.05E-03	0.3973	1.12E-16	1.70E-16
	Min	1.82E-16	1.74E-16	8.16E-12	1.91E-12	0	5.97E-03	0.7294	5.66E-04	0
	Q_1	3.24E-16	3.11E-16	1.14E-11	2.34E-12	0.995	0.01274	1.283	5.66E-04	2.22E-16
	Median	4.03E-16	3.83E-16	1.26E-11	2.45E-12	1.99	0.01502	1.552	5.66E-04	3.33E-16
	Q_3	4.61E-16	4.75E-16	1.41E-11	2.70E-12	2.736	0.02195	1.773	5.66E-04	4.44E-16
	Max	5.77E-16	6.74E-16	1.93E-11	3.24E-12	2.985	0.02799	2.397	5.66E-04	7.77E-16
	IQR	1.37E-16	1.63E-16	2.71E-12	3.58E-13	1.741	9.21E-03	0.4892	2.00E-16	2.22E-16
	Time (s)	2.66	3.05	4.56	3.24	3.51	3.36	6.30	3.81	3.91

Table A24: Results on the classical functions F1–F18 (continued, part 6/8)

No.par = 30										
Algorithm	Metric	F_{10}	F_{11}	F_{12}	F_{13}	F_{14}	F_{15}	F_{16}	F_{17}	F_{18}
DLPSO	Mean	0.5909	0.1032	1.33E-37	426.6	0.01813	0.6222	5.655	3.68E-41	0.2917
	Std	0.7287	0.01795	5.82E-37	192.8	0.08258	0.1663	7.829	1.75E-40	0.1792
	Min	0.02867	0.09987	9.90E-57	118.4	1.50E-32	1.97E-31	0	1.25E-53	4.91E-03
	Q_1	0.189	0.09987	1.85E-45	266.5	1.50E-32	0.6667	5.68E-14	2.44E-47	0.1641
	Median	0.5321	0.09987	1.56E-41	355.3	1.50E-32	0.6667	5.68E-14	1.35E-44	0.2962
	Q_3	0.719	0.09987	1.77E-39	592.2	1.50E-32	0.6667	14.14	1.72E-42	0.3676
	Max	4.154	0.1999	3.21E-36	832.1	0.4543	0.6667	28.27	9.74E-40	0.7766
	IQR	0.53	1.54E-14	1.77E-39	325.7	0	0	14.14	1.72E-42	0.2035
	Time (s)	3.61	2.59	3.47	2.55	5.79	3.34	3.46	2.74	3.65
PSO	Mean	0.4148	0.1532	1.053	1201	2.98E-03	0.6	43.35	8.90E-84	1.154
	Std	1.201	0.04989	5.669	318.7	0.01607	0.2	22.18	2.66E-83	0.6903
	Min	8.37E-04	0.09987	1.54E-98	454	1.50E-32	1.97E-31	5.68E-14	4.10E-87	0.04667
	Q_1	5.02E-03	0.09987	5.43E-96	943.7	1.50E-32	0.6667	28.27	1.67E-85	0.5756
	Median	8.86E-03	0.1999	5.75E-95	1146	1.50E-32	0.6667	42.41	6.65E-85	1.099
	Q_3	0.01276	0.1999	2.32E-93	1476	1.50E-32	0.6667	56.55	4.45E-84	1.6
	Max	4.06	0.1999	31.58	1777	0.08953	0.6667	84.82	1.45E-82	2.626
	IQR	7.73E-03	0.1	2.31E-93	531.9	0	0	28.27	4.28E-84	1.024
	Time (s)	2.47	1.57	2.46	1.55	4.66	2.36	2.50	1.71	2.68
BA	Mean	7.96	1.843	0.01355	1789	8.92	0.6796	55.7	0.1056	3.921
	Std	1.865	0.3373	5.58E-03	369	4.192	6.47E-03	18.21	0.03726	0.5355
	Min	4.755	1.3	2.42E-03	978.4	2.19	0.6695	14.27	0.04106	2.674
	Q_1	6.838	1.6	9.80E-03	1576	6.512	0.6742	42.48	0.07701	3.708
	Median	7.816	1.8	0.01269	1844	7.51	0.6782	56.63	0.1026	3.909
	Q_3	9.207	2	0.01816	2101	11.14	0.6848	70.75	0.1327	4.172
	Max	11.43	2.6	0.02474	2393	17.82	0.6968	84.92	0.1728	5.192
	IQR	2.369	0.4	8.37E-03	524.2	4.628	0.01057	28.27	0.05566	0.4639
	Time (s)	2.59	1.74	2.60	1.74	4.89	2.50	2.70	1.75	2.90
DE	Mean	5.939	0.09987	0.03525	460.9	1.50E-32	0.6668	3.77	3.24E-78	0.1767
	Std	1.554	1.69E-08	0.1731	275	0	9.70E-04	7.239	1.31E-77	0.1317
	Min	1.726	0.09987	4.90E-63	0	1.50E-32	0.6667	5.68E-14	1.84E-80	7.99E-14
	Q_1	5.387	0.09987	9.76E-14	188.4	1.50E-32	0.6667	5.68E-14	1.01E-79	0.07993
	Median	6.138	0.09987	3.54E-11	475.3	1.50E-32	0.6667	5.68E-14	3.69E-79	0.1781
	Q_3	7.01	0.09987	3.97E-06	678.2	1.50E-32	0.6667	5.68E-14	1.19E-78	0.2412
	Max	8.187	0.09987	0.9652	1017	1.50E-32	0.6721	28.27	7.35E-77	0.5605
	IQR	1.622	2.60E-09	3.97E-06	489.7	0	0	0	1.09E-78	0.1613
	Time (s)	3.13	2.22	3.09	2.06	5.36	2.94	3.06	2.28	3.28
CMA-ES	Mean	5.43E-16	0.1325	3.73E-16	1041	4.06E-16	0.6667	3.79E-15	2.42E-11	0.1553
	Std	1.73E-16	0.03928	8.87E-17	209.5	1.18E-16	1.52E-16	1.42E-14	3.99E-12	0.08976
	Min	2.75E-16	0.09987	2.37E-16	355.3	2.12E-16	0.6667	0	1.31E-11	0.03761
	Q_1	4.21E-16	0.1005	2.97E-16	927.8	3.14E-16	0.6667	0	2.15E-11	0.08193
	Median	4.81E-16	0.1088	3.73E-16	1086	4.22E-16	0.6667	0	2.39E-11	0.1449
	Q_3	6.55E-16	0.1616	4.18E-16	1179	4.85E-16	0.6667	0	2.75E-11	0.2031
	Max	9.05E-16	0.1999	5.94E-16	1382	6.30E-16	0.6667	5.68E-14	3.23E-11	0.3575
	IQR	2.34E-16	0.06114	1.21E-16	251.7	1.71E-16	1.11E-16	0	6.08E-12	0.1212
	Time (s)	3.85	2.97	3.98	3.04	6.18	3.81	4.02	3.09	4.10

Table A25: Results on the classical functions F1–F18 (continued, part 7/8)

No.par = 50										
Algorithm	Metric	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9
DLPSO	Mean	3.39E-46	1.72E-43	4.00E-15	3.71E-18	1.094	0.0488	0.2152	8.73E-04	5.10E-03
	Std	1.83E-45	9.01E-43	0	1.95E-17	0.9732	0.02249	0.08894	4.26E-04	6.39E-03
	Min	2.38E-60	1.66E-57	4.00E-15	1.40E-50	0	0.01328	0.09716	5.66E-04	0
	Q_1	2.02E-57	2.42E-54	4.00E-15	1.00E-34	0.2487	0.03677	0.1522	5.66E-04	0
	Median	1.08E-54	8.59E-52	4.00E-15	3.00E-28	0.995	0.04316	0.2072	7.16E-04	8.85E-12
	Q_3	6.94E-52	2.38E-50	4.00E-15	1.26E-23	0.995	0.06065	0.2827	1.03E-03	9.86E-03
	Max	1.02E-44	5.02E-42	4.00E-15	1.09E-16	3.98	0.1144	0.4261	2.58E-03	0.02214
	IQR	6.94E-52	2.37E-50	0	1.26E-23	0.7462	0.02388	0.1305	4.63E-04	9.86E-03
	Time (s)	1.71	2.09	3.51	2.30	2.55	2.54	5.58	2.86	2.95
PSO	Mean	4.45E-104	4.38E-101	4.00E-15	9.05E-16	7.064	0.1345	0.5226	8.51E-04	0.07282
	Std	8.87E-104	1.24E-100	0	8.59E-16	3.054	0.05526	0.5455	2.59E-04	0.04607
	Min	9.03E-108	4.88E-105	4.00E-15	4.71E-77	2.985	0.03862	0.09388	5.66E-04	7.40E-03
	Q_1	5.83E-106	7.52E-103	4.00E-15	5.55E-17	4.975	0.08794	0.1522	6.37E-04	0.03939
	Median	4.67E-105	2.46E-102	4.00E-15	8.33E-16	6.965	0.1213	0.2072	8.04E-04	0.06522
	Q_3	5.13E-104	1.56E-101	4.00E-15	1.26E-15	7.96	0.1864	1.027	1.06E-03	0.1045
	Max	3.84E-103	6.02E-100	4.00E-15	3.33E-15	15.92	0.2424	2.098	1.59E-03	0.1871
	IQR	5.07E-104	1.48E-101	0	1.21E-15	2.985	0.09846	0.8753	4.21E-04	0.0651
	Time (s)	1.06	1.52	2.88	1.63	1.94	1.89	4.85	2.17	2.31
BA	Mean	5.11E-03	261.3	2.997	0.8122	51.99	0.2481	3.561	2.80E-03	39.53
	Std	1.99E-03	472.1	2.926	0.5921	15.43	0.1171	0.326	1.80E-04	16.03
	Min	1.28E-03	7.92E-05	0.08666	0.215	27.06	0.1017	2.789	2.10E-03	14.17
	Q_1	3.92E-03	6.43E-04	0.1221	0.4581	40.07	0.1776	3.408	2.77E-03	28.63
	Median	4.83E-03	0.6111	2.62	0.6272	50.04	0.2063	3.619	2.85E-03	38.3
	Q_3	6.05E-03	249.6	6.09	1.017	60.97	0.2832	3.775	2.91E-03	49.62
	Max	0.01109	1740	9.66	3.224	85.24	0.6957	4.1	3.01E-03	78.73
	IQR	2.13E-03	249.6	5.968	0.5586	20.89	0.1057	0.3672	1.44E-04	20.99
	Time (s)	1.13	1.58	3.04	1.75	2.00	1.96	4.94	2.24	2.48
DE	Mean	1.55E-85	2.32E-82	3.64E-15	1.02E-34	0.6985	0.1105	0.995	1.35E-03	0.01975
	Std	2.65E-85	6.76E-82	1.07E-15	4.81E-34	0.8217	0.03365	0.7246	3.86E-04	0.01587
	Min	1.57E-88	3.72E-85	4.44E-16	6.91E-105	0	0.02379	0.0587	0	0
	Q_1	1.70E-86	5.02E-84	4.00E-15	1.69E-68	0	0.09893	0.3554	1.11E-03	8.01E-03
	Median	4.53E-86	3.15E-83	4.00E-15	1.60E-55	3.47E-04	0.1186	0.7694	1.43E-03	0.016
	Q_3	9.91E-86	8.77E-83	4.00E-15	4.42E-41	0.9951	0.1285	1.684	1.63E-03	0.02646
	Max	1.15E-84	3.58E-81	4.00E-15	2.67E-33	2.051	0.1688	2.184	1.92E-03	0.05906
	IQR	8.21E-86	8.27E-83	0	4.42E-41	0.9951	0.0296	1.329	5.27E-04	0.01845
	Time (s)	1.42	1.80	3.31	1.96	2.25	2.29	5.24	2.51	2.71
CMA-ES	Mean	6.13E-16	6.20E-16	1.25E-11	2.50E-12	1.079	9.83E-03	0.8838	5.66E-04	6.29E-16
	Std	2.44E-16	2.41E-16	2.59E-12	4.85E-13	0.6299	3.70E-03	0.3478	2.12E-16	2.40E-16
	Min	1.11E-16	2.04E-16	7.02E-12	1.67E-12	0	1.30E-03	0.3172	5.66E-04	2.22E-16
	Q_1	4.57E-16	4.82E-16	1.05E-11	2.11E-12	0.995	7.52E-03	0.6295	5.66E-04	4.72E-16
	Median	5.75E-16	6.09E-16	1.26E-11	2.45E-12	0.995	0.01003	0.8372	5.66E-04	6.11E-16
	Q_3	7.58E-16	7.55E-16	1.43E-11	2.72E-12	1.37	0.01221	1.084	5.66E-04	7.49E-16
	Max	1.29E-15	1.22E-15	1.85E-11	3.95E-12	1.99	0.01939	1.709	5.66E-04	1.22E-15
	IQR	3.00E-16	2.73E-16	3.84E-12	6.14E-13	0.3747	4.69E-03	0.4543	2.50E-16	2.78E-16
	Time (s)	2.10	2.54	3.92	2.66	3.01	2.86	5.91	3.31	3.35

Table A26: Results on the classical functions F1–F18 (continued, part 8/8)

No.par = 50										
Algorithm	Metric	F_{10}	F_{11}	F_{12}	F_{13}	F_{14}	F_{15}	F_{16}	F_{17}	F_{18}
DLPSO	Mean	0.7634	0.09987	1.08E-25	304	1.50E-32	0.6667	0.9424	2.73E-25	0.1736
	Std	0.2812	4.27E-13	3.97E-25	142.4	0	1.69E-14	3.526	1.47E-24	0.1193
	Min	0.2279	0.09987	9.72E-32	9.09E-13	1.50E-32	0.6667	0	4.32E-34	0.04111
	Q_1	0.5729	0.09987	5.13E-29	236.9	1.50E-32	0.6667	0	1.74E-31	0.04667
	Median	0.7567	0.09987	3.93E-28	355.3	1.50E-32	0.6667	5.68E-14	8.34E-30	0.1908
	Q_3	0.9475	0.09987	3.34E-27	355.3	1.50E-32	0.6667	5.68E-14	3.52E-29	0.248
	Max	1.516	0.09987	2.10E-24	592.2	1.50E-32	0.6667	14.14	8.17E-24	0.4837
	IQR	0.3746	1.05E-14	3.29E-27	118.4	0	5.55E-16	5.68E-14	3.51E-29	0.2013
	Time (s)	2.97	2.18	3.00	2.10	5.25	2.92	2.99	2.24	3.17
PSO	Mean	0.4628	0.1032	5.36E-62	1029	1.50E-32	0.6	41.94	1.29E-54	1.021
	Std	1.203	0.01795	1.82E-61	272.6	0	0.2	19.14	3.24E-54	0.7185
	Min	8.85E-03	0.09987	2.24E-66	533	1.50E-32	1.97E-31	5.68E-14	1.86E-56	0.1451
	Q_1	0.02913	0.09987	7.29E-65	809.3	1.50E-32	0.6667	28.27	7.62E-56	0.4329
	Median	0.04359	0.09987	2.67E-63	968	1.50E-32	0.6667	42.41	2.16E-55	0.9208
	Q_3	0.06032	0.09987	2.09E-62	1298	1.50E-32	0.6667	42.41	3.58E-55	1.398
	Max	4.125	0.1999	1.01E-60	1461	1.50E-32	0.6667	84.82	1.67E-53	2.822
	IQR	0.03119	0	2.08E-62	488.6	0	0	14.14	2.82E-55	0.965
	Time (s)	2.39	1.51	2.31	1.41	4.62	2.25	2.38	1.57	2.57
BA	Mean	8.776	1.387	0.02692	1629	7.593	0.6853	47.74	0.1641	4.03
	Std	1.838	0.3922	7.53E-03	302.1	4.618	0.01756	16.92	0.03611	0.4947
	Min	5.221	0.3999	5.18E-03	1051	2.29E-03	0.6103	14.26	0.07576	2.902
	Q_1	7.366	1.125	0.02365	1372	3.5	0.6807	42.51	0.1415	3.738
	Median	8.754	1.4	0.02881	1628	7.465	0.6848	42.58	0.1691	3.953
	Q_3	9.591	1.6	0.03066	1797	10.3	0.6921	56.75	0.1929	4.377
	Max	14.34	2.1	0.03768	2311	17.93	0.7176	84.98	0.2236	4.996
	IQR	2.226	0.475	7.01E-03	426	6.8	0.01143	14.24	0.05138	0.6385
	Time (s)	2.45	1.60	2.38	1.58	4.75	2.35	2.47	1.63	2.73
DE	Mean	3.038	0.09987	2.36E-59	263.4	1.50E-32	0.6667	5.68E-14	1.58E-42	0.05928
	Std	1.317	2.33E-09	4.57E-59	191.7	0	0	0	1.30E-42	0.06964
	Min	0.3399	0.09987	3.18E-62	0	1.50E-32	0.6667	5.68E-14	2.07E-43	7.64E-14
	Q_1	2.515	0.09987	1.35E-60	118.4	1.50E-32	0.6667	5.68E-14	6.33E-43	0.01309
	Median	3.323	0.09987	5.64E-60	240.7	1.50E-32	0.6667	5.68E-14	1.27E-42	0.04176
	Q_3	3.854	0.09987	2.03E-59	371.3	1.50E-32	0.6667	5.68E-14	2.01E-42	0.04667
	Max	5.619	0.09987	2.26E-58	741.5	1.50E-32	0.6667	5.68E-14	6.25E-42	0.3201
	IQR	1.339	9.20E-10	1.90E-59	252.9	0	0	0	1.38E-42	0.03359
	Time (s)	2.69	1.81	2.74	1.71	4.85	2.58	2.74	1.90	2.84
CMA-ES	Mean	9.00E-16	0.1022	6.32E-16	939.6	6.73E-16	0.6667	1.71E-14	2.37E-11	0.08834
	Std	3.34E-16	6.44E-03	1.96E-16	202.3	1.95E-16	1.99E-16	2.60E-14	4.78E-12	0.05829
	Min	2.77E-16	0.09987	2.32E-16	533	2.02E-16	0.6667	0	1.36E-11	4.91E-03
	Q_1	7.09E-16	0.09987	4.94E-16	834	5.68E-16	0.6667	0	2.10E-11	0.04667
	Median	8.30E-16	0.09996	6.04E-16	937.7	6.35E-16	0.6667	0	2.26E-11	0.08049
	Q_3	1.05E-15	0.1004	7.45E-16	1101	8.19E-16	0.6667	5.68E-14	2.59E-11	0.1087
	Max	1.71E-15	0.1284	1.00E-15	1263	1.16E-15	0.6667	5.68E-14	3.99E-11	0.2828
	IQR	3.42E-16	4.98E-04	2.51E-16	266.5	2.51E-16	3.33E-16	5.68E-14	4.94E-12	0.06202
	Time (s)	3.35	2.57	3.58	2.49	5.43	3.35	3.50	2.62	3.38

Mean (used in all comparisons), Std, Min, Q_1, Median, Q_3, Max and IQR $= Q_3 - Q_1$ over the runs; Time: mean CPU time (s).

A.5 CEC2017, $D = 10$ and $D = 30$ (Section 3.5)

This part supports Section 3.5. Tables **??–??** ($D = 10$) and **??–??** ($D = 30$) give the per-function results. Figures **??** and **??** show the distributions of the best values and the convergence curves for $D = 10$, and Figure **??** the distributions of the CPU times for $D = 30$.

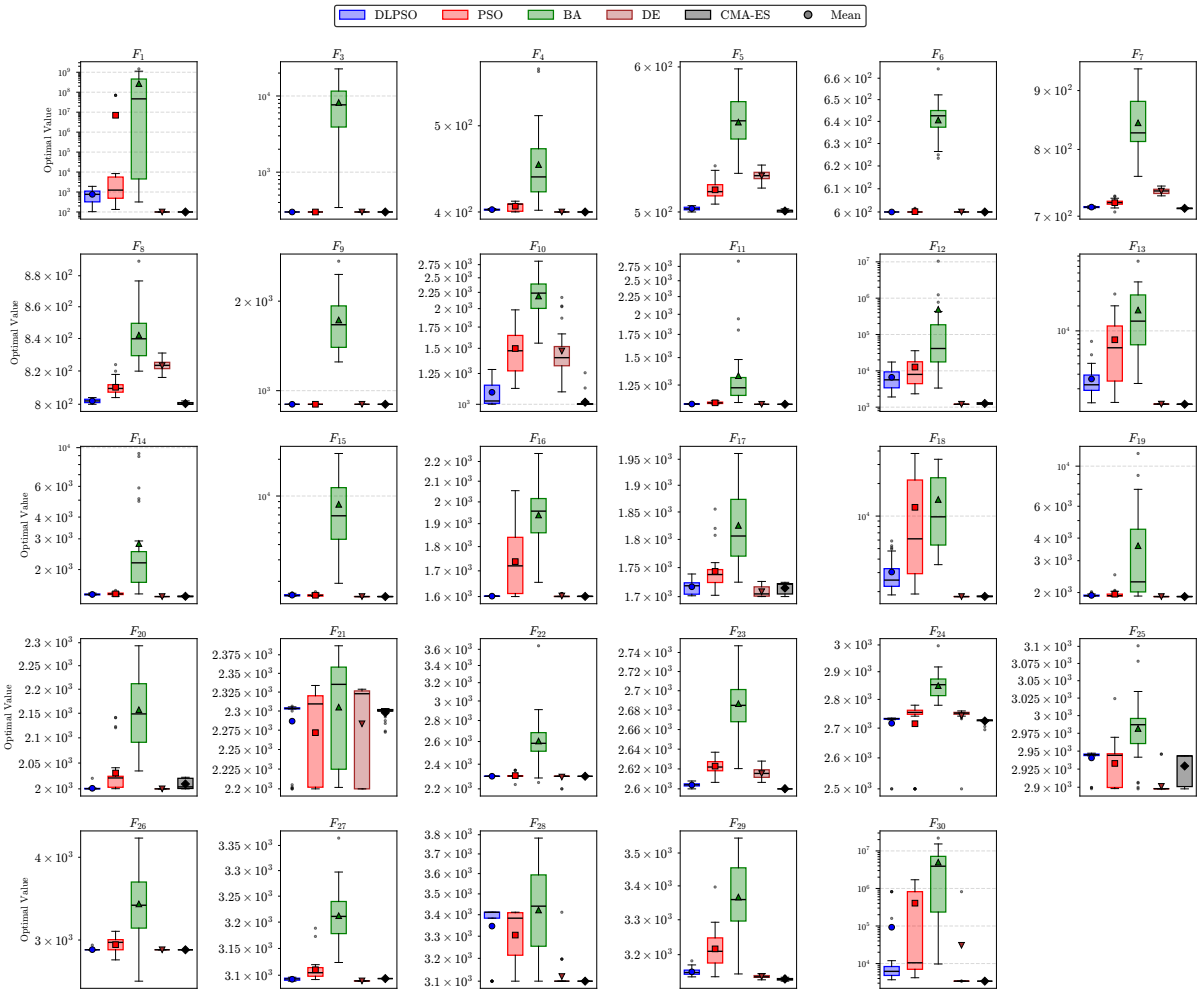


Fig. A18: Distribution over 30 runs of the best value $f(x_{best})$ of DLPSO, PSO, BA, DE and CMA-ES on the CEC2017 functions ($D = 10$, box: first quartile Q_1 to third quartile Q_3, line: median, marker: mean, whiskers: most extreme runs within 1.5 IQR of the box, points: other runs)

Table A27: Results on CEC2017, $D = 10$ (best value $f(x_{best})$, $F_i^* = 100\,i$; 30 runs, MaxFE $= 10^4 D$, $N = 100$) (part 1/4)

Algorithm	Metric	F_1	F_3	F_4	F_5	F_6	F_7	F_8	F_9
DLPSO	Mean	765.898294	300	402.464817	502.156303	600	712.828718	801.939722	900
	Std	514.8	1.18E-07	0.5559	1.022	3.36E-13	0.9125	1.14	2.08E-14
	Min	103.699884	300	401.810474	500	600	710.69188	800	900
	Q_1	321.376495	300	401.952747	501.373095	600	712.207826	800.994963	900
	Median	766.231668	300	402.397784	501.989919	600	712.859637	801.989918	900
	Q_3	1111.049923	300	402.860417	502.984877	600	713.542352	802.984877	900
	Max	1922.654669	300	403.57464	503.97991	600	714.491782	803.986905	900
	IQR	789.7	1.69E-08	0.9077	1.612	0	1.335	1.99	0
	Time (s)	0.51	0.52	0.55	0.53	0.57	0.58	0.53	0.59
PSO	Mean	7.0077E+06	300	405.995832	514.119685	600.151901	719.155274	810.148575	900
	Std	2.10E+07	2.08E-14	3.971	6.008	0.4098	4.713	4.562	2.94E-14
	Min	134.051063	300	400.039409	504.974795	600	705.929847	803.979836	900
	Q_1	486.648191	300	400.731064	510.209156	600	716.991395	807.213452	900
	Median	1246.323383	300	408.303365	512.934437	600	719.2375	809.452106	900
	Q_3	5631.975313	300	408.303365	517.461272	600	721.251691	811.69076	900
	Max	7.0058E+07	300	411.331838	529.848701	601.57973	728.840342	823.878992	900
	IQR	5145	0	7.572	7.252	1.14E-13	4.26	4.477	0
	Time (s)	0.17	0.18	0.19	0.18	0.21	0.24	0.18	0.23
BA	Mean	2.7031E+08	8214.023068	452.246426	559.766284	640.614707	843.763338	842.060075	1728.569299
	Std	3.84E+08	5457	44.97	18.53	8.784	48.23	16.64	394.1
	Min	317.613016	343.266938	401.79844	524.882122	623.4083	758.074842	819.905793	1249.213842
	Q_1	4506.645158	3912.680696	421.435038	548.011234	637.36528	812.804544	829.358526	1398.459722
	Median	4.6476E+07	7652.823338	438.020278	560.698203	642.573573	826.970284	839.803206	1666.05708
	Q_3	4.5778E+08	11589.527721	471.162255	574.380288	644.977245	880.728321	849.503602	1927.836547
	Max	1.4724E+09	22678.391533	579.342927	598.508566	664.400515	939.867098	889.550579	2725.016956
	IQR	4.58E+08	7677	49.73	26.37	7.612	67.92	20.15	529.4
	Time (s)	0.23	0.25	0.26	0.22	0.25	0.28	0.22	0.27
DE	Mean	100	300	400.000313	523.231457	600	735.984859	823.248365	900
	Std	0	0	1.26E-03	3.691	4.34E-13	3.633	3.085	0
	Min	100	300	400	515.249366	600	729.081953	816.08222	900
	Q_1	100	300	400	521.394647	600	732.534443	821.419997	900
	Median	100	300	400	523.355448	600	736.260295	823.428314	900
	Q_3	100	300	400	525.555733	600	738.740445	825.276827	900
	Max	100	300	400.00653	530.367063	600	743.502941	830.983974	900
	IQR	0	0	2.37E-08	4.161	7.96E-13	6.206	3.857	0
	Time (s)	0.34	0.35	0.37	0.34	0.39	0.40	0.35	0.42
CMA-ES	Mean	100	300	400	500.630141	600	711.101562	800.411767	900
	Std	0	0	0	0.6541	1.24E-11	0.4653	0.546	0
	Min	100	300	400	500	600	710.435888	800	900
	Q_1	100	300	400	500	600	710.686132	800	900
	Median	100	300	400	500.994959	600	711.051325	800	900
	Q_3	100	300	400	500.994959	600	711.355793	800.994959	900
	Max	100	300	400	501.989918	600	712.188196	801.989918	900
	IQR	0	0	0	0.995	1.99E-11	0.6697	0.995	0
	Time (s)	0.93	0.94	0.94	0.98	0.97	1.00	0.94	1.05

Table A28: Results on CEC2017, $D = 10$ (continued, part 2/4)

Algorithm	Metric	F_{10}	F_{11}	F_{12}	F_{13}	F_{14}	F_{15}	F_{16}	F_{17}
DLPSO	Mean	1090.726874	1101.875207	6633.026652	2625.854803	1439.886078	1540.647042	1601.583871	1716.893519
	Std	97.17	1.022	3959	1212	11.23	28.62	0.7339	11.07
	Min	1000.249822	1100.36013	1906.860296	1355.013638	1407.703907	1508.716755	1600.809172	1701.469634
	Q_1	1007.048536	1101.149688	3405.064234	1913.571451	1431.734541	1522.528615	1601.128009	1704.146031
	Median	1024.23088	1101.645789	5652.829482	2235.16605	1439.461282	1528.56742	1601.321383	1718.57017
	Q_3	1148.328212	1102.529461	9360.333438	2928.860285	1447.575136	1553.372901	1601.785573	1723.679396
	Max	1286.808917	1104.111414	17466.614629	7467.147442	1465.713407	1614.620444	1603.968344	1738.882998
	IQR	141.3	1.38	5955	1015	15.84	30.84	0.6576	19.53
	Time (s)	0.66	0.63	0.68	0.65	0.67	0.63	0.78	0.87
PSO	Mean	1498.007864	1110.838647	12719.420534	7812.719604	1451.103566	1538.053663	1737.497088	1743.414834
	Std	228	7.059	1.012e+04	6358	22.28	31.51	123.6	31.31
	Min	1122.040659	1101.990541	2330.794054	1368.669091	1408.76014	1502.5246	1600.029342	1702.302091
	Q_1	1273.963973	1105.69138	4431.859321	2478.576246	1437.667459	1512.535615	1611.493156	1724.165094
	Median	1473.670677	1110.168681	7973.51012	6252.390621	1449.735067	1535.924461	1720.133479	1737.939426
	Q_3	1646.828495	1115.889744	17758.28278	11459.075672	1465.072663	1548.706841	1839.850465	1746.565112
	Max	1981.908909	1125.988916	35752.072422	27855.983062	1518.491892	1643.598695	2053.642076	1855.319922
	IQR	372.9	10.2	1.333e+04	8980	27.41	36.17	228.4	22.4
	Time (s)	0.29	0.26	0.31	0.32	0.32	0.29	0.41	0.51
BA	Mean	2190.970194	1330.157222	4.9190E+05	17827.844149	2817.329727	8542.241922	1939.347577	1825.31344
	Std	295.8	337.4	1.86E+06	1.412e+04	1978	5463	131	67.22
	Min	1557.318938	1112.857225	3355.851181	2317.773463	1449.938535	1926.853844	1654.606715	1724.748602
	Q_1	2003.028397	1166.709617	17594.28757	6792.806777	1689.533619	4420.958497	1859.623305	1770.216874
	Median	2237.706216	1227.541239	41171.3587	13110.795481	2185.869147	6886.31851	1956.975554	1806.129671
	Q_3	2390.425311	1314.094497	1.8547E+05	27153.532327	2534.351668	11712.806876	2015.918594	1873.41853
	Max	2819.995991	2846.874993	1.0427E+07	69371.87766	9259.669032	22308.185763	2241.838108	1961.279512
	IQR	387.4	147.4	1.68E+05	2.036e+04	844.8	7292	156.3	103.2
	Time (s)	0.34	0.32	0.37	0.35	0.36	0.33	0.50	0.57
DE	Mean	1471.18996	1100.037112	1203.136816	1305.945359	1400.033184	1500.071002	1602.246663	1708.415758
	Std	253.4	0.1787	5.047	2.052	0.1786	0.157	4.07	8.194
	Min	1094.274655	1100	1200.00026	1300.151857	1400	1500.000016	1600.010981	1700.123074
	Q_1	1321.939303	1100	1200.209819	1305.03952	1400	1500.000464	1600.24317	1700.830412
	Median	1401.749523	1100	1200.416289	1305.642266	1400	1500.002611	1600.459028	1704.642691
	Q_3	1519.397571	1100.000066	1200.62798	1307.373605	1400	1500.014327	1600.689077	1716.757478
	Max	2168.690729	1100.994959	1212.280065	1309.20539	1400.994963	1500.498081	1611.655714	1726.011565
	IQR	197.5	6.58E-05	0.4182	2.334	3.10E-10	0.01386	0.4459	15.93
	Time (s)	0.48	0.43	0.49	0.51	0.48	0.46	0.58	0.70
CMA-ES	Mean	1015.325324	1100	1262.110495	1301.35752	1405.222773	1500.447825	1600.697544	1714.704593
	Std	49.97	0	58.36	2.3	8.451	0.5359	0.2016	9.282
	Min	1000.062454	1100	1200	1300	1400	1500.021482	1600.29896	1700.124375
	Q_1	1000.327886	1100	1200.468322	1300	1400	1500.078369	1600.549448	1704.524729
	Median	1003.571097	1100	1264.805645	1300	1400.000014	1500.276398	1600.716285	1721.31175
	Q_3	1003.711619	1100	1318.646478	1300.994992	1408.267521	1500.5	1600.796902	1722.24433
	Max	1256.527694	1100	1329.61129	1306.698101	1420.994908	1502.018181	1601.137407	1724.477351
	IQR	3.384	0	118.2	0.995	8.268	0.4216	0.2475	17.72
	Time (s)	1.00	1.01	1.08	1.09	1.02	0.96	1.08	1.23

Table A29: Results on CEC2017, $D = 10$ (continued, part 3/4)

Algorithm	Metric	F_{18}	F_{19}	F_{20}	F_{21}	F_{22}	F_{23}	F_{24}	F_{25}
DLPSO	Mean	3036.428561	1928.604623	2001.118462	2286.680856	2300.183768	2603.856374	2716.657027	2940.515321
	Std	1141	25.39	3.606	38.05	0.3592	1.759	57.92	13.86
	Min	1864.91209	1905.254145	2000	2200.000062	2300	2600.00002	2500.000875	2897.940444
	Q_1	2238.286307	1913.104636	2000	2302.760879	2300	2602.93494	2730.808672	2943.811316
	Median	2556.767863	1921.391319	2000.312173	2303.354184	2300.000003	2604.059857	2732.611592	2944.993963
	Q_3	3263.53256	1934.551729	2000.902337	2304.117917	2300.222986	2605.003977	2733.131769	2945.877123
	Max	5881.929418	2019.871032	2020.312182	2306.327524	2301.585856	2607.888162	2734.525721	2947.088442
	IQR	1025	21.45	0.9023	1.357	0.223	2.069	2.323	2.066
	Time (s)	0.70	0.89	0.88	0.95	1.10	1.28	1.14	1.24
PSO	Mean	12054.140188	1958.441047	2030.494547	2271.760873	2305.632186	2622.801566	2715.068712	2932.85041
	Std	1.245e+04	109.9	41.13	57.91	19.38	7.422	96.55	28.44
	Min	1901.320043	1902.181444	2000.312173	2200	2233.925573	2606.503911	2500	2898.098499
	Q_1	2930.700098	1911.785955	2003.048309	2202.080532	2301.463129	2618.174428	2747.680622	2899.630272
	Median	6160.107508	1922.311711	2020.965719	2309.324973	2302.04554	2621.929305	2754.371826	2944.161379
	Q_3	21562.968093	1961.030627	2024.213669	2320.087346	2303.130105	2627.35195	2761.828405	2946.267644
	Max	37937.829471	2506.98657	2141.381167	2333.771772	2349.697421	2636.894505	2779.543148	3024.266798
	IQR	1.863e+04	49.24	21.17	118	1.667	9.178	14.15	46.64
	Time (s)	0.33	0.53	0.50	0.60	0.73	0.86	0.76	0.86
BA	Mean	14234.596272	3629.729269	2157.251453	2305.050823	2604.754457	2686.600659	2849.326514	2981.775367
	Std	9595	2410	70.73	65.47	237.3	30.32	46.47	45.35
	Min	3552.732188	1912.572744	2034.586129	2201.836658	2249.84298	2620.349974	2779.40193	2897.79451
	Q_1	5395.259102	2015.983008	2091.13531	2224.790484	2512.547543	2668.012103	2813.489896	2960.581841
	Median	9839.630439	2291.564913	2148.643867	2335.249185	2583.604448	2684.850571	2852.678753	2987.114744
	Q_3	22629.025938	4477.197501	2211.614799	2358.35065	2683.004938	2701.240603	2873.30772	2995.981361
	Max	33541.52977	11748.701315	2293.298973	2387.253451	3644.980572	2747.074559	2996.836449	3100.848702
	IQR	1.723e+04	2461	120.5	133.6	170.5	33.23	59.82	35.4
	Time (s)	0.37	0.59	0.56	0.65	0.79	0.94	0.81	0.94
DE	Mean	1800.022446	1900.004421	2000.052029	2283.250035	2293.700041	2615.370567	2742.289333	2901.133678
	Std	0.08917	7.82E-03	0.1163	58.92	25.05	5.134	45.38	11.94
	Min	1800.000016	1900	2000	2200	2200	2606.492894	2500	2897.742869
	Q_1	1800.000488	1900	2000	2200	2300	2611.551433	2747.84052	2897.809399
	Median	1800.002577	1900	2000	2322.808648	2300.00001	2615.472742	2751.704938	2898.008989
	Q_3	1800.00675	1900.002347	2000	2326.530699	2300.344912	2619.172819	2754.379346	2898.032924
	Max	1800.499997	1900.019432	2000.312173	2328.782497	2302.739753	2627.914755	2759.409889	2945.816999
	IQR	6.26E-03	2.35E-03	0	126.5	0.3449	7.621	6.539	0.2235
	Time (s)	0.51	0.71	0.69	0.77	0.91	1.08	0.94	1.05
CMA-ES	Mean	1806.628896	1900.382188	2009.367615	2297.766882	2300	2600.096054	2724.641025	2929.29148
	Std	9.27	0.4171	9.4	7.828	3.58E-12	0.5173	7.091	20.27
	Min	1800.009734	1900.019432	2000	2272.360601	2300	2600	2694.526263	2897.742869
	Q_1	1800.47323	1900.110003	2000.717	2300	2300	2600	2725.331042	2900.873872
	Median	1800.5	1900.169589	2003.842155	2300	2300	2600	2727.164706	2943.32995
	Q_3	1820.371843	1900.805468	2020.308812	2301.899435	2300	2600	2727.919976	2943.359238
	Max	1821.485245	1901.281012	2022.304198	2303.242253	2300	2602.881632	2728.31883	2943.373121
	IQR	19.9	0.6955	19.59	1.899	5.46E-12	3.18E-12	2.589	42.49
	Time (s)	1.01	1.26	1.22	1.44	1.55	1.80	1.66	1.67

Table A30: Results on CEC2017, $D = 10$ (continued, part 4/4)

Algorithm	Metric	F_{26}	F_{27}	F_{28}	F_{29}	F_{30}
DLPSO	Mean	2901.571645	3092.52478	3346.6498	3151.98322	92688.102296
	Std	8.464	2.385	123.6	10.35	2.44E+05
	Min	2900	3089.005527	3100	3138.042498	3697.04549
	Q_1	2900	3090.751899	3383.745226	3145.262705	4817.673977
	Median	2900	3093.126069	3411.821808	3149.984013	6195.517904
	Q_3	2900	3094.523083	3411.821808	3156.66341	8329.036134
	Max	2947.149355	3096.257544	3411.821808	3182.596847	8.2060E+05
	IQR	1.82E-12	3.771	28.08	11.4	3511
	Time (s)	1.41	1.48	1.46	1.61	1.55
PSO	Mean	2952.207687	3110.143274	3305.162135	3216.354704	4.0576E+05
	Std	81.79	20.53	114.4	50.91	5.18E+05
	Min	2800	3091.996464	3100	3138.314942	4186.469132
	Q_1	2900	3098.098184	3213.833993	3176.549926	7028.435758
	Median	2975.751254	3104.652555	3383.754758	3209.484441	10475.326506
	Q_3	3004.090086	3114.142794	3409.589386	3247.693455	8.2058E+05
	Max	3092.344153	3188.331044	3412.052573	3396.65364	1.7051E+06
	IQR	104.1	16.04	195.8	71.14	8.14E+05
	Time (s)	1.03	1.10	1.06	1.25	1.19
BA	Mean	3401.197114	3212.233845	3422.517435	3366.687988	4.9984E+06
	Std	411.8	52.63	193.1	106.2	5.25E+06
	Min	2600.274573	3123.669655	3100.070929	3146.101929	9727.139206
	Q_1	3126.95286	3178.070465	3254.243052	3296.287734	2.3534E+05
	Median	3385.579416	3210.884917	3440.774784	3359.234125	3.9134E+06
	Q_3	3670.072027	3239.651057	3593.67197	3454.614477	7.1593E+06
	Max	4275.769703	3364.776529	3782.424744	3546.350386	2.1972E+07
	IQR	543.1	61.58	339.4	158.3	6.92E+06
	Time (s)	1.11	1.20	1.17	1.30	1.28
DE	Mean	2900	3089.474061	3120.050872	3139.04256	30634.945642
	Std	0	0.1769	61.41	4.369	1.47E+05
	Min	2900	3089.005527	3100	3129.83987	3394.503218
	Q_1	2900	3089.308077	3100	3136.6031	3394.553838
	Median	2900	3089.51799	3100	3138.167173	3394.647084
	Q_3	2900	3089.51799	3100	3141.578835	3394.761103
	Max	2900	3089.954938	3411.821808	3146.749523	8.2058E+05
	IQR	0	0.2099	0	4.976	0.2073
	Time (s)	1.22	1.31	1.27	1.43	1.37
CMA-ES	Mean	2900	3093.805968	3100	3131.79181	3394.512507
	Std	5.26E-11	0.06891	6.71E-10	3.554	0.02919
	Min	2900	3093.434866	3100	3126.129146	3394.500772
	Q_1	2900	3093.818765	3100	3129.296264	3394.500772
	Median	2900	3093.818765	3100	3131.080247	3394.503149
	Q_3	2900	3093.818765	3100	3133.346007	3394.514841
	Max	2900	3093.818765	3100	3140.032638	3394.665121
	IQR	6.23E-11	0	1.86E-10	4.05	0.01407
	Time (s)	1.87	2.12	2.06	1.98	2.23

Mean (used in all comparisons), Std, Min, Q_1, Median, Q_3, Max and IQR $= Q_3 - Q_1$ over the runs; Time: mean CPU time (s).

Table A31: Results on CEC2017, $D = 30$ (best value $f(x_{best})$, $F_i^* = 100i$; 30 runs, MaxFE $= 10^4 D$, $N = 100$) (part 1/4)

Algorithm	Metric	F_1	F_3	F_4	F_5	F_6	F_7	F_8	F_9
DLPSO	Mean	2556.141267	644.851638	491.67588	520.513364	600.000815	750.230354	821.605908	900.294909
	Std	2204	267.2	15.79	4.35	1.12E-03	4.593	5.793	0.4866
	Min	121.596577	303.909149	458.562699	512.934463	600.000014	739.067459	810.944605	900
	Q_1	429.792113	423.639688	483.673099	517.053278	600.000106	747.127567	818.819944	900
	Median	2661.623167	565.455178	487.263952	520.338491	600.000412	750.13609	819.940668	900.089528
	Q_3	4099.685718	802.369746	508.225423	522.884053	600.000779	754.071617	824.522806	900.268585
	Max	7826.222256	1236.312729	516.443797	529.848757	600.004305	759.494282	835.280972	902.085881
	IQR	3670	378.7	24.55	5.831	6.73E-04	6.944	5.703	0.2686
	Time (s)	2.01	1.98	2.12	2.07	2.15	2.28	2.06	2.25
PSO	Mean	6.2392E+08	300.570217	535.955603	606.655301	605.320855	817.247746	898.568239	1620.133947
	Std	1.03E+09	2.403	89.2	26.11	5.546	24.39	21.13	1322
	Min	107.508917	300.000026	458.561557	564.678784	600.062611	777.068529	868.652054	907.904962
	Q_1	318.81789	300.004224	489.069229	587.686814	602.113184	800.838177	884.571242	1021.408869
	Median	5590.376533	300.025907	516.067093	607.816419	603.815258	810.883105	897.08941	1155.22137
	Q_3	1.1447E+09	300.105243	540.952526	619.11588	606.791548	832.410627	907.206358	1503.658045
	Max	4.1030E+09	313.442	888.936369	688.501815	624.743686	879.935909	970.340187	6694.887146
	IQR	1.14E+09	0.101	51.88	31.43	4.678	31.57	22.64	482.2
	Time (s)	0.93	0.90	1.06	0.95	1.03	1.14	0.95	1.20
BA	Mean	1.5235E+10	77134.924987	2682.944405	779.075167	666.89213	1753.699689	1009.713837	8454.512008
	Std	4.48E+09	3.051e+04	1066	52.71	10.68	190.1	42.83	1997
	Min	3.1840E+09	20994.286787	1128.097672	657.203689	646.410138	1433.030094	903.476567	4453.435719
	Q_1	1.3306E+10	53682.631489	1871.977189	750.728145	660.913113	1620.653028	985.062515	7053.414665
	Median	1.5184E+10	76871.684063	2436.088313	768.140019	663.673943	1783.111222	1005.458931	8846.366751
	Q_3	1.8477E+10	92890.02674	3177.493175	810.845065	672.882915	1830.280061	1037.543154	9682.770636
	Max	2.2380E+10	1.4575E+05	5119.211003	907.927592	690.525352	2165.445043	1085.550315	13066.514579
	IQR	5.17E+09	3.921e+04	1306	60.12	11.97	209.6	52.48	2629
	Time (s)	1.08	1.14	1.18	1.09	1.23	1.28	1.11	1.29
DE	Mean	100	358.125827	458.561557	677.206995	600	911.135627	977.658994	900
	Std	1.04E-14	42.91	1.19E-10	10.64	2.64E-08	9.478	9.291	0
	Min	100	307.408408	458.561557	649.580788	600	893.251453	960.157024	900
	Q_1	100	320.225992	458.561557	669.938233	600	904.832762	969.046086	900
	Median	100	347.835495	458.561557	676.033123	600	910.825363	980.188639	900
	Q_3	100	385.12603	458.561557	685.305896	600	915.159983	983.714315	900
	Max	100	464.963458	458.561557	695.039385	600	931.641974	996.8583	900
	IQR	0	64.9	4.18E-11	15.37	1.59E-08	10.33	14.67	0
	Time (s)	1.32	1.38	1.47	1.45	1.53	1.66	1.45	1.62
CMA-ES	Mean	100	300	401.93864	505.969754	600	735.764988	806.500399	900
	Std	0	0	10.44	1.359	1.07E-11	1.002	1.353	0
	Min	100	300	400	503.979836	600	733.527028	803.979836	900
	Q_1	100	300	400	504.974795	600	735.08134	805.969754	900
	Median	100	300	400	505.969754	600	735.653767	805.969754	900
	Q_3	100	300	400	506.964713	600	736.409181	807.710933	900
	Max	100	300	458.157587	507.959672	600	738.539198	809.949591	900
	IQR	0	0	5.68E-14	1.99	1.03E-11	1.328	1.741	0
	Time (s)	6.57	7.15	11.34	7.33	6.91	7.07	7.24	7.87

Table A32: Results on CEC2017, $D = 30$ (continued, part 2/4)

Algorithm	Metric	F_{10}	F_{11}	F_{12}	F_{13}	F_{14}	F_{15}	F_{16}	F_{17}
DLPSO	Mean	3651.371542	1135.891452	39014.312957	10933.014986	2950.930121	2863.92521	1746.488037	1748.446942
	Std	330.3	28.64	2.619e+04	6490	1988	2150	130.1	13.32
	Min	3022.589754	1109.411218	9591.834129	1885.872479	1545.11714	1505.885226	1607.286067	1729.747099
	Q_1	3478.298645	1112.368663	23171.656937	5994.628219	1771.053047	1622.12927	1623.796612	1736.416522
	Median	3615.699592	1117.277	33800.762673	10317.513432	1944.947563	1998.364079	1724.572365	1746.649679
	Q_3	3818.259235	1170.466225	42452.631845	13455.916304	3401.374188	2971.678225	1860.674324	1759.49502
	Max	4778.069659	1182.816907	1.5131E+05	29085.427953	9307.155221	11863.880847	1967.165119	1781.443852
	IQR	340	58.1	1.928e+04	7461	1630	1350	236.9	23.08
	Time (s)	2.87	2.39	2.58	2.50	2.59	2.36	2.90	3.40
PSO	Mean	4183.368287	1256.416753	9.4882E+06	3.5922E+07	17957.38447	15952.28722	2523.716255	2135.797678
	Std	709	61.28	3.58E+07	1.90E+08	4.797e+04	1.589e+04	309.5	218.6
	Min	2694.140433	1158.287946	15144.314121	2915.744229	1976.376658	1743.220341	2058.027831	1824.344514
	Q_1	3738.773165	1205.24023	64677.802745	10359.566647	5999.075391	5106.066051	2267.106258	1951.982396
	Median	4131.127758	1255.49051	4.6772E+05	15109.731996	7412.118327	11907.723723	2510.17933	2087.330944
	Q_3	4631.637299	1305.314885	1.6144E+06	39865.721739	9929.914407	18094.025554	2781.476779	2283.167477
	Max	5544.764754	1400.754559	1.9994E+08	1.0593E+09	2.7429E+05	73355.858456	3258.38185	2798.022124
	IQR	892.9	100.1	1.55E+06	2.951e+04	3931	1.299e+04	514.4	331.2
	Time (s)	1.63	1.24	1.51	1.42	1.50	1.36	1.74	2.26
BA	Mean	5727.560777	3417.108797	8.5755E+08	3.0396E+07	2.6830E+05	54567.370604	3440.900211	2869.249851
	Std	467.5	1966	7.12E+08	1.45E+08	3.80E+05	3.272e+04	438.9	442.8
	Min	4970.051273	1371.215577	5.6832E+06	24795.338027	1912.113775	4356.814436	2624.864022	1808.611206
	Q_1	5370.486616	1903.950605	3.3249E+08	59548.224534	15104.496462	36042.278004	3110.283132	2524.250913
	Median	5729.19616	2554.155085	6.9903E+08	1.0037E+05	1.1147E+05	54143.696484	3439.980963	2902.952796
	Q_3	5983.284235	4908.382517	1.2805E+09	1.3960E+05	3.3393E+05	66921.038743	3782.934573	3177.56291
	Max	6894.995054	8194.682408	2.6920E+09	8.0910E+08	1.4859E+06	1.5811E+05	4334.683165	3660.094283
	IQR	612.8	3004	9.48E+08	8.005e+04	3.19E+05	3.088e+04	672.7	653.3
	Time (s)	1.82	1.44	1.69	1.52	1.67	1.37	1.95	2.38
DE	Mean	7429.899135	1169.437701	10442.501394	1383.387749	1464.650068	1535.141052	2350.356232	1784.458379
	Std	738.9	29.13	5416	14.23	5.735	9.08	368	20.38
	Min	4892.601211	1114.794887	4288.233812	1367.111214	1446.74215	1505.349414	1625.105172	1760.933226
	Q_1	7336.416602	1149.258549	5863.510025	1377.419357	1461.090387	1531.521613	2076.143214	1772.394888
	Median	7546.919746	1154.475118	8469.410339	1380.797935	1466.399627	1538.015383	2433.865488	1777.883501
	Q_3	7964.67979	1202.502725	16024.181147	1385.478558	1468.016343	1540.88855	2614.455053	1789.185377
	Max	8112.739327	1228.662908	20419.318912	1451.737405	1475.073632	1549.110529	3054.104239	1845.738768
	IQR	628.3	53.24	1.016e+04	8.059	6.926	9.367	538.3	16.79
	Time (s)	2.24	1.68	1.95	1.82	1.93	1.73	2.27	2.74
CMA-ES	Mean	1766.516244	1123.490568	2021.515254	1310.070113	1496.674748	1527.951195	1683.817487	1743.958908
	Std	341.5	13.39	367.2	5.999	19.87	38.53	113.2	6.702
	Min	1127.127682	1104.974795	1433.939051	1300.994959	1456.803675	1503.342694	1603.062323	1729.137181
	Q_1	1509.258005	1111.156409	1766.817624	1303.233617	1483.179415	1508.201047	1607.273822	1740.48696
	Median	1798.28514	1124.873891	2001.915796	1312.268098	1499.096526	1510.216349	1620.139493	1743.771127
	Q_3	1955.541859	1134.581511	2214.534656	1315.643281	1512.529661	1522.138039	1727.980264	1749.09924
	Max	2854.825935	1164.67202	3169.112055	1317.421411	1535.414568	1642.413566	2123.956626	1753.713429
	IQR	446.3	23.43	447.7	12.41	29.35	13.94	120.7	8.612
	Time (s)	5.83	7.19	6.47	6.45	6.51	5.49	6.20	7.49

Table A33: Results on CEC2017, $D = 30$ (continued, part 3/4)

Algorithm	Metric	F_{18}	F_{19}	F_{20}	F_{21}	F_{22}	F_{23}	F_{24}	F_{25}
DLPSO	Mean	77507.466624	6419.946265	2086.965553	2321.029265	2300.491099	2672.091493	2840.093194	2887.353362
	Std	4.162e+04	3862	61.25	4.201	0.9822	7.417	4.28	0.8895
	Min	24779.710469	2041.591076	2009.981574	2313.295797	2300	2658.64298	2829.074113	2883.57313
	Q_1	43937.836644	3172.872805	2031.910472	2317.995089	2300	2665.202177	2837.521594	2887.053098
	Median	73992.812703	5762.453871	2057.529765	2320.049124	2300	2672.34895	2840.422001	2887.286159
	Q_3	1.0221E+05	8309.957019	2151.962314	2323.857576	2300	2678.026281	2842.014387	2887.742533
	Max	1.9801E+05	16736.655959	2181.176855	2329.752908	2302.459765	2683.676751	2852.789584	2888.980943
	IQR	5.827e+04	5137	120.1	5.862	4.55E-13	12.82	4.493	0.6894
	Time (s)	2.50	3.88	3.40	3.96	4.78	5.69	4.92	5.45
PSO	Mean	1.9159E+05	21592.047773	2380.656101	2392.204688	3719.613969	2840.4272	3025.273295	2907.475984
	Std	2.53E+05	3.396e+04	116.3	27.81	1890	64.48	58.37	31.61
	Min	11710.614202	2041.410984	2184.38257	2343.835424	2300	2695.169237	2937.388588	2886.92634
	Q_1	47479.121316	4297.412447	2281.18029	2374.575373	2416.186958	2807.418894	2974.666795	2888.754136
	Median	98046.065511	10961.00351	2382.810451	2392.590112	2617.976929	2824.794786	3014.923579	2891.071579
	Q_3	2.3579E+05	23751.861719	2488.643866	2406.748093	5949.445572	2864.215614	3068.883237	2917.550502
	Max	1.2395E+06	1.5758E+05	2607.32991	2460.979898	7474.812839	3048.532499	3144.031196	3003.534037
	IQR	1.88E+05	1.945e+04	207.5	32.17	3533	56.8	94.22	28.8
	Time (s)	1.45	2.78	2.26	2.76	3.66	4.43	3.68	4.25
BA	Mean	1.1301E+06	2.4497E+05	2838.350196	2566.557194	7093.19834	3348.728399	3670.348932	3406.539996
	Std	3.35E+06	3.09E+05	216.2	66.94	770.8	156.6	152.5	263.6
	Min	42309.65418	39681.358124	2386.198218	2459.809985	5752.216832	2995.444622	3428.121851	3086.219875
	Q_1	86620.917285	72182.376298	2744.583354	2511.078884	6527.880634	3298.074995	3582.363656	3214.231966
	Median	1.6066E+05	1.0552E+05	2798.741247	2559.847497	6989.710555	3355.796673	3633.416783	3330.774254
	Q_3	4.9305E+05	2.8188E+05	2957.562366	2613.433213	7510.651093	3446.462602	3767.789129	3507.240349
	Max	1.8571E+07	1.4333E+06	3330.98361	2735.804968	9379.192244	3688.129601	4001.945281	4025.009108
	IQR	4.06E+05	2.10E+05	213	102.4	982.8	148.4	185.4	293
	Time (s)	1.60	2.98	2.42	3.00	3.96	4.68	3.93	4.56
DE	Mean	1836.825719	1917.722476	2053.243656	2470.343611	4329.287096	2825.754524	2994.815238	2886.749874
	Std	4.428	6.627	61.25	9.395	3104	10.87	9.039	0.02211
	Min	1823.006377	1907.254134	2001.617664	2445.790847	2300	2783.468407	2975.847035	2886.692848
	Q_1	1834.498179	1911.089518	2030.797359	2464.71629	2300	2820.121666	2988.441176	2886.738606
	Median	1837.151832	1917.491757	2039.774815	2471.445717	2300	2828.299849	2996.507914	2886.74969
	Q_3	1839.066762	1922.821477	2048.303431	2475.818561	8667.349055	2830.163292	3000.657891	2886.767631
	Max	1844.957231	1930.113285	2344.975415	2486.212241	9461.367084	2844.661225	3011.447668	2886.803822
	IQR	4.569	11.73	17.51	11.1	6367	10.04	12.22	0.02902
	Time (s)	1.84	3.23	2.74	3.27	4.17	5.07	4.25	4.77
CMA-ES	Mean	1910.50059	1972.187797	2074.22099	2306.720258	2300	2639.491925	2809.619266	2886.709117
	Std	53.59	19.61	46.66	2.215	6.27E-11	4.13	2.666	0.01394
	Min	1842.365155	1918.055004	2031.067849	2301.738557	2300	2623.96749	2803.971812	2886.691862
	Q_1	1863.809782	1961.941047	2042.308728	2305.53839	2300	2638.403716	2807.760489	2886.699162
	Median	1896.657945	1975.256379	2058.490949	2306.908262	2300	2639.635637	2810.178044	2886.706322
	Q_3	1939.277444	1983.813794	2073.578174	2307.741929	2300	2641.617182	2811.583257	2886.720271
	Max	2044.386144	2011.906133	2178.72043	2314.014944	2300	2646.272292	2815.132647	2886.742353
	IQR	75.47	21.87	31.27	2.204	1.02E-10	3.213	3.823	0.02111
	Time (s)	5.71	7.37	7.03	9.08	10.34	12.09	12.35	28.72

Table A34: Results on CEC2017, $D = 30$ (continued, part 4/4)

Algorithm	Metric	F_{26}	F_{27}	F_{28}	F_{29}	F_{30}
DLPSO	Mean	3834.394275	3210.103311	3182.30373	3402.981881	5947.556331
	Std	84.8	6.592	38.76	38.25	1120
	Min	3715.028794	3197.869979	3100.000011	3334.067427	5126.964162
	Q_1	3765.14777	3206.628217	3189.334977	3381.885448	5296.347189
	Median	3832.621257	3208.71479	3192.045763	3398.096416	5675.186994
	Q_3	3895.860101	3211.464039	3207.260216	3416.94738	6148.868312
	Max	4060.562972	3229.801135	3240.126519	3512.130775	11234.011694
	IQR	130.7	4.836	17.93	35.06	852.5
	Time (s)	6.62	6.92	6.67	6.64	6.83
PSO	Mean	4964.500249	3261.09015	3294.833313	3788.553852	1.0663E+05
	Std	633.3	36.45	88.6	148.1	3.30E+05
	Min	2800	3208.353926	3209.324545	3498.808847	5122.007025
	Q_1	4851.349781	3235.489903	3235.067448	3671.576713	8448.954673
	Median	5049.575948	3254.307427	3266.272657	3813.727133	15571.69393
	Q_3	5269.947072	3272.133472	3308.284989	3856.630194	37882.598662
	Max	6030.573746	3350.651858	3620.572107	4162.890731	1.8121E+06
	IQR	418.6	36.64	73.22	185.1	2.943e+04
	Time (s)	5.34	5.76	5.38	5.39	5.68
BA	Mean	9359.375651	4135.599	4427.85785	5349.43147	2.3097E+07
	Std	1324	257.5	480.6	577.5	4.56E+07
	Min	6738.552611	3517.783113	3714.394303	4037.069432	1.0851E+06
	Q_1	8541.619392	4012.43306	4085.191318	4965.35458	2.6360E+06
	Median	9559.846023	4139.066204	4339.682617	5321.940365	5.8693E+06
	Q_3	10300.594117	4258.025649	4728.384071	5626.321718	1.2773E+07
	Max	11569.768139	4653.519034	5405.622167	6698.013672	1.9627E+08
	IQR	1759	245.6	643.2	661	1.01E+07
	Time (s)	5.69	6.08	5.68	5.60	5.88
DE	Mean	5221.440058	3197.544739	3133.742848	3471.735086	5016.092191
	Std	130.9	8.643	52.23	167.4	54.75
	Min	4835.734954	3186.044949	3100	3302.231272	4946.658548
	Q_1	5189.907484	3191.022341	3100	3376.189991	4963.939842
	Median	5227.681249	3194.70169	3100	3404.711577	4991.420583
	Q_3	5303.395628	3203.956393	3203.285685	3468.780487	5061.792077
	Max	5451.495794	3214.220156	3253.930304	3949.104516	5090.308316
	IQR	113.5	12.93	103.3	92.59	97.85
	Time (s)	6.01	6.29	5.89	5.99	6.14
CMA-ES	Mean	3454.116745	3203.022294	3100.124071	3332.03814	5003.252507
	Std	108.6	3.581	0.6681	10.68	24.98
	Min	3168.331283	3196.30284	3100	3311.545583	4967.612354
	Q_1	3382.540377	3200.552095	3100	3324.947362	4988.118017
	Median	3484.584756	3202.878484	3100	3331.941209	5004.260104
	Q_3	3513.609999	3204.48448	3100	3338.16816	5017.824935
	Max	3638.857118	3211.94719	3103.722116	3358.026754	5057.728753
	IQR	131.1	3.932	2.50E-11	13.22	29.71
	Time (s)	9.97	25.67	11.89	10.64	17.03

Mean (used in all comparisons), Std, Min, Q_1, Median, Q_3, Max and IQR $= Q_3 - Q_1$ over the runs; Time: mean CPU time (s).

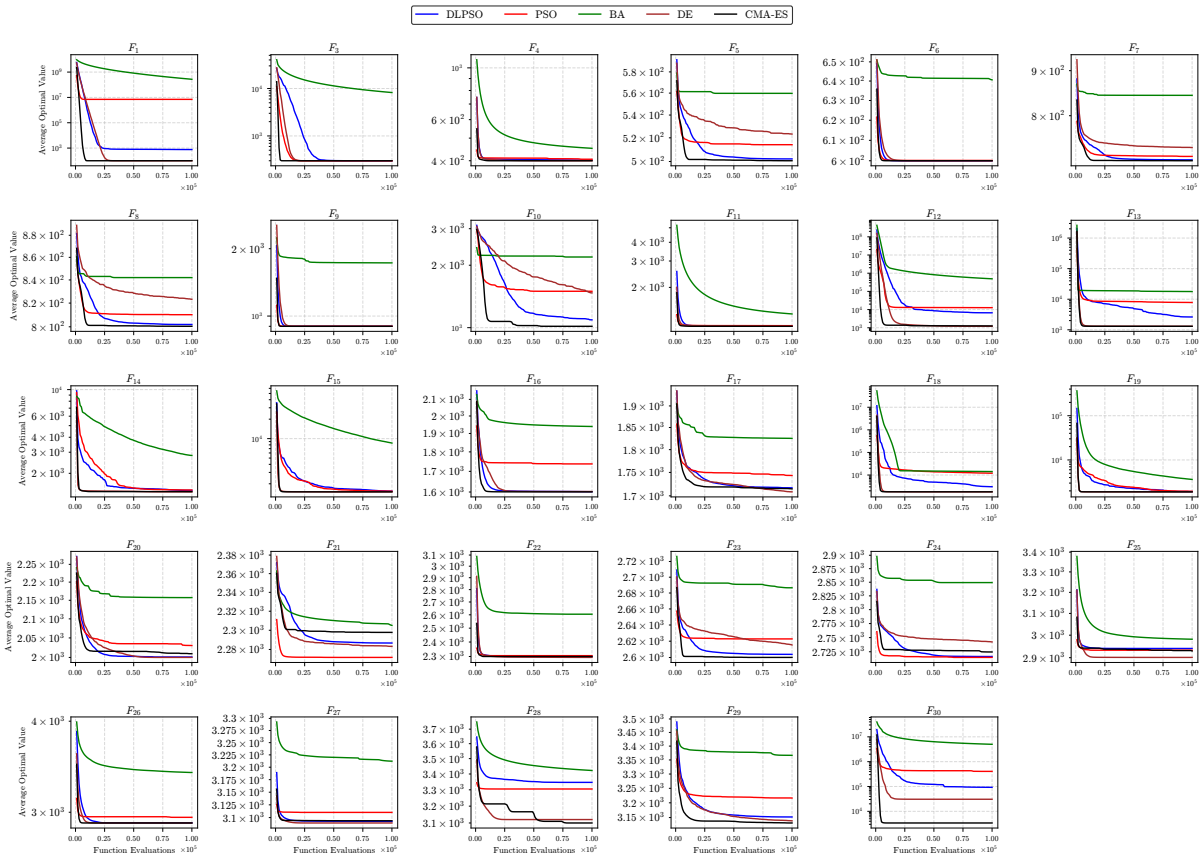


Fig. A19: Convergence curves of DLPSO, PSO, BA, DE and CMA-ES on all the CEC2017 functions ($D = 10$, mean over 30 runs of the best-so-far value $f(x_{best})$ as a function of the number of function evaluations)

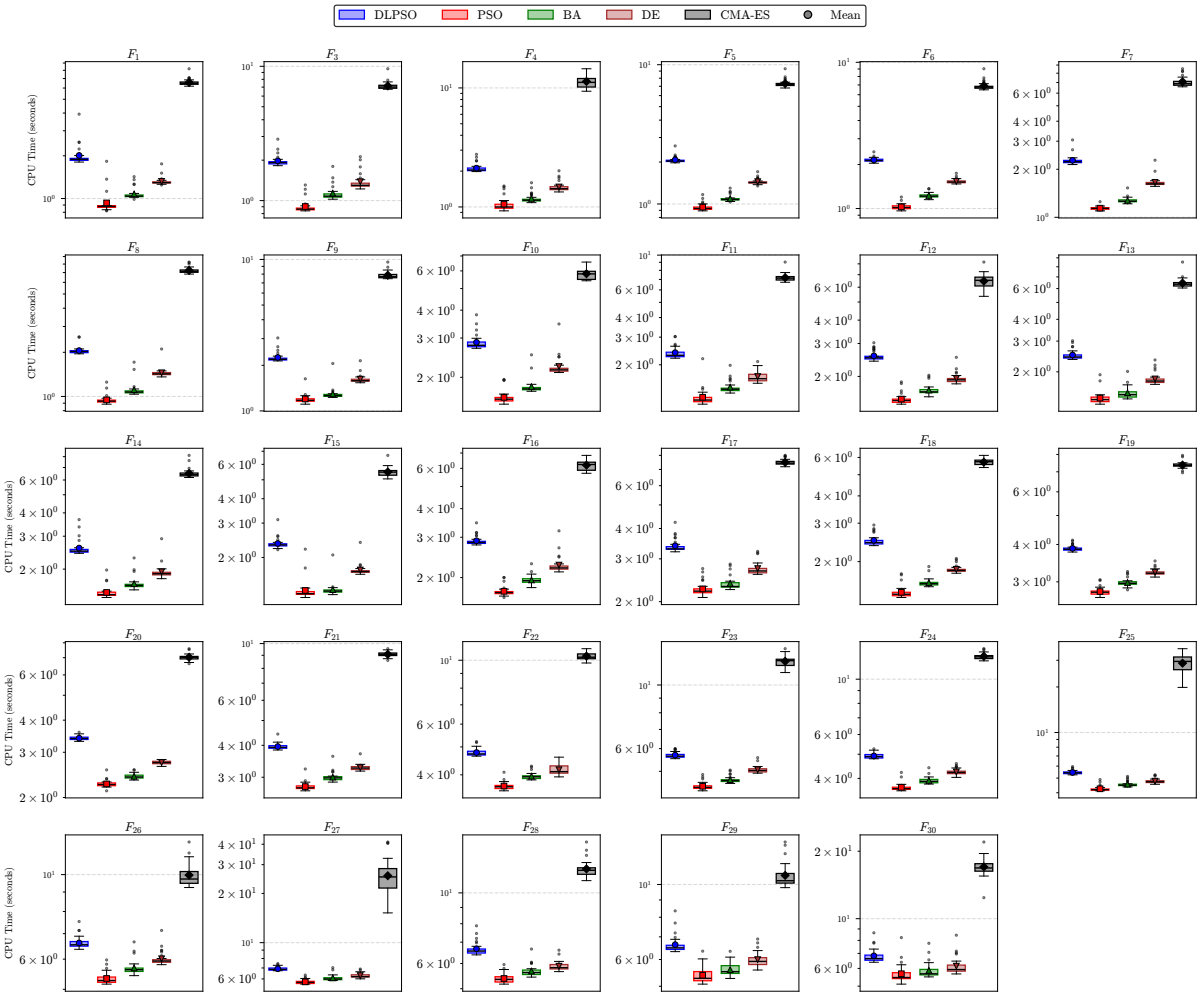


Fig. A20: Distribution of the CPU time (seconds, 30 runs) of one run of DLPSO, PSO, BA, DE and CMA-ES on the CEC2017 functions ($D = 30$, MaxFE $= 3 \cdot 10^5$, box: first quartile Q_1 to third quartile Q_3, line: median, marker: mean, whiskers: most extreme runs within 1.5 IQR of the box, points: other runs)

A.6 CEC2017, $D = 50$ and $D = 100$ (Section 3.6)

This part supports Section 3.6. Tables **??–??** ($D = 50$) and **??–??** ($D = 100$) give the per-function results. Figures **??** and **??** show the convergence curves, and Figure **??** the distributions of the CPU times for $D = 100$.

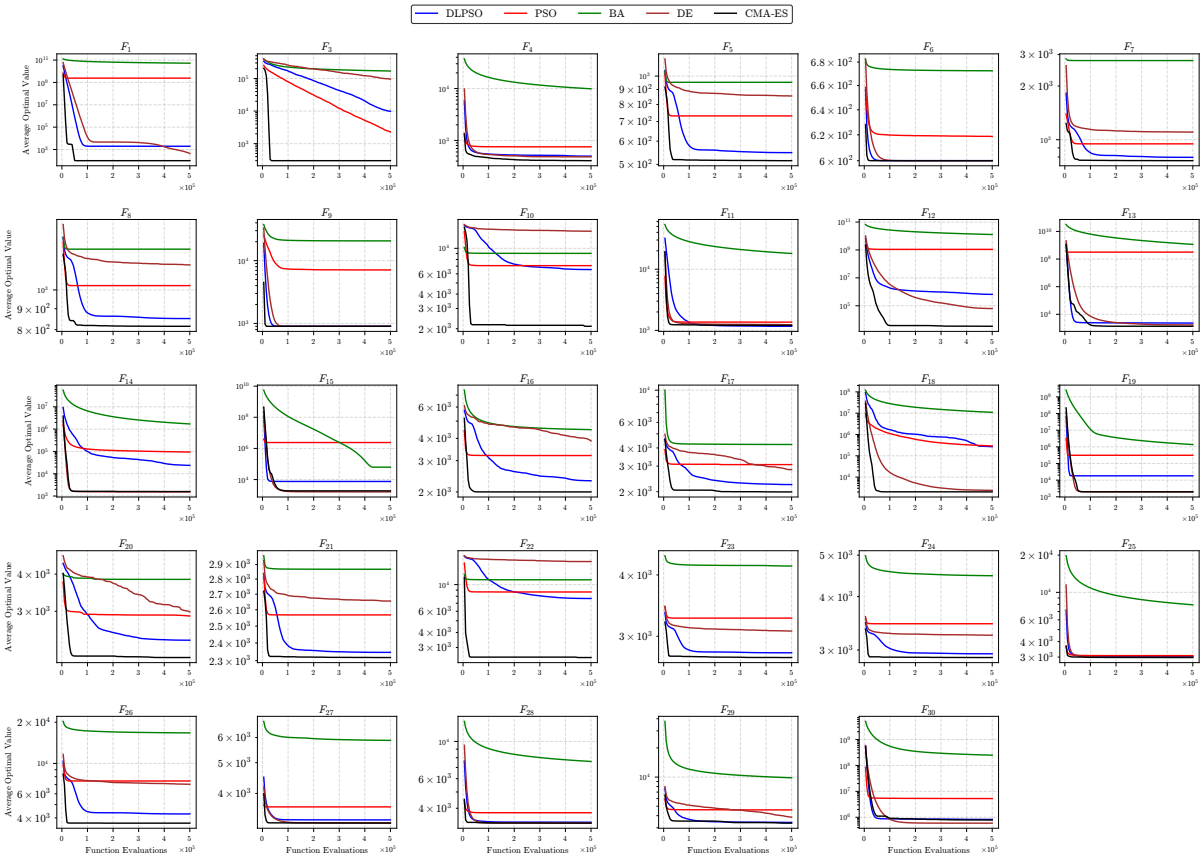


Fig. A21: Convergence curves of DLPSO, PSO, BA, DE and CMA-ES on all the CEC2017 functions ($D = 50$, mean over 30 runs of the best-so-far value $f(x_{best})$)

Table A35: Results on CEC2017, $D = 50$ (best value $f(x_{best})$, $F_i^* = 100\,i$; 30 runs, MaxFE $= 10^4 D$, $N = 100$) (part 1/4)

Algorithm	Metric	F_1	F_3	F_4	F_5	F_6	F_7	F_8	F_9
DLPSO	Mean	1954.337561	9851.403811	497.947553	548.438607	600.007749	796.538677	851.346144	902.403522
	Std	2180	4960	55.92	8.548	6.91E-03	11.08	8.912	1.761
	Min	100.019941	3164.959747	428.512713	534.823557	600.000675	773.337361	834.823557	900.089528
	Q_1	324.645129	5867.089103	429.566067	543.083565	600.002734	788.790691	845.029181	901.177233
	Median	796.005475	8957.280707	497.290685	548.589043	600.006113	796.254159	852.195903	901.765849
	Q_3	3105.769064	12672.480106	541.342087	551.784778	600.010397	801.888362	856.416229	903.575517
	Max	7144.747706	24446.696794	618.136397	575.864846	600.031512	830.757496	870.975948	906.597952
	IQR	2781	6805	111.8	8.701	7.66E-03	13.1	11.39	2.398
	Time (s)	4.26	4.31	4.48	4.62	4.19	4.95	4.59	4.90
PSO	Mean	2.4207E+09	2279.482415	753.491306	731.613199	618.727019	947.964992	1024.608662	7110.453849
	Std	1.95E+09	1625	196.5	54.1	8.92	53.48	46.35	2912
	Min	2361.648794	640.033066	484.994603	657.203041	604.852408	866.355478	952.228337	2088.793259
	Q_1	8.7442E+08	1376.345056	619.757938	690.161978	612.270267	904.627039	987.373986	5200.677287
	Median	1.8264E+09	1914.220503	712.567383	717.446714	614.98797	939.344799	1014.743357	6584.349603
	Q_3	3.8639E+09	2446.757824	817.809712	762.667655	625.346274	974.577072	1050.479144	8903.225738
	Max	7.1956E+09	9002.880607	1254.840938	892.011305	640.661215	1077.481687	1116.394856	14042.899619
	IQR	2.99E+09	1070	198.1	72.51	13.08	69.95	63.11	3703
	Time (s)	2.56	2.53	2.81	2.67	2.40	3.04	2.70	3.23
BA	Mean	5.2752E+10	1.6967E+05	9886.389462	952.337285	672.505048	2776.64714	1257.543891	20618.508555
	Std	1.11E+10	5.234e+04	2900	54.35	10.04	326.1	65.46	4716
	Min	3.2172E+10	78800.567274	2828.18071	859.178129	656.31634	2132.901843	1145.248735	11671.341461
	Q_1	4.4228E+10	1.4114E+05	8690.833287	913.899681	664.890902	2625.155351	1204.945487	18181.521136
	Median	5.3355E+10	1.6579E+05	10111.14866	946.235354	673.320853	2776.469498	1249.717536	19902.334513
	Q_3	5.9107E+10	2.0401E+05	11590.566202	975.088598	680.620182	2966.135633	1302.946153	24106.76815
	Max	7.6077E+10	2.8277E+05	15943.25775	1091.990719	691.261057	3504.087866	1390.997721	33211.506284
	IQR	1.49E+10	6.287e+04	2900	61.19	15.73	341	98	5925
	Time (s)	2.76	2.82	2.94	2.98	2.68	3.29	3.01	3.33
DE	Mean	444.031268	94850.069709	483.107632	855.472019	600	1102.817679	1150.757828	900.027081
	Std	905.6	1.319e+04	49.53	11.78	9.43E-07	10.64	13.54	0.08495
	Min	100.000307	70583.405591	415.791877	825.523227	600	1083.639511	1115.53945	900
	Q_1	102.272821	85869.025393	428.512712	850.115597	600	1095.684015	1142.861139	900
	Median	108.377413	92745.80709	497.413123	856.354235	600	1100.088796	1150.767897	900
	Q_3	158.762494	99513.008187	529.510574	863.296305	600	1112.327881	1159.802078	900
	Max	3905.701308	1.2846E+05	552.927823	875.366175	600.000005	1122.205434	1182.821147	900.454324
	IQR	56.49	1.364e+04	101	13.18	3.64E-08	16.64	16.94	0
	Time (s)	3.10	3.13	3.32	3.48	2.97	3.84	3.47	3.66
CMA-ES	Mean	100	300	407.739298	514.89122	600	762.859279	815.123376	900
	Std	2.03E-14	1.47E-14	12.55	2.319	7.98E-08	1.586	2.374	5.08E-14
	Min	100	300	400	508.954632	600	760.2869	808.954632	900
	Q_1	100	300	400	513.183208	600	761.738277	814.178165	900
	Median	100	300	400	514.924386	600	762.492576	814.924386	900
	Q_3	100	300	422.385014	516.914304	600	763.963098	815.919345	900
	Max	100	300	428.528712	518.904222	600	766.123432	820.894135	900
	IQR	4.26E-14	0	22.39	3.731	2.49E-11	2.225	1.741	0
	Time (s)	23.09	23.72	45.17	25.25	23.40	23.58	25.29	27.19

Table A36: Results on CEC2017, $D = 50$ (continued, part 2/4)

Algorithm	Metric	F_{10}	F_{11}	F_{12}	F_{13}	F_{14}	F_{15}	F_{16}	F_{17}
DLPSO	Mean	6541.358239	1165.754814	6.5372E+05	2328.527533	23689.109622	7538.967892	2311.639713	2238.213517
	Std	585.2	24.51	3.84E+05	1073	1.917e+04	4049	224.8	156.1
	Min	5425.078128	1135.721637	43206.176282	1358.755547	3065.380714	1590.588365	1852.378575	1949.461149
	Q_1	6147.971833	1152.592263	3.6679E+05	1553.952511	8461.088611	3751.334335	2149.687158	2143.594671
	Median	6509.84984	1164.718409	5.2774E+05	1988.911123	17270.600942	7056.714944	2302.051809	2225.394282
	Q_3	7030.962351	1173.347961	7.6517E+05	2771.504187	31240.16275	9516.587186	2441.609373	2326.37426
	Max	7516.57199	1266.455948	1.6221E+06	6294.139098	89102.437286	17025.688875	2808.959257	2668.340717
	IQR	883	20.76	3.98E+05	1218	2.278e+04	5765	291.9	182.8
	Time (s)	6.32	4.88	5.42	5.23	5.41	4.95	6.23	7.19
PSO	Mean	7069.263102	1363.383544	1.1016E+09	3.1710E+08	93724.877164	2.3904E+06	3202.934696	3071.308957
	Std	937.1	85.92	1.10E+09	7.16E+08	1.05E+05	1.28E+07	396.2	368
	Min	5709.336637	1236.712741	1.8748E+05	6934.152579	6221.129631	2732.584004	2412.749239	2331.87457
	Q_1	6357.281317	1310.490158	1.4017E+08	38052.783334	20761.653167	6734.490853	2943.799569	2853.703076
	Median	7035.619675	1339.861225	8.2357E+08	9.2726E+06	70073.126129	15456.109148	3124.351331	3171.541389
	Q_3	7609.997426	1399.318122	1.9663E+09	1.9204E+08	1.1442E+05	22665.123165	3509.976498	3283.903249
	Max	9376.866103	1654.987037	3.3215E+09	3.0001E+09	5.1923E+05	7.1231E+07	3984.538272	3745.514475
	IQR	1253	88.83	1.83E+09	1.92E+08	9.366e+04	1.593e+04	566.2	430.2
	Time (s)	4.27	3.11	3.77	3.45	3.84	3.35	4.36	5.19
BA	Mean	9038.968574	18078.003401	1.2779E+10	1.1515E+09	1.7055E+06	62567.527389	4478.772944	4225.387349
	Std	894.6	8663	4.51E+09	1.31E+09	2.63E+06	4.651e+04	526	468.8
	Min	7229.590388	3957.367539	3.7526E+09	58090.468953	2881.465439	20815.321358	3583.702234	3433.405636
	Q_1	8452.348339	11754.595973	9.7648E+09	3.3636E+07	82638.760711	28151.243626	4139.552775	3862.898108
	Median	8844.034179	17198.578056	1.3225E+10	7.8550E+08	6.1090E+05	45972.399428	4494.570799	4184.145269
	Q_3	9581.455061	24169.599738	1.6266E+10	1.6154E+09	1.9672E+06	75516.176826	4760.908594	4516.762595
	Max	10848.38349	36227.427911	2.0433E+10	4.8002E+09	1.1086E+07	1.8422E+05	6074.253808	5280.232984
	IQR	1129	1.242e+04	6.50E+09	1.58E+09	1.88E+06	4.736e+04	621.4	653.9
	Time (s)	4.64	3.44	3.96	3.67	4.01	3.39	4.74	5.45
DE	Mean	14093.738944	1235.761769	62900.314247	1706.852603	1528.291639	1608.316399	3867.416159	2835.612744
	Std	215.7	30.63	3.69e+04	375.2	7.726	9.583	883	445.6
	Min	13627.215034	1135.278371	19761.011907	1473.921241	1505.94895	1585.703804	1887.722114	2054.385766
	Q_1	13933.572759	1232.530744	31564.708326	1566.336993	1524.757949	1603.128132	3354.717978	2549.321054
	Median	14091.712931	1240.385302	59662.197211	1594.833717	1528.913318	1609.811848	4204.313716	2736.621736
	Q_3	14248.603201	1243.39128	72930.382627	1647.089632	1533.611018	1613.967166	4513.208235	3228.363385
	Max	14482.707692	1328.495902	1.8947E+05	3488.870779	1539.190452	1626.597533	4741.25073	3579.575472
	IQR	315	10.86	4.137e+04	80.75	8.853	10.84	1158	679
	Time (s)	5.21	3.72	4.30	3.97	4.30	3.84	4.94	6.01
CMA-ES	Mean	2099.988053	1205.878693	3370.542604	1359.356323	1584.265302	1878.864239	2000.716338	1991.544084
	Std	595.5	14.07	423.7	14.29	26.07	100.6	158	67.15
	Min	1252.319457	1167.017745	2407.59095	1327.858828	1532.411784	1732.985431	1728.158711	1904.74176
	Q_1	1769.926396	1199.104956	3140.740779	1350.996654	1571.232755	1788.270111	1876.624347	1948.785713
	Median	1942.845009	1204.328441	3425.615183	1358.965891	1583.669416	1873.575469	1966.683314	1971.116501
	Q_3	2489.568323	1216.268161	3631.642315	1365.897683	1593.86742	1943.640471	2151.077869	2013.110529
	Max	3754.868484	1233.679536	4234.68873	1407.894515	1662.767054	2078.191627	2323.266303	2233.439069
	IQR	719.6	17.16	490.9	14.9	22.63	155.4	274.5	64.32
	Time (s)	19.74	29.02	20.66	21.16	21.47	19.18	20.19	23.08

Table A37: Results on CEC2017, $D = 50$ (continued, part 3/4)

Algorithm	Metric	F_{18}	F_{19}	F_{20}	F_{21}	F_{22}	F_{23}	F_{24}	F_{25}
DLPSO	Mean	2.6733E+05	18144.102908	2398.719353	2346.346374	7662.837992	2776.321519	2939.984623	3041.733348
	Std	2.58E+05	5862	131.8	10.3	630.6	10.9	9.436	32.88
	Min	32639.50683	7413.932676	2157.659357	2330.852917	5915.233201	2755.130168	2920.24245	2961.788705
	Q_1	1.0455E+05	14311.184481	2312.735644	2338.19456	7271.64528	2768.635319	2934.265656	3027.699277
	Median	1.3863E+05	17915.721267	2411.446592	2345.342991	7543.73651	2775.658266	2938.002579	3062.971437
	Q_3	3.7739E+05	23215.858231	2464.538024	2351.209313	8102.958512	2785.206066	2947.543586	3063.418986
	Max	1.0325E+06	28398.848327	2655.341656	2370.056424	9097.296005	2803.00211	2960.388773	3082.741807
	IQR	2.73E+05	8905	151.8	13.01	831.3	16.57	13.28	35.72
	Time (s)	5.33	9.05	7.68	9.79	12.26	14.27	12.38	13.98
PSO	Mean	2.8871E+05	3.1158E+05	2892.948118	2568.235442	8681.223485	3266.812974	3456.207483	3081.304474
	Std	2.09E+05	9.04E+05	281.8	68.75	1414	138.3	122	46.43
	Min	39682.863659	2430.318851	2308.379784	2450.559912	2693.495563	3068.256703	3209.484062	3002.676567
	Q_1	1.3984E+05	16720.096729	2716.252342	2525.316211	8225.860443	3142.50201	3368.324102	3035.136278
	Median	2.7401E+05	26521.468432	2905.257768	2542.599031	8813.539779	3243.532521	3437.897099	3093.445244
	Q_3	3.7561E+05	78952.013272	3063.257179	2587.034838	9245.73863	3374.506339	3562.049633	3117.682126
	Max	9.5527E+05	4.8923E+06	3532.878408	2759.890409	11271.230673	3579.510871	3711.086278	3173.306788
	IQR	2.36E+05	6.223e+04	347	61.72	1020	232	193.7	82.55
	Time (s)	3.57	7.11	5.55	7.76	9.96	12.14	10.46	12.16
BA	Mean	1.0974E+07	1.3713E+06	3840.208458	2867.112303	10962.130115	4169.511882	4475.837358	7931.627929
	Std	1.28E+07	1.77E+06	382.7	82.55	1034	140.3	140.7	1460
	Min	2.7484E+05	57743.857687	2795.004491	2677.600029	9048.226459	3970.293864	4154.321245	4918.153788
	Q_1	2.1016E+06	1.8237E+05	3601.792484	2814.624875	10312.328395	4040.054345	4384.820948	7106.755745
	Median	5.9675E+06	4.0289E+05	3892.165434	2859.795048	10886.188268	4137.104634	4455.257113	7901.352263
	Q_3	1.4553E+07	2.3245E+06	4046.734625	2933.482364	11886.626308	4263.360221	4590.644938	8656.863956
	Max	5.1012E+07	7.2907E+06	4697.278299	3100.592119	13477.600165	4499.31302	4720.613774	11911.805789
	IQR	1.25E+07	2.14E+06	444.9	118.9	1574	223.3	205.8	1550
	Time (s)	3.89	7.40	5.75	8.01	10.38	12.63	10.87	12.45
DE	Mean	2391.231515	1961.995007	2985.137356	2655.507775	15541.614243	3074.67602	3249.445276	2993.078276
	Std	551	9.272	458.4	13.35	373.5	15.12	10.93	26.34
	Min	1963.610482	1923.027206	2269.057782	2612.969777	14638.504422	3031.818904	3215.816897	2980.235007
	Q_1	2114.644398	1957.658137	2590.696429	2647.690845	15284.898687	3064.734835	3246.017859	2980.275808
	Median	2254.082138	1962.300384	2886.075512	2657.789904	15626.961528	3074.718211	3250.721505	2980.31729
	Q_3	2406.157561	1968.952324	3423.446812	2665.579463	15825.368128	3084.353342	3255.974798	2980.3897
	Max	4676.66806	1970.980512	3756.26279	2680.080811	16382.491485	3101.370832	3272.228531	3062.697256
	IQR	291.5	11.29	832.8	17.89	540.5	19.62	9.957	0.1139
	Time (s)	4.17	7.81	6.29	8.48	11.05	13.23	11.17	12.83
CMA-ES	Mean	2016.983864	1990.699221	2097.646601	2317.355493	2475.297855	2715.255281	2881.139926	2973.754625
	Std	73.48	22.71	21.26	1.899	547.9	6.211	4.133	9.929
	Min	1887.39438	1945.884426	2061.803344	2314.311116	2300	2700.54705	2874.182094	2955.042534
	Q_1	1977.50864	1977.036276	2082.806216	2315.426994	2300	2710.889077	2878.885926	2959.660659
	Median	2016.307627	1989.518254	2095.470384	2317.018399	2300	2715.609905	2880.504761	2980.234149
	Q_3	2054.485057	2008.519355	2106.240686	2318.518566	2300	2720.149546	2882.484588	2980.23581
	Max	2270.994858	2025.331186	2157.414896	2320.71477	4740.123878	2727.887261	2889.630847	2980.239197
	IQR	76.98	31.48	23.43	3.092	2.54E-10	9.26	3.599	20.58
	Time (s)	19.64	23.37	22.18	31.03	30.45	35.76	41.33	66.81

Table A38: Results on CEC2017, $D = 50$ (continued, part 4/4)

Algorithm	Metric	F_{26}	F_{27}	F_{28}	F_{29}	F_{30}
DLPSO	Mean	4273.471555	3300.936251	3302.87616	3406.139519	8.2634E+05
	Std	172	29.2	9.899	90.28	4.894e+04
	Min	4059.482855	3247.52405	3263.729018	3263.179392	7.4715E+05
	Q_1	4171.776914	3282.957728	3301.437288	3339.528458	7.8834E+05
	Median	4221.702319	3297.788981	3305.155503	3394.201435	8.2561E+05
	Q_3	4322.236814	3327.163263	3307.694695	3456.090255	8.6385E+05
	Max	4797.760916	3355.98725	3323.374797	3656.277362	9.3094E+05
	IQR	150.5	44.21	6.257	116.6	7.551e+04
	Time (s)	16.68	17.92	16.97	14.50	15.35
PSO	Mean	7440.29614	3627.934685	3761.174067	4586.801689	5.2739E+06
	Std	1169	170.2	646.7	463.5	5.99E+06
	Min	3324.823167	3337.092461	3279.218886	3606.403026	8.8260E+05
	Q_1	6697.928682	3498.020742	3358.008862	4307.062602	2.1625E+06
	Median	7383.470463	3647.786824	3448.640641	4577.63503	3.4948E+06
	Q_3	8089.031083	3739.050924	3892.828109	4831.642513	5.7317E+06
	Max	9936.961869	3996.984524	5891.239864	5502.504795	3.2759E+07
	IQR	1391	241	534.8	524.6	3.57E+06
	Time (s)	14.64	16.23	14.95	12.50	13.46
BA	Mean	16710.766306	5876.537832	7599.073714	9833.156106	2.4911E+08
	Std	1719	660.7	1095	2670	1.29E+08
	Min	13389.203919	4348.670894	5575.578246	6721.588408	7.7729E+07
	Q_1	15373.171367	5425.357394	6937.039353	7897.796293	1.5493E+08
	Median	16871.113495	5818.863263	7666.551777	8829.067742	2.2536E+08
	Q_3	17547.548714	6272.143314	7973.986623	10959.995063	3.0087E+08
	Max	20878.168244	7613.893734	10989.567775	17674.101137	5.4311E+08
	IQR	2174	846.8	1037	3062	1.46E+08
	Time (s)	15.23	16.63	15.27	13.02	13.68
DE	Mean	7045.037219	3234.60601	3269.694995	3861.29412	5.9300E+05
	Std	141.2	30.34	19.74	547.8	1.992e+04
	Min	6723.376506	3196.453364	3258.848723	3309.888287	5.8242E+05
	Q_1	6957.443747	3213.855053	3258.848723	3364.723055	5.8242E+05
	Median	7058.920292	3224.049746	3258.848723	3559.862513	5.8242E+05
	Q_3	7124.450145	3242.294013	3258.848723	4434.229017	6.0126E+05
	Max	7363.221584	3320.407572	3307.694693	4868.083715	6.5745E+05
	IQR	167	28.44	1.27E-11	1070	1.884e+04
	Time (s)	15.65	16.68	15.64	13.29	13.87
CMA-ES	Mean	3659.549611	3224.265494	3260.47697	3346.820955	7.8519E+05
	Std	116.8	5.399	8.768	55.18	7.193e+04
	Min	3339.491507	3210.482292	3258.848723	3268.323649	6.6490E+05
	Q_1	3597.782966	3220.928258	3258.848723	3308.99368	7.3006E+05
	Median	3677.159688	3224.209881	3258.848758	3329.691719	7.8088E+05
	Q_3	3714.232203	3229.125004	3258.848823	3377.201842	8.2729E+05
	Max	3962.784375	3235.190447	3307.694708	3477.969027	9.6380E+05
	IQR	116.4	8.197	1.00E-04	68.21	9.723e+04
	Time (s)	30.32	56.08	64.10	29.96	54.77

Mean (used in all comparisons), Std, Min, Q_1, Median, Q_3, Max and IQR $= Q_3 - Q_1$ over the runs; Time: mean CPU time (s).

Table A39: Results on CEC2017, $D = 100$ (best value $f(x_{best})$, $F_i^* = 100\,i$; 30 runs, MaxFE $= 10^4 D$, $N = 100$) (part 1/4)

Algorithm	Metric	F_1	F_3	F_4	F_5	F_6	F_7	F_8	F_9
DLPSO	Mean	4844.094254	87901.534427	615.381018	625.21525	600.069298	933.397099	925.223791	914.754426
	Std	4005	1.846e+04	26.81	22.11	0.05846	21.77	19.44	11.03
	Min	100.193587	60820.632472	578.315054	592.531127	600.004033	876.505791	887.556352	902.795424
	Q_1	1321.32423	72816.370799	597.38695	607.619475	600.026046	916.053447	914.253375	907.788696
	Median	3119.859498	84922.31448	601.648264	622.059861	600.047349	931.985204	923.872266	910.819293
	Q_3	7843.978855	1.0174E+05	628.915735	636.960098	600.087698	951.24478	938.796607	919.582514
	Max	13802.056154	1.3087E+05	701.753166	676.151704	600.20778	974.299621	967.152886	954.653481
	IQR	6523	2.892e+04	31.53	29.34	0.06165	35.19	24.54	11.79
	Time (s)	19.59	18.15	18.77	19.43	11.04	20.00	19.10	20.19
PSO	Mean	1.6570E+10	69105.449985	2042.645686	1172.324416	641.037068	1619.522481	1454.761036	21022.960805
	Std	9.17E+09	2.541e+04	1173	95.14	7.64	180.9	94	5336
	Min	2.5246E+09	31631.870102	867.915228	1037.104466	623.334376	1275.856736	1198.640776	10459.160296
	Q_1	9.4034E+09	47888.833629	1217.843911	1090.060686	636.107619	1475.972819	1412.996529	18576.759983
	Median	1.4773E+10	68730.501122	1644.526898	1172.416926	643.182868	1606.896484	1465.686888	20212.505954
	Q_3	2.2112E+10	82833.63294	2245.716817	1205.503865	646.199974	1713.97222	1498.644326	22473.20203
	Max	4.1882E+10	1.5914E+05	6261.841099	1444.571408	654.277197	2034.28816	1633.158536	32851.519037
	IQR	1.27E+10	3.494e+04	1028	115.4	10.09	238	85.65	3896
	Time (s)	15.79	14.50	15.73	15.54	7.68	16.24	15.55	16.90
BA	Mean	1.7184E+11	3.6095E+05	36637.674913	1561.086285	671.32559	6133.211957	2008.611546	34219.785701
	Std	2.03E+10	6.756e+04	8025	111.2	5.5	546.8	123.6	6022
	Min	1.3541E+11	1.9654E+05	21589.830727	1313.869205	659.860165	4932.917632	1755.147137	24130.299485
	Q_1	1.5465E+11	3.2075E+05	31273.192933	1511.110051	668.155981	5712.038691	1916.329078	29605.694059
	Median	1.7050E+11	3.7315E+05	35619.475209	1578.020802	670.359213	6134.634048	1981.996471	33625.025231
	Q_3	1.8428E+11	4.0757E+05	39403.209484	1645.427688	672.601457	6476.391063	2104.123991	37922.363136
	Max	2.1074E+11	4.6089E+05	62562.417295	1750.633685	685.940336	7334.326011	2255.595946	49829.87865
	IQR	2.96E+10	8.682e+04	8130	134.3	4.445	764.4	187.8	8317
	Time (s)	15.08	15.03	15.54	16.01	8.10	16.70	15.97	17.01
DE	Mean	8081.584769	4.2412E+05	615.688852	1312.632811	600.001236	1625.878158	1619.305493	907.02952
	Std	9219	4.136e+04	22.51	21.92	2.03E-03	24.18	18.62	7.759
	Min	110.016599	3.5869E+05	564.324151	1258.869863	600.000021	1568.039187	1562.441278	900.543852
	Q_1	1425.391781	3.9608E+05	597.436396	1297.140272	600.000133	1612.399489	1607.466996	902.788073
	Median	3347.756498	4.2471E+05	615.887532	1317.905131	600.000232	1627.647538	1624.895725	904.525534
	Q_3	11749.120701	4.5413E+05	636.462705	1326.194502	600.001404	1643.090448	1630.02083	908.248046
	Max	33832.881312	5.1671E+05	655.129902	1354.129424	600.008753	1666.74985	1646.100847	938.186082
	IQR	1.032e+04	5.804e+04	39.03	29.05	1.27E-03	30.69	22.55	5.46
	Time (s)	15.64	15.50	15.95	16.81	7.94	17.28	16.96	17.02
CMA-ES	Mean	100	300	420.139914	541.191677	600.000001	836.06907	842.352085	900
	Std	1.10E-14	0	53.68	3.414	5.77E-07	2.815	3.799	0
	Min	100	300	400	531.838685	600	830.831351	834.823567	900
	Q_1	100	300	400	539.05214	600	834.761767	840.047101	900
	Median	100	300	400	541.290793	600.000001	836.427069	842.28576	900
	Q_3	100	300	400	543.53003	600.000001	837.677405	844.773151	900
	Max	100	300	595.110221	547.758025	600.000002	844.109613	850.742887	900
	IQR	0	0	0	4.478	6.95E-07	2.916	4.726	0
	Time (s)	168.67	164.08	271.98	191.49	171.97	175.96	194.18	198.81

Table A40: Results on CEC2017, $D = 100$ (continued, part 2/4)

Algorithm	Metric	F_{10}	F_{11}	F_{12}	F_{13}	F_{14}	F_{15}	F_{16}	F_{17}
DLPSO	Mean	13951.897046	2033.418309	1.9395E+06	3975.546426	3.9118E+05	2429.922713	4049.826818	3426.528903
	Std	922.9	165.7	1.21E+06	2203	1.64E+05	1100	389	328
	Min	11754.72234	1769.022582	3.6436E+05	1547.244329	1.4845E+05	1587.034652	3164.874162	2837.239579
	Q_1	13559.047083	1907.081376	1.1557E+06	2366.215392	2.5391E+05	1759.750855	3805.193984	3237.542707
	Median	14091.396811	1979.551036	1.5502E+06	3472.305682	3.9456E+05	2025.37087	4031.27493	3413.164176
	Q_3	14723.273765	2173.852434	2.3178E+06	4517.911986	4.7734E+05	2328.46434	4277.704761	3562.127982
	Max	15286.126386	2421.485386	5.6829E+06	10528.293285	8.1469E+05	6733.469436	4967.159994	4358.721299
	IQR	1164	266.8	1.16E+06	2152	2.23E+05	568.7	472.5	324.6
	Time (s)	25.87	19.66	21.61	20.66	21.77	20.25	23.64	27.89
PSO	Mean	14780.769896	2623.230608	6.8828E+09	8.1926E+08	8.5510E+05	1.7340E+08	5793.648885	5537.556145
	Std	1179	1475	5.97E+09	1.02E+09	1.85E+06	4.02E+08	733.1	644
	Min	11606.331574	1839.461245	2.9425E+07	17241.530739	55588.171689	5429.448564	4103.253276	4270.694349
	Q_1	14206.873163	2162.654584	3.2617E+09	8.7509E+06	2.2765E+05	11228.975123	5371.802648	5094.400412
	Median	14831.962699	2362.877509	5.2653E+09	3.5901E+08	3.6097E+05	21698.285214	5697.290288	5501.968056
	Q_3	15429.278558	2563.258155	8.0025E+09	1.2213E+09	6.8363E+05	1.2113E+05	6119.485259	5725.641981
	Max	17423.823434	10425.711851	2.5474E+10	3.4780E+09	1.0316E+07	1.4659E+09	7532.345657	7348.342089
	IQR	1222	400.6	4.74E+09	1.21E+09	4.56E+05	1.10E+05	747.7	631.2
	Time (s)	21.26	16.30	18.75	17.70	19.09	17.36	19.89	23.51
BA	Mean	17187.154254	1.7099E+05	6.6181E+10	7.5379E+09	7.9164E+06	1.0120E+09	8781.941151	15269.65598
	Std	1513	6.194e+04	1.44E+10	3.18E+09	6.44E+06	8.50E+08	1182	1.357e+04
	Min	13856.35813	89862.095355	3.5927E+10	2.2405E+09	8.2431E+05	51858.767492	7087.620914	8091.177309
	Q_1	16148.832475	1.3021E+05	5.4922E+10	5.1303E+09	2.8408E+06	3.1911E+08	7800.426054	9970.849676
	Median	17645.634476	1.5656E+05	6.7208E+10	7.6947E+09	6.3617E+06	7.9911E+08	8398.195384	11858.130768
	Q_3	18306.850271	1.9102E+05	7.8066E+10	9.3264E+09	1.0372E+07	1.6374E+09	9813.149684	14673.796433
	Max	19607.1724	3.9461E+05	8.9517E+10	1.6924E+10	2.6855E+07	3.1577E+09	10779.819471	85084.128727
	IQR	2158	6.081e+04	2.31E+10	4.20E+09	7.53E+06	1.32E+09	2013	4703
	Time (s)	22.04	16.71	18.58	17.50	18.82	16.99	20.27	24.15
DE	Mean	31035.962042	1719.36237	2.4134E+05	7129.662911	13052.049992	7522.160233	9265.560954	6054.262016
	Std	404.3	68.67	9.384e+04	4843	1.111e+04	7116	376.8	826.5
	Min	30204.327795	1576.613261	96030.10348	1683.432136	2193.260275	1618.303676	7967.600411	3766.141662
	Q_1	30732.187417	1690.415895	1.6203E+05	3728.546986	5082.687467	2354.524382	9134.767334	5953.996979
	Median	31087.450591	1715.721826	2.4080E+05	5742.957468	7084.595345	3785.969529	9315.267186	6300.233835
	Q_3	31320.604119	1748.364315	2.8114E+05	9377.745983	17600.41414	11033.523387	9507.52207	6621.651086
	Max	31799.988922	1926.303058	4.7969E+05	19502.506014	42741.763243	26198.408978	9855.298372	7027.024188
	IQR	588.4	57.95	1.19E+05	5649	1.252e+04	8679	372.8	667.7
	Time (s)	22.90	16.91	18.96	17.70	19.03	17.32	20.75	24.41
CMA-ES	Mean	4480.549665	1908.31928	6380.936993	4931.848007	1740.177641	1777.141599	2436.328205	2719.718397
	Std	773	109.2	698.1	596	56.76	51	411.4	261.7
	Min	2815.341325	1630.073	4927.991953	3966.274084	1631.92457	1679.56124	1858.748503	2360.177922
	Q_1	3980.013786	1840.974807	5981.731055	4560.407042	1711.422481	1750.362143	2107.362672	2449.514402
	Median	4688.162885	1910.759561	6424.777345	4889.675513	1735.443339	1776.946076	2429.153147	2706.463344
	Q_3	5068.009221	1964.666369	6897.648814	5361.762746	1758.033297	1801.330769	2625.122353	2885.683744
	Max	5685.146104	2148.421675	8017.283962	6126.019147	1899.561939	1910.289796	3799.53474	3334.819351
	IQR	1088	123.7	915.9	801.4	46.61	50.97	517.8	436.2
	Time (s)	147.78	167.89	147.26	144.55	150.18	144.87	147.12	155.27

Table A41: Results on CEC2017, $D = 100$ (continued, part 3/4)

Algorithm	Metric	F_{18}	F_{19}	F_{20}	F_{21}	F_{22}	F_{23}	F_{24}	F_{25}
DLPSO	Mean	8.8434E+05	2549.946263	3760.565684	2464.245626	15908.827403	2964.641514	3408.600357	3271.74631
	Std	3.86E+05	860.2	322.6	20.1	948.9	21.4	30.92	44.16
	Min	2.2547E+05	1940.45462	3229.16436	2431.116362	13245.60302	2927.822094	3369.500868	3198.488802
	Q_1	6.0474E+05	2006.300648	3559.038986	2450.171688	15594.275678	2948.474511	3383.043563	3259.819954
	Median	7.7281E+05	2194.30036	3686.826124	2465.127482	16129.796376	2966.120008	3410.546081	3263.942273
	Q_3	1.1521E+06	2677.704559	3989.555151	2475.342209	16382.209234	2980.367524	3429.978809	3287.674626
	Max	1.6317E+06	5478.906188	4518.764316	2507.094357	17357.132886	3000.217559	3483.796811	3383.165639
	IQR	5.47E+05	671.4	430.5	25.17	787.9	31.89	46.94	27.85
	Time (s)	21.01	33.88	28.66	49.64	58.49	70.28	64.07	74.92
PSO	Mean	7.6363E+05	2.6304E+08	4968.151823	3203.889983	18476.450535	4405.131344	5279.157902	3644.363341
	Std	5.81E+05	4.66E+08	496.3	141.1	1408	234.1	407.1	605.6
	Min	2.8775E+05	3074.148768	4149.533773	2876.940354	16193.284689	4050.16073	4581.099399	3288.335813
	Q_1	3.9633E+05	2.3084E+05	4615.725929	3125.439016	17214.935802	4263.218274	4968.856238	3379.532745
	Median	5.2872E+05	9.0399E+06	4796.761046	3199.511691	18462.613468	4382.735201	5273.701958	3442.775577
	Q_3	9.1486E+05	4.6255E+08	5242.045447	3255.337838	19228.481878	4488.045501	5553.978138	3658.139212
	Max	3.0158E+06	1.7370E+09	5910.124449	3524.200107	21762.480696	4969.452933	6407.420037	6590.212143
	IQR	5.19E+05	4.62E+08	626.3	129.9	2014	224.8	585.1	278.6
	Time (s)	17.69	29.29	23.95	44.33	52.69	66.41	60.32	70.79
BA	Mean	7.1895E+06	1.0283E+09	6305.987125	4240.512553	20375.723865	5856.263742	9465.626955	17546.616811
	Std	7.84E+06	8.79E+08	464.3	197.8	1293	320.3	502.2	2377
	Min	1.4374E+05	5.3090E+06	5329.986898	3696.531862	17731.065116	5354.334486	8242.798801	13053.926357
	Q_1	1.9618E+06	4.6279E+08	5949.421813	4142.197388	19584.185042	5568.573297	9149.056572	16195.541879
	Median	4.5055E+06	7.4671E+08	6282.726329	4267.569685	20533.647366	5809.85817	9388.743498	17522.218473
	Q_3	9.9900E+06	1.3690E+09	6653.597518	4342.623422	21277.271237	6033.023922	9831.067014	19138.067291
	Max	3.4549E+07	3.1730E+09	7537.767691	4553.350043	23543.697587	6781.476656	10528.646485	22194.572896
	IQR	8.03E+06	9.06E+08	704.2	200.4	1693	464.5	682	2943
	Time (s)	17.67	30.46	24.05	45.04	53.96	66.85	60.59	71.40
DE	Mean	1.1850E+05	8433.487956	6429.790633	3146.275148	32638.169073	3109.819532	3993.914927	3250.182169
	Std	7.405e+04	5990	730.6	25.11	509.3	249.7	177.1	49.93
	Min	31019.569762	2138.212128	4470.407893	3071.400648	31328.368791	2889.300173	3335.235051	3152.656537
	Q_1	58010.677231	4069.709521	6401.888009	3140.395205	32428.628905	2919.912625	3992.810879	3203.766261
	Median	1.0632E+05	6395.137742	6678.385952	3153.056381	32697.922596	2943.731708	4041.198802	3262.104648
	Q_3	1.5362E+05	11297.778535	6841.542323	3161.321338	33100.7525	3322.507532	4070.979688	3292.485906
	Max	3.3622E+05	25160.477954	7037.840647	3174.932449	33309.508984	3641.376101	4123.612719	3359.904865
	IQR	9.561e+04	7228	439.7	20.93	672.1	402.6	78.17	88.72
	Time (s)	17.94	30.74	24.61	45.93	55.05	67.47	60.79	71.65
CMA-ES	Mean	2055.441896	2164.565125	2402.742544	2364.45004	6319.85977	2881.708364	3294.578389	3158.011879
	Std	60.91	49.96	84.45	3.815	1665	9.209	17.45	36.32
	Min	1951.719052	2069.197387	2284.153367	2356.830728	2300	2863.6183	3253.594113	3077.000986
	Q_1	2017.219947	2147.016081	2336.960503	2361.862029	5893.238464	2875.914332	3284.249644	3137.191021
	Median	2068.247957	2162.195863	2387.884226	2364.28662	6435.045393	2879.204681	3293.604014	3149.782024
	Q_3	2093.985812	2193.370434	2450.132991	2366.733744	7166.085275	2885.544161	3308.375991	3197.755287
	Max	2212.328969	2286.776895	2586.188441	2371.899233	9789.65265	2900.264835	3322.099183	3203.847806
	IQR	76.77	46.35	113.2	4.872	1273	9.63	24.13	60.56
	Time (s)	145.04	152.96	153.81	228.26	177.19	212.38	188.44	358.49

Table A42: Results on CEC2017, $D = 100$ (continued, part 4/4)

Algorithm	Metric	F_{26}	F_{27}	F_{28}	F_{29}	F_{30}
DLPSO	Mean	7073.931653	3378.114065	3373.621708	4727.897963	9673.719169
	Std	269.8	20.82	28.16	403.4	2260
	Min	6630.102616	3331.281623	3327.946083	3983.602816	6595.150675
	Q_1	6810.164216	3364.765168	3356.620908	4412.510536	7636.708885
	Median	7072.18743	3378.5285	3368.046242	4666.672656	9620.319436
	Q_3	7232.443517	3388.321843	3386.994523	4984.576799	11387.047578
	Max	7631.933628	3424.519875	3433.162745	5630.783746	14139.974674
	IQR	422.3	23.56	30.37	572.1	3750
	Time (s)	84.42	95.79	92.79	64.35	67.99
PSO	Mean	18846.567943	3982.749223	6029.972254	7176.343419	9.2016E+08
	Std	2716	284.6	2654	462.7	9.65E+08
	Min	13104.381371	3572.717272	3583.694061	6429.145415	5.5935E+05
	Q_1	17226.244543	3771.204517	4072.787507	6815.32937	1.9835E+07
	Median	18523.28463	3904.183386	5029.036142	7123.913305	5.9668E+08
	Q_3	20505.176541	4107.238746	6996.987129	7373.954026	1.3920E+09
	Max	24611.349618	4723.655984	13319.816253	8378.337042	3.7317E+09
	IQR	3279	336	2924	558.6	1.37E+09
	Time (s)	80.39	92.38	88.98	58.47	61.59
BA	Mean	40583.705823	10210.971447	21484.484868	18901.643122	7.4602E+09
	Std	3291	1259	2423	4337	2.84E+09
	Min	30615.47217	7292.589417	17226.380587	10775.940944	2.0348E+09
	Q_1	38910.363066	9292.786197	19677.717757	15335.594528	5.6623E+09
	Median	40814.390741	10131.029432	21546.854339	18593.936163	7.5244E+09
	Q_3	42477.130095	11191.684403	23341.300656	20181.674191	8.8728E+09
	Max	45456.088918	12658.248929	26673.563446	29637.797294	1.5797E+10
	IQR	3567	1899	3664	4846	3.21E+09
	Time (s)	81.52	93.33	90.24	59.95	62.92
DE	Mean	12172.855474	3325.743858	3353.252003	7439.472931	6663.175865
	Std	2558	28.72	37.01	1380	1229
	Min	6182.025083	3266.148678	3279.356107	4049.602013	5361.987905
	Q_1	12875.018348	3302.680953	3325.073173	7370.567278	5915.724102
	Median	13269.029569	3322.197821	3354.061489	8099.118915	6223.553997
	Q_3	13584.41807	3350.265166	3379.156892	8322.641233	6920.393442
	Max	14166.23985	3386.773732	3430.744707	8655.350288	10365.108751
	IQR	709.4	47.58	54.08	952.1	1005
	Time (s)	82.60	93.11	90.67	60.98	62.49
CMA-ES	Mean	5693.805666	3324.343761	3209.970303	4579.707654	5567.097106
	Std	151.8	18.26	96.94	337.7	155.4
	Min	5411.56942	3288.811362	3100	3909.756983	5222.057297
	Q_1	5588.891053	3315.169957	3100	4341.818504	5443.686112
	Median	5715.448947	3321.759953	3278.20606	4557.683263	5564.476453
	Q_3	5810.916864	3336.099527	3293.900194	4760.027078	5672.907585
	Max	6006.22061	3369.676575	3321.705014	5485.299907	5888.50695
	IQR	222	20.93	193.9	418.2	229.2
	Time (s)	212.84	330.47	368.65	213.33	232.32

Mean (used in all comparisons), Std, Min, Q_1, Median, Q_3, Max and IQR $= Q_3 - Q_1$ over the runs; Time: mean CPU time (s).

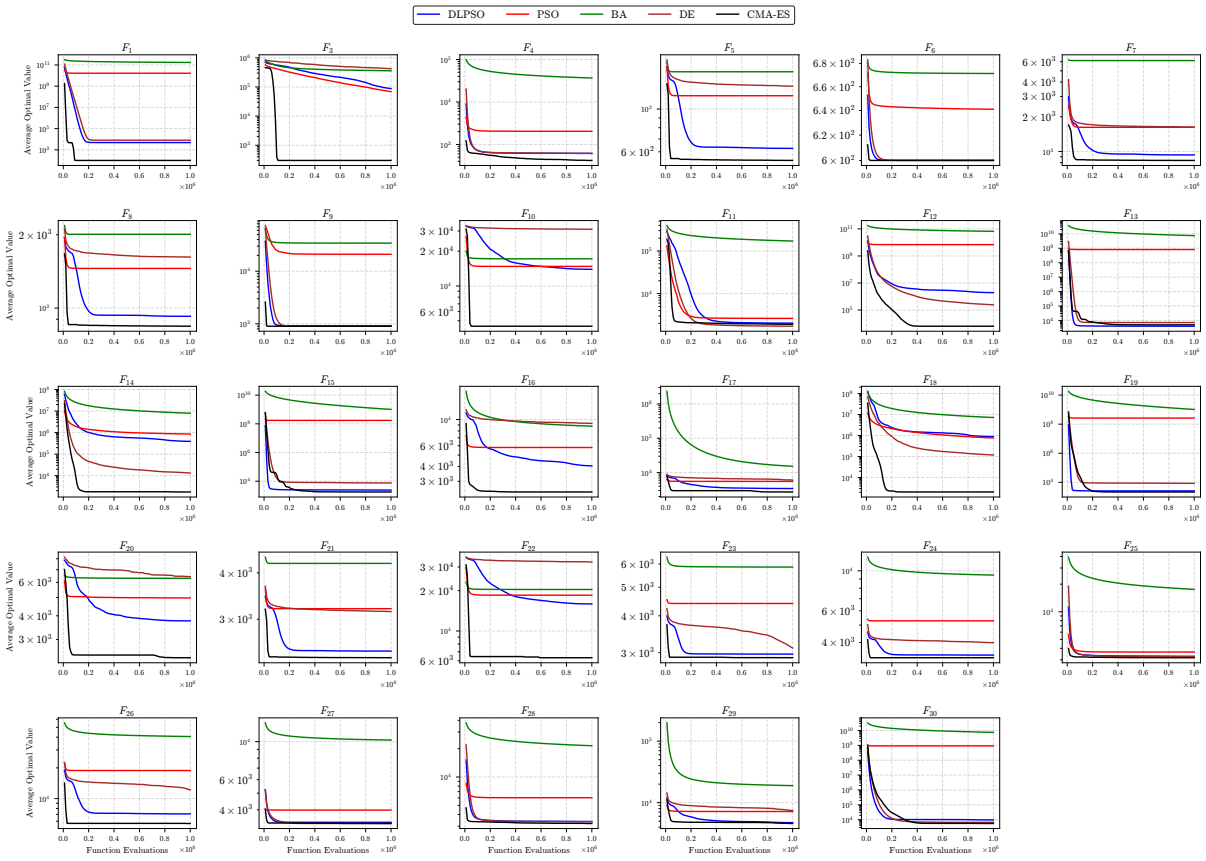


Fig. A22: Convergence curves of DLPSO, PSO, BA, DE and CMA-ES on all the CEC2017 functions ($D = 100$, mean over 30 runs of the best-so-far value $f(x_{best})$)

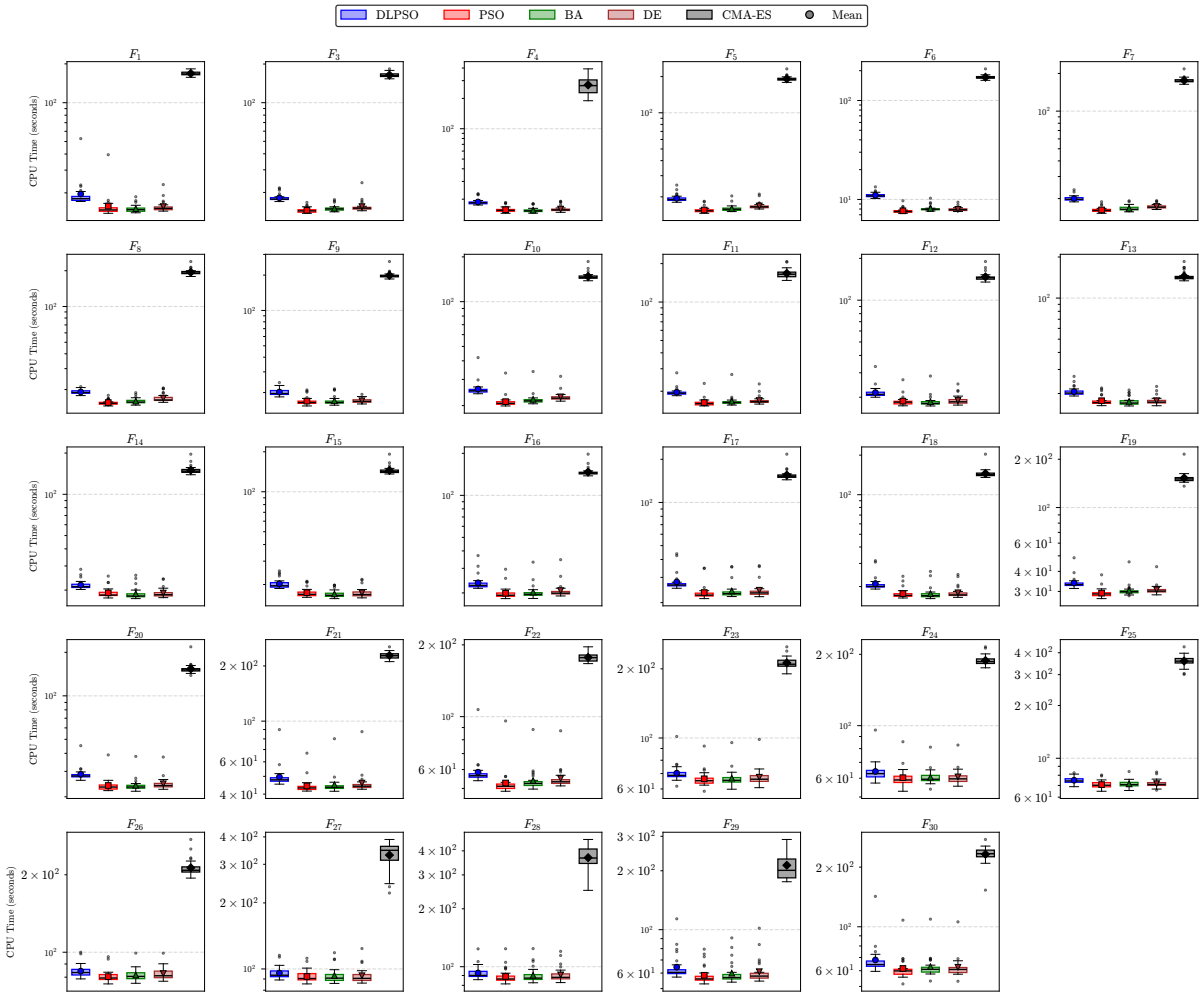


Fig. A23: Distribution of the CPU time (seconds, 30 runs) of one run on the CEC2017 functions ($D = 100$, MaxFE $= 10^6$, box: first quartile Q_1 to third quartile Q_3, line: median, marker: mean, whiskers: most extreme runs within 1.5IQR of the box, points: other runs)

A.7 CEC2013 LSGO, $D = 1000$ (Section 3.7)

This part supports Section 3.7. Tables **??–??** give the per-function results of the four algorithms, and Figure **??** shows the distributions of their CPU times.

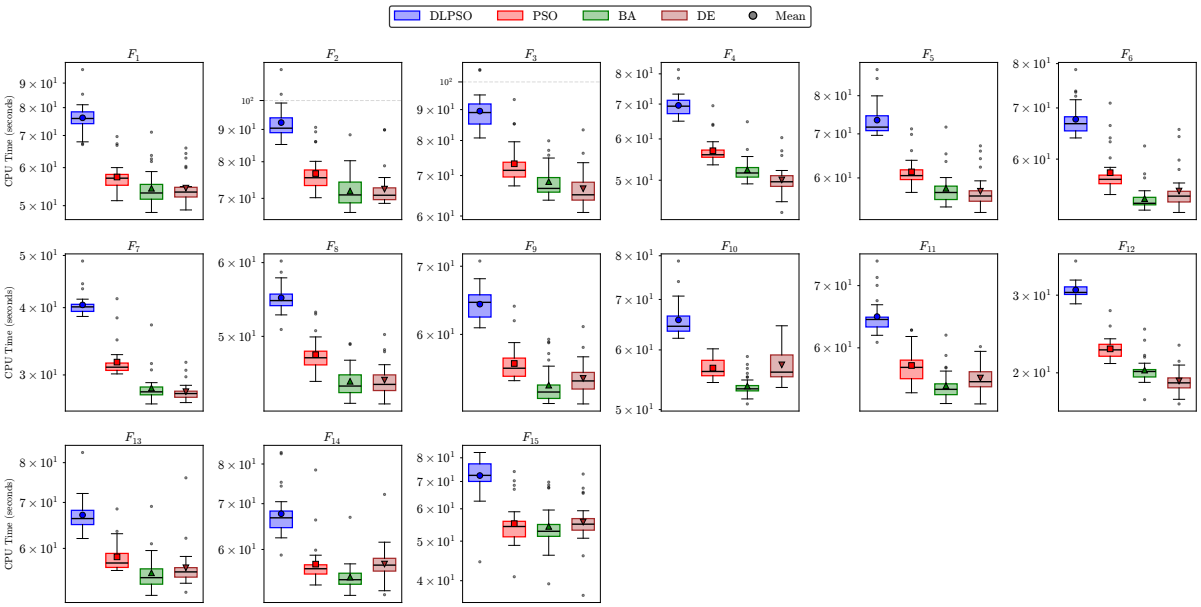


Fig. A24: Distribution of the CPU time (seconds, 30 runs) of the four algorithms on the CEC2013 LSGO functions ($D = 1000$, MaxFE $= 1.2 \cdot 10^5$, box: first quartile Q_1 to third quartile Q_3, line: median, marker: mean, whiskers: most extreme runs within 1.5 IQR of the box, points: other runs)

Table A43: Results on CEC2013 LSGO, $D = 1000$ ($D = 905$ for F_{13} and F_{14}; $F^* = 0$; 30 runs, MaxFE $= 1.2 \cdot 10^5$, $N = 100$) (part 1/2)

Algorithm	Metric	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8
DLPSO	Mean	2.1235E+09	22971.115763	21.564745	2.3027E+11	3.6287E+06	1.0574E+06	1.9350E+09	1.6137E+15
	Std	4.46E+08	1454	0.02364	4.31E+10	6.68E+05	3670	1.26E+09	8.86E+14
	Min	1.5927E+09	20349.437879	21.519073	1.6301E+11	2.7426E+06	1.0503E+06	8.5927E+08	3.1791E+14
	Q_1	1.8322E+09	22079.64394	21.549152	1.9857E+11	3.1301E+06	1.0546E+06	1.1895E+09	1.0290E+15
	Median	1.9710E+09	22782.156114	21.564688	2.2205E+11	3.5070E+06	1.0563E+06	1.6271E+09	1.2841E+15
	Q_3	2.2637E+09	24034.246746	21.586544	2.5516E+11	3.9799E+06	1.0600E+06	2.0538E+09	1.9614E+15
	Max	3.5023E+09	26246.785753	21.603965	3.4710E+11	5.3873E+06	1.0641E+06	7.0338E+09	4.4822E+15
	IQR	4.31E+08	1955	0.03739	5.66E+10	8.50E+05	5454	8.64E+08	9.32E+14
	Time (s)	76.20	92.28	89.48	69.58	73.45	67.64	40.47	55.00
PSO	Mean	3.6798E+10	30935.128998	21.219256	3.4974E+11	9.5653E+06	1.0613E+06	5.8524E+09	2.8920E+15
	Std	9.97E+09	1914	0.05721	2.12E+11	2.01E+06	2251	3.39E+09	2.25E+15
	Min	2.1117E+10	28051.175492	21.087462	1.5572E+11	4.9935E+06	1.0539E+06	1.6309E+09	3.5381E+14
	Q_1	3.1513E+10	29385.096338	21.186047	2.6890E+11	8.0019E+06	1.0602E+06	3.1394E+09	1.3051E+15
	Median	3.4603E+10	31179.793465	21.224644	3.0494E+11	9.7317E+06	1.0615E+06	4.7520E+09	1.8933E+15
	Q_3	4.2017E+10	32013.908547	21.261639	3.6888E+11	1.0748E+07	1.0632E+06	7.0623E+09	4.2185E+15
	Max	7.3288E+10	35614.313216	21.32402	1.4053E+12	1.3211E+07	1.0653E+06	1.5359E+10	9.8591E+15
	IQR	1.05E+10	2629	0.07559	1.00E+11	2.75E+06	2950	3.92E+09	2.91E+15
	Time (s)	57.36	76.63	73.22	56.96	61.29	57.60	31.69	47.79
BA	Mean	1.8588E+11	37738.952501	21.617596	4.0870E+12	1.6754E+07	1.0635E+06	4.8071E+13	1.5546E+17
	Std	1.05E+10	1811	7.53E-03	2.12E+12	3.40E+06	1190	3.27E+13	8.02E+16
	Min	1.5634E+11	34177.624526	21.603737	1.4782E+12	1.0962E+07	1.0610E+06	6.4694E+12	3.9285E+16
	Q_1	1.8198E+11	36676.091903	21.614752	2.8734E+12	1.4111E+07	1.0626E+06	2.4003E+13	9.1306E+16
	Median	1.8585E+11	37790.553055	21.619227	3.2959E+12	1.7006E+07	1.0635E+06	4.1251E+13	1.4238E+17
	Q_3	1.9037E+11	38776.191541	21.622665	4.7262E+12	1.8511E+07	1.0643E+06	5.6823E+13	2.0073E+17
	Max	2.0648E+11	41007.489187	21.628247	1.0520E+13	2.7406E+07	1.0655E+06	1.3710E+14	3.1700E+17
	IQR	8.39E+09	2100	7.91E-03	1.85E+12	4.40E+06	1657	3.28E+13	1.09E+17
	Time (s)	54.35	71.89	68.40	52.37	57.85	53.27	28.37	44.74
DE	Mean	6.2665E+09	22038.002422	21.57086	3.7383E+11	9.7564E+06	1.0635E+06	7.2769E+09	7.2020E+14
	Std	7.10E+08	657.8	7.84E-03	4.95E+10	3.94E+05	1361	1.47E+09	4.32E+14
	Min	4.4938E+09	20426.932427	21.552882	2.5298E+11	8.7666E+06	1.0596E+06	4.0121E+09	2.7150E+14
	Q_1	5.6751E+09	21767.432478	21.565248	3.4208E+11	9.4867E+06	1.0630E+06	6.1170E+09	3.8187E+14
	Median	6.2337E+09	22113.990181	21.574113	3.6557E+11	9.8937E+06	1.0639E+06	7.3310E+09	5.9596E+14
	Q_3	6.7034E+09	22372.450052	21.577344	3.9970E+11	1.0017E+07	1.0643E+06	8.3468E+09	8.9618E+14
	Max	7.8001E+09	23667.328922	21.581336	4.7838E+11	1.0550E+07	1.0653E+06	9.6063E+09	2.0349E+15
	IQR	1.03E+09	605	0.0121	5.76E+10	5.30E+05	1317	2.23E+09	5.14E+14
	Time (s)	54.38	72.40	66.58	50.07	57.27	54.50	27.93	44.88

Table A44: Results on CEC2013 LSGO, $D = 1000$ (continued, part 2/2)

Algorithm	Metric	F_9	F_{10}	F_{11}	F_{12}	F_{13}	F_{14}	F_{15}
DLPSO	Mean	3.6075E+08	9.3846E+07	1.2212E+11	1.0196E+11	3.2507E+10	4.3762E+11	1.0794E+08
	Std	5.07E+07	6.20E+05	7.37E+10	2.64E+10	7.74E+09	1.61E+11	3.21E+07
	Min	2.5602E+08	9.2462E+07	4.1255E+10	5.0467E+10	1.9992E+10	1.7312E+11	6.6981E+07
	Q_1	3.2283E+08	9.3416E+07	8.0230E+10	8.3011E+10	2.7260E+10	3.0633E+11	8.5082E+07
	Median	3.4387E+08	9.3829E+07	1.0445E+11	9.6229E+10	3.2622E+10	3.9903E+11	9.4461E+07
	Q_3	4.0274E+08	9.4271E+07	1.3741E+11	1.2257E+11	3.6792E+10	5.4548E+11	1.2535E+08
	Max	4.7363E+08	9.4903E+07	3.7478E+11	1.6684E+11	4.8572E+10	8.6074E+11	2.0045E+08
	IQR	7.99E+07	8.55E+05	5.72E+10	3.96E+10	9.53E+09	2.39E+11	4.03E+07
	Time (s)	64.25	65.76	64.83	30.83	67.11	67.67	72.38
PSO	Mean	8.4217E+08	9.3862E+07	6.2244E+11	4.6326E+11	8.4815E+10	8.5535E+11	5.2257E+13
	Std	1.11E+08	5.95E+05	3.51E+11	7.85E+10	2.92E+10	3.14E+11	4.60E+13
	Min	6.6024E+08	9.2615E+07	1.7866E+11	3.3366E+11	5.0406E+10	2.7190E+11	2.2364E+12
	Q_1	7.4115E+08	9.3467E+07	3.7661E+11	4.0195E+11	6.6838E+10	6.4516E+11	1.0328E+13
	Median	8.5251E+08	9.3851E+07	5.2360E+11	4.5875E+11	7.4614E+10	8.3695E+11	3.4602E+13
	Q_3	8.9462E+08	9.4237E+07	7.9376E+11	5.1327E+11	9.0966E+10	1.0781E+12	9.3909E+13
	Max	1.0536E+09	9.5180E+07	1.4326E+12	6.1168E+11	1.6641E+11	1.7155E+12	1.5441E+14
	IQR	1.53E+08	7.69E+05	4.17E+11	1.11E+11	2.41E+10	4.33E+11	8.36E+13
	Time (s)	56.18	56.76	57.44	22.67	58.33	57.11	55.19
BA	Mean	1.3815E+09	9.4619E+07	3.3712E+15	2.9318E+12	3.8475E+15	7.8946E+15	3.7545E+15
	Std	1.46E+08	2.78E+05	2.05E+15	2.31E+11	2.19E+15	7.24E+15	1.81E+15
	Min	1.0249E+09	9.3889E+07	8.3911E+14	2.5706E+12	7.9824E+14	2.2433E+15	1.7550E+15
	Q_1	1.2916E+09	9.4406E+07	1.9455E+15	2.7942E+12	2.3602E+15	4.3652E+15	2.4083E+15
	Median	1.4064E+09	9.4646E+07	2.8964E+15	2.9185E+12	3.5235E+15	5.5381E+15	3.3207E+15
	Q_3	1.4852E+09	9.4831E+07	4.0512E+15	3.0183E+12	4.9723E+15	9.3997E+15	4.4331E+15
	Max	1.6864E+09	9.5180E+07	1.0215E+16	3.5660E+12	9.7828E+15	4.1958E+16	9.6332E+15
	IQR	1.94E+08	4.25E+05	2.11E+15	2.24E+11	2.61E+15	5.03E+15	2.02E+15
	Time (s)	53.49	53.74	54.63	20.29	55.28	54.68	54.33
DE	Mean	7.7828E+08	9.4661E+07	5.0297E+11	4.9770E+11	1.5275E+11	1.5530E+12	2.7049E+11
	Std	3.52E+07	2.61E+05	2.29E+11	3.84E+10	3.51E+10	3.97E+11	2.13E+11
	Min	6.8560E+08	9.4079E+07	2.0246E+11	4.2504E+11	9.6731E+10	7.0894E+11	1.8137E+10
	Q_1	7.5785E+08	9.4484E+07	3.4012E+11	4.7074E+11	1.2129E+11	1.2951E+12	9.8229E+10
	Median	7.7438E+08	9.4697E+07	4.0809E+11	4.9813E+11	1.4815E+11	1.5753E+12	2.0268E+11
	Q_3	8.0244E+08	9.4816E+07	6.2753E+11	5.1763E+11	1.8163E+11	1.8501E+12	3.7848E+11
	Max	8.3795E+08	9.5213E+07	1.0240E+12	5.9535E+11	2.2660E+11	2.2259E+12	8.3355E+11
	IQR	4.46E+07	3.32E+05	2.87E+11	4.69E+10	6.03E+10	5.55E+11	2.80E+11
	Time (s)	54.32	57.32	55.67	19.17	56.22	57.12	55.68

Mean (used in all comparisons), Std, Min, Q_1, Median, Q_3, Max and IQR $= Q_3 - Q_1$ over the runs; Time: mean CPU time (s).